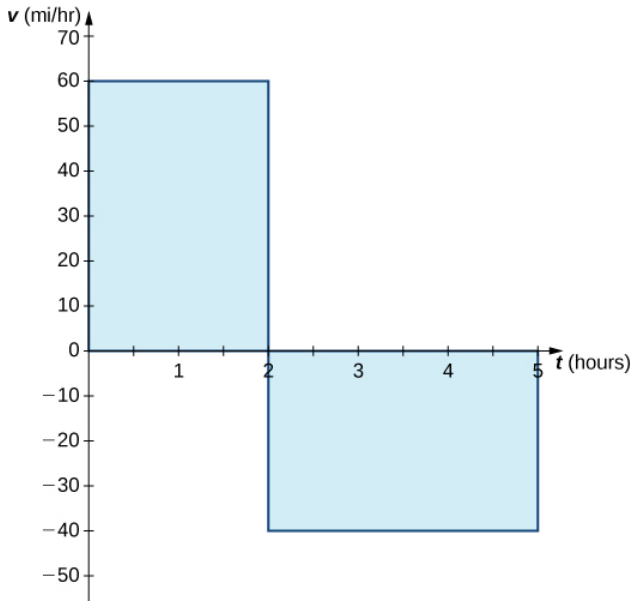


Figure 5.2.6: The area above the axis and the area below the axis are equal, so the net signed area is zero.

Suppose we want to know how far the car travels overall, regardless of direction. In this case, we want to know the area between the curve and the t-axis, regardless of whether that area is above or below the axis. This is called the **total area.**

Graphically, it is easiest to think of calculating total area by adding the areas above the axis and the areas below the axis (rather than subtracting the areas below the axis, as we did with net signed area). To accomplish this mathematically, we use the absolute value function. Thus, the total distance traveled by the car is

$$\int_0^2 |60|\,dt + \int_2^5 |-40|\,dt = \int_0^2 60\,dt + \int_2^5 40\,dt = 120 + 120 = 240.$$

Bringing these ideas together formally, we state the following definitions.

Definition: Net Signed Area

Let $f(x)$ be an integrable function defined on an interval $[a, b]$. Let A_1 represent the area between $f(x)$ and the x-axis that lies above the axis and let A_2 represent the area between $f(x)$ and the x-axis that lies below the axis. Then, the *net signed area* between $f(x)$ and the x-axis is given by

$$\int_a^b f(x)\,dx = A_1 - A_2.$$

The total area between $f(x)$ and the x-axis is given by

$$\int_a^b |f(x)|\,dx = A_1 + A_2.$$

Example 5.2.4: Finding the Total Area

Find the total area between $f(x) = x - 2$ and the x-axis over the interval $[0, 6]$.

Solution

Calculate the x-intercept as $(2, 0)$ (set $y = 0$, solve for x). To find the total area, take the area below the x-axis over the subinterval $[0, 2]$ and add it to the area above the x-axis on the subinterval $[2, 6]$ (Figure 5.2.7).

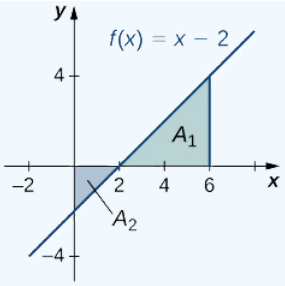


Figure 5.2.7: The total area between the line and the x-axis over $[0, 6]$ is A_2 plus A_1.

We have

$$\int_0^6 |(x-2)|\,dx = A_2 + A_1.$$

Then, using the formula for the area of a triangle, we obtain

$$A_2 = \frac{1}{2}bh = \frac{1}{2}\cdot 2\cdot 2 = 2$$

$$A_1 = \frac{1}{2}bh = \frac{1}{2}\cdot 4\cdot 4 = 8.$$

The total area, then, is

$$A_1 + A_2 = 8 + 2 = 10\,\text{units}^2.$$

? Exercise 5.2.4

Find the total area between the function $f(x) = 2x$ and the x-axis over the interval $[-3, 3]$.

Hint

Review the solving strategy in Example 5.2.4.

Answer

$18\,\text{units}^2$

Properties of the Definite Integral

The properties of indefinite integrals apply to definite integrals as well. Definite integrals also have properties that relate to the limits of integration. These properties, along with the rules of integration that we examine later in this chapter, help us manipulate expressions to evaluate definite integrals.

Rule: Properties of the Definite Integral

1.

$$\int_a^a f(x)\,dx = 0 \tag{5.2.1}$$

If the limits of integration are the same, the integral is just a line and contains no area.

2.

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$

If the limits are reversed, then place a negative sign in front of the integral.

3.

$$\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

The integral of a sum is the sum of the integrals.

4.

$$\int_a^b [f(x) - g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

The integral of a difference is the difference of the integrals.

5.

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$$

for constant c. The integral of the product of a constant and a function is equal to the constant multiplied by the integral of the function.

6.

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Although this formula normally applies when c is between a and b, the formula holds for all values of a, b, and c, provided $f(x)$ is integrable on the largest interval.

Example 5.2.5: Using the Properties of the Definite Integral

Use the properties of the definite integral to express the definite integral of $f(x) = -3x^3 + 2x + 2$ over the interval $[-2, 1]$ as the sum of three definite integrals.

Solution

Using integral notation, we have $\int_{-2}^{1} (-3x^3 + 2x + 2)\,dx$. We apply properties 3. and 5. to get

$$\begin{aligned}\int_{-2}^{1} (-3x^3 + 2x + 2)\,dx &= \int_{-2}^{1} -3x^3\,dx + \int_{-2}^{1} 2x\,dx + \int_{-2}^{1} 2\,dx \\ &= -3\int_{-2}^{1} x^3\,dx + 2\int_{-2}^{1} x\,dx + \int_{-2}^{1} 2\,dx.\end{aligned}$$

Exercise 5.2.5

Use the properties of the definite integral to express the definite integral of $f(x) = 6x^3 - 4x^2 + 2x - 3$ over the interval $[1, 3]$ as the sum of four definite integrals.

Hint

Use the solving strategy from Example 5.2.5 and the properties of definite integrals.

Answer

$$6\int_1^3 x^3\,dx - 4\int_1^3 x^2\,dx + 2\int_1^3 x\,dx - \int_1^3 3\,dx$$

Example 5.2.6: Using the Properties of the Definite Integral

If it is known that $\int_0^8 f(x)\,dx = 10$ and $\int_0^5 f(x)\,dx = 5$, find the value of $\int_5^8 f(x)\,dx$.

Solution

By property 6,

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Thus,

$$\int_0^8 f(x)\,dx = \int_0^5 f(x)\,dx + \int_5^8 f(x)\,dx$$

$$10 = 5 + \int_5^8 f(x)\,dx$$

$$5 = \int_5^8 f(x)\,dx.$$

Exercise 5.2.6

If it is known that $\int_1^5 f(x)\,dx = -3$ and $\int_2^5 f(x)\,dx = 4$, find the value of $\int_1^2 f(x)\,dx$.

Hint

Use the solving strategy from Example 5.2.6 and the rule on properties of definite integrals.

Answer

-7

Comparison Properties of Integrals

A picture can sometimes tell us more about a function than the results of computations. Comparing functions by their graphs as well as by their algebraic expressions can often give new insight into the process of integration. Intuitively, we might say that if a function $f(x)$ is above another function $g(x)$, then the area between $f(x)$ and the x-axis is greater than the area between $g(x)$ and the x-axis. This is true depending on the interval over which the comparison is made. The properties of definite integrals are valid whether $a < b$, $a = b$, or $a > b$. The following properties, however, concern only the case $a \le b$, and are used when we want to compare the sizes of integrals.

Comparison Theorem

i. If $f(x) \ge 0$ for $a \le x \le b$, then

$$\int_a^b f(x)\,dx \ge 0.$$

ii. If $f(x) \ge g(x)$ for $a \le x \le b$, then

$$\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx.$$

iii. If m and M are constants such that $m \le f(x) \le M$ for $a \le x \le b$, then

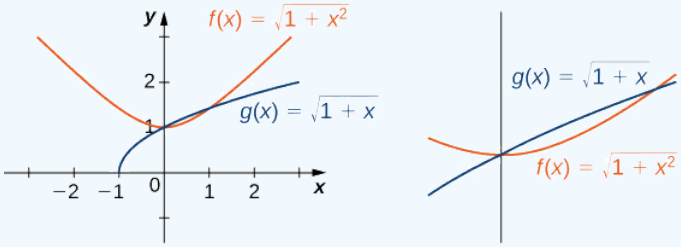
$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

Example 5.2.7: Comparing Two Functions over a Given Interval

Compare $f(x) = \sqrt{1+x^2}$ and $g(x) = \sqrt{1+x}$ over the interval $[0,1]$.

Solution

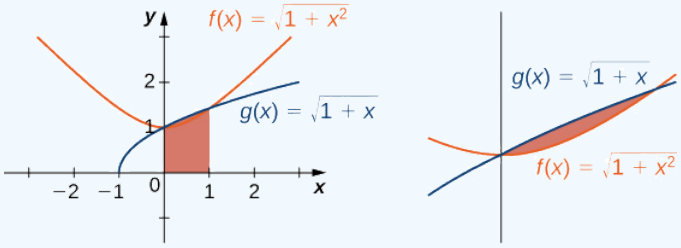
Graphing these functions is necessary to understand how they compare over the interval $[0, 1]$. Initially, when graphed on a graphing calculator, $f(x)$ appears to be above $g(x)$ everywhere. However, on the interval $[0, 1]$, the graphs appear to be on top of each other. We need to zoom in to see that, on the interval $[0, 1]$, $g(x)$ is above $f(x)$. The two functions intersect at $x = 0$ and $x = 1$ (Figure 5.2.8).



(a) (b)

Figure 5.2.8: (a) The function $f(x)$ appears above the function $g(x)$ except over the interval $[0, 1]$ (b) Viewing the same graph with a greater zoom shows this more clearly.

We can see from the graph that over the interval $[0, 1]$, $g(x) \geq f(x)$. Comparing the integrals over the specified interval $[0, 1]$, we also see that $\int_0^1 g(x)\,dx \geq \int_0^1 f(x)\,dx$ (Figure 5.2.9). The thin, red-shaded area shows just how much difference there is between these two integrals over the interval $[0, 1]$.



(a) (b)

Figure 5.2.9: (a) The graph shows that over the interval $[0, 1], g(x) \geq f(x)$, where equality holds only at the endpoints of the interval. (b) Viewing the same graph with a greater zoom shows this more clearly.

Average Value of a Function

We often need to find the average of a set of numbers, such as an average test grade. Suppose you received the following test scores in your algebra class: 89, 90, 56, 78, 100, and 69. Your semester grade is your average of test scores and you want to know what grade to expect. We can find the average by adding all the scores and dividing by the number of scores. In this case, there are six test scores. Thus,

$$\frac{89 + 90 + 56 + 78 + 100 + 69}{6} = \frac{482}{6} \approx 80.33.$$

Therefore, your average test grade is approximately 80.33, which translates to a B− at most schools.

Suppose, however, that we have a function $v(t)$ that gives us the speed of an object at any time t, and we want to find the object's average speed. The function $v(t)$ takes on an infinite number of values, so we can't use the process just described. Fortunately, we can use a definite integral to find the average value of a function such as this.

Let $f(x)$ be continuous over the interval $[a, b]$ and let $[a, b]$ be divided into n subintervals of width $\Delta x = (b - a)/n$. Choose a representative x_i^* in each subinterval and calculate $f(x_i^*)$ for $i = 1, 2, \ldots, n$. In other words, consider each $f(x_i^*)$ as a sampling of the function over each subinterval. The average value of the function may then be approximated as

$$f_{ave} \approx \frac{f(x_1^*) + f(x_2^*) + \cdots + f(x_n^*)}{n},$$

which is basically the same expression used to calculate the average of discrete values.

But we know $\Delta x = \dfrac{b-a}{n}$, so $n = \dfrac{b-a}{\Delta x}$, and we get

$$f_{ave} \approx \frac{f(x_1^*) + f(x_2^*) + \cdots + f(x_n^*)}{n} = \frac{f(x_1^*) + f(x_2^*) + \cdots + f(x_n^*)}{\left(\dfrac{b-a}{\Delta x}\right)}.$$

Following through with the algebra, the numerator is a sum that is represented as $\sum_{i=1}^{n} f(x_i^*)$, and we are dividing by a fraction. To divide by a fraction, invert the denominator and multiply. Thus, an approximate value for the average value of the function is given by

$$\frac{\sum_{i=1}^{n} f(x_i^*)}{\left(\dfrac{b-a}{\Delta x}\right)} = \left(\frac{\Delta x}{b-a}\right)\sum_{i=1}^{n} f(x_i^*) = \left(\frac{1}{b-a}\right)\sum_{i=1}^{n} f(x_i^*)\Delta x.$$

This is a Riemann sum. Then, to get the exact average value, take the limit as n goes to infinity. Thus, the average value of a function is given by

$$\frac{1}{b-a}\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\Delta x = \frac{1}{b-a}\int_a^b f(x)dx.$$

Definition: Average Value of a Function

Let $f(x)$ be continuous over the interval $[a, b]$. Then, the *average value of the function* $f(x)$ (or f_{ave}) on $[a, b]$ is given by

$$f_{ave} = \frac{1}{b-a}\int_a^b f(x)\,dx. \tag{5.2.2}$$

Example 5.2.8: Finding the Average Value of a Linear Function

Find the average value of $f(x) = x + 1$ over the interval $[0, 5]$.

Solution

First, graph the function on the stated interval, as shown in Figure 5.2.10.

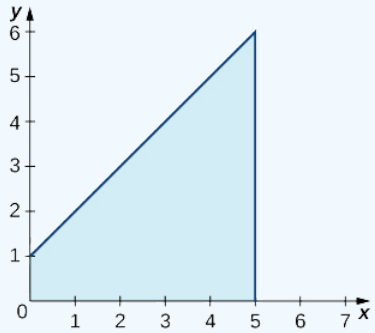


Figure 5.2.10:The graph shows the area under the function $(x) = x + 1$ over $[0, 5]$.

The region is a trapezoid lying on its side, so we can use the area formula for a trapezoid $A = \dfrac{1}{2}h(a + b)$, where h represents height, and a and b represent the two parallel sides. Then,

$$\int_0^5 x + 1\,dx = \frac{1}{2}h(a+b) = \frac{1}{2}\cdot 5\cdot(1+6) = \frac{35}{2}.$$

Thus the average value of the function is

$$\frac{1}{5-0}\int_0^5 x + 1\,dx = \frac{1}{5}\cdot\frac{35}{2} = \frac{7}{2}.$$

Exercise 5.2.7

Find the average value of $f(x) = 6 - 2x$ over the interval $[0, 3]$.

Hint

Use the average value formula (Equation 5.2.2), and use geometry to evaluate the integral.

Answer

3

Key Concepts

- The definite integral can be used to calculate net signed area, which is the area above the x-axis less the area below the x-axis. Net signed area can be positive, negative, or zero.
- The component parts of the definite integral are the integrand, the variable of integration, and the limits of integration.
- Continuous functions on a closed interval are integrable. Functions that are not continuous may still be integrable, depending on the nature of the discontinuities.
- The properties of definite integrals can be used to evaluate integrals.
- The area under the curve of many functions can be calculated using geometric formulas.
- The average value of a function can be calculated using definite integrals.

Key Equations

- **Definite Integral**

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

- **Properties of the Definite Integral**

$$\int_a^a f(x)\,dx = 0$$

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$

$$\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

$$\int_a^b [f(x) - g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx, \text{ for constant } c$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Glossary

average value of a function

(or f_{ave}) the average value of a function on an interval can be found by calculating the definite integral of the function and dividing that value by the length of the interval

definite integral

a primary operation of calculus; the area between the curve and the x-axis over a given interval is a definite integral

integrable function

a function is integrable if the limit defining the integral exists; in other words, if the limit of the Riemann sums as n goes to infinity exists

integrand

the function to the right of the integration symbol; the integrand includes the function being integrated

limits of integration

these values appear near the top and bottom of the integral sign and define the interval over which the function should be integrated

net signed area

the area between a function and the x-axis such that the area below the x-axis is subtracted from the area above the x-axis; the result is the same as the definite integral of the function

total area

total area between a function and the x-axis is calculated by adding the area above the x-axis and the area below the x-axis; the result is the same as the definite integral of the absolute value of the function

variable of integration

indicates which variable you are integrating with respect to; if it is x, then the function in the integrand is followed by dx

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5.2E: EXERCISES FOR SECTION 5.2

In exercises 1 - 4, express the limits as integrals.

1) $\lim_{n\to\infty}\sum_{i=1}^{n}(x_i^*)\Delta x$ over $[1,3]$

2) $\lim_{n\to\infty}\sum_{i=1}^{n}(5(x_i^*)^2-3(x_i^*)^3)\Delta x$ over $[0,2]$

Answer

$$\int_0^2 (5x^2-3x^3)\,dx$$

3) $\lim_{n\to\infty}\sum_{i=1}^{n}\sin^2(2\pi x_i^*)\Delta x$ over $[0,1]$

4) $\lim_{n\to\infty}\sum_{i=1}^{n}\cos^2(2\pi x_i^*)\Delta x$ over $[0,1]$

Answer

$$\int_0^1 \cos^2(2\pi x)\,dx$$

In exercises 5 - 10, given L_n or R_n as indicated, express their limits as $n\to\infty$ as definite integrals, identifying the correct intervals.

5) $L_n=\frac{1}{n}\sum_{i=1}^{n}\frac{i-1}{n}$

6) $R_n=\frac{1}{n}\sum_{i=1}^{n}\frac{i}{n}$

Answer

$$\int_0^1 x\,dx$$

7) $Ln=\frac{2}{n}\sum_{i=1}^{n}(1+2\frac{i-1}{n})$

8) $R_n=\frac{3}{n}\sum_{i=1}^{n}(3+3\frac{i}{n})$

Answer

$$\int_3^6 x\,dx$$

9) $L_n=\frac{2\pi}{n}\sum_{i=1}^{n}2\pi\frac{i-1}{n}\cos(2\pi\frac{i-1}{n})$

10 $R_n=\frac{1}{n}\sum_{i=1}^{n}(1+\frac{i}{n})\log((1+\frac{i}{n})^2)$

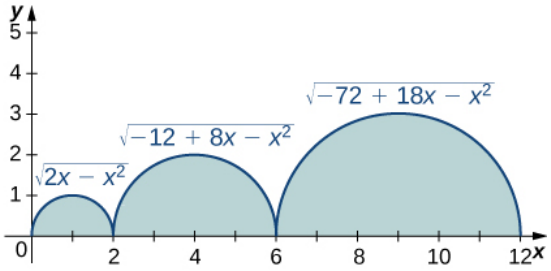
Answer

$$\int_1^2 x\log(x^2)\,dx$$

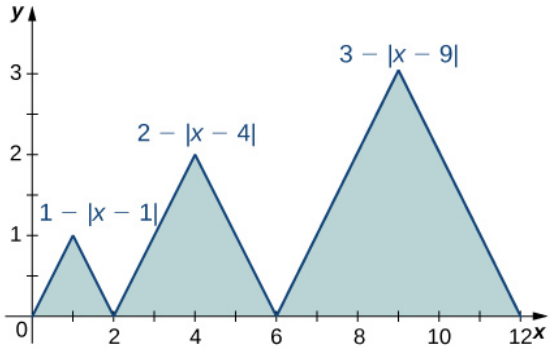
In exercises 11 - 16, evaluate the integrals of the functions graphed using the formulas for areas of triangles and circles, and subtracting the areas below the x-axis.

11)

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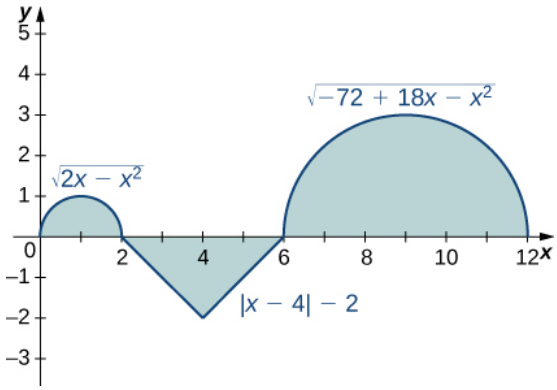
12)



Answer

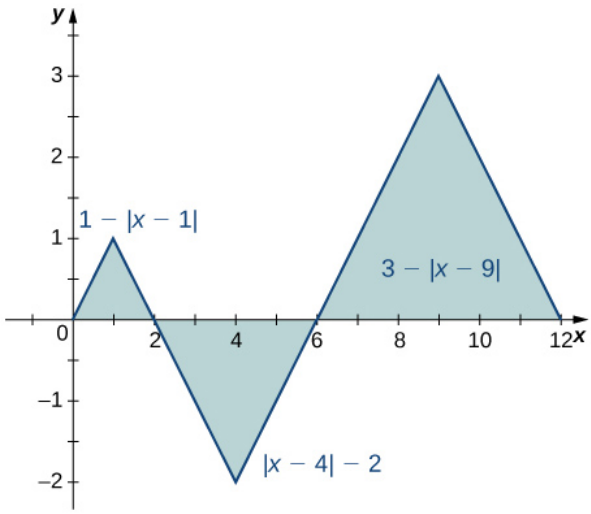
$1 + 2 \cdot 2 + 3 \cdot 3 = 14$

13)



14)

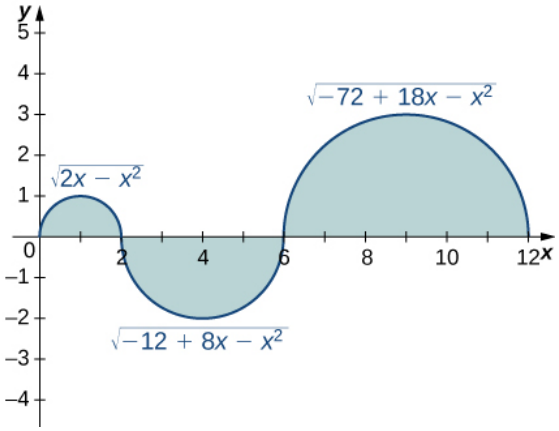
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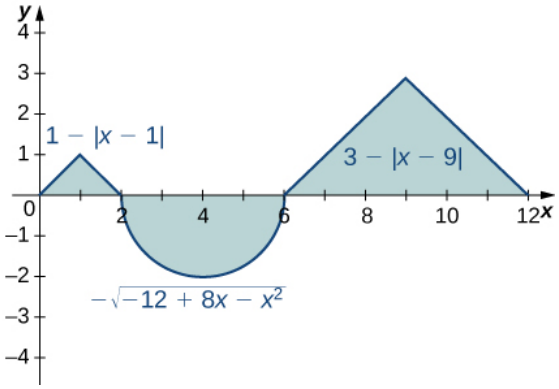
Answer

$1-4+9=6$

15)



16)



Answer

$1-2\pi+9=10-2\pi$

In exercises 17 - 24, evaluate the integral using area formulas.

17) $\int_0^3 (3-x)\,dx$

18) $\int_2^3 (3-x)\,dx$

Answer

The integral is the area of the triangle, $\frac{1}{2}$.

19) $\int_{-3}^3 (3-|x|)\,dx$

20) $\int_0^6 (3-|x-3|)\,dx$

Answer

The integral is the area of the triangle, $\mathbf{9}$.

21) $\int_{-2}^2 \sqrt{4-x^2}\,dx$

22) $\int_1^5 \sqrt{4-(x-3)^2}\,dx$

Answer

The integral is the area $\frac{1}{2}\pi r^2 = 2\pi$.

23) $\int_0^{12} \sqrt{36-(x-6)^2}\,dx$

24) $\int_{-2}^3 (3-|x|)\,dx$

Answer

The integral is the area of the "big" triangle less the "missing" triangle, $9-\frac{1}{2}$.

In exercises 25 - 28, use averages of values at the left (L) and right (R) endpoints to compute the integrals of the piecewise linear functions with graphs that pass through the given list of points over the indicated intervals.

25) $(0,0),(2,1),(4,3),(5,0),(6,0),(8,3)$ over $[0,8]$

26) $(0,2),(1,0),(3,5),(5,5),(6,2),(8,0)$ over $[0,8]$

Answer

$L = 2+0+10+5+4 = 21,\ R = 0+10+10+2+0 = 22,\ \dfrac{L+R}{2} = 21.5$

27) $(-4,-4),(-2,0),(0,-2),(3,3),(4,3)$ over $[-4,4]$

28) $(-4,0),(-2,2),(0,0),(1,2),(3,2),(4,0)$ over $[-4,4]$

Answer

$L = 0+4+0+4+2 = 10,\ R = 4+0+2+4+0 = 10,\ \dfrac{L+R}{2} = 10$

Suppose that $\int_0^4 f(x)\,dx = 5$ and $\int_0^2 f(x)\,dx = -3$, and $\int_0^4 g(x)\,dx = -1$ and $\int_0^2 g(x)\,dx = 2$. In exercises 29 - 34, compute the integrals.

29) $\int_0^4 (f(x)+g(x))\,dx$

30) $\int_2^4 (f(x)+g(x))\,dx$

Answer

$\int_2^4 f(x)\,dx + \int_2^4 g(x)\,dx = 8-3 = 5$

31) $\int_0^2 (f(x)-g(x))\,dx$

32) $\int_2^4 (f(x)-g(x))\,dx$

Answer

$$\int_2^4 f(x)\,dx - \int_2^4 g(x)\,dx = 8 + 3 = 11$$

33) $\int_0^2 (3f(x) - 4g(x))\,dx$

34) $\int_2^4 (4f(x) - 3g(x))\,dx$

Answer

$$4\int_2^4 f(x)\,dx - 3\int_2^4 g(x)\,dx = 32 + 9 = 41$$

In exercises 35 - 38, use the identity $\int_{-A}^{A} f(x)\,dx = \int_{-A}^{0} f(x)\,dx + \int_0^A f(x)\,dx$ to compute the integrals.

35) $\int_{-\pi}^{\pi} \frac{\sin t}{1+t^2}\,dt$ (Hint: $\sin(-t) = -\sin(t)$)

36) $\int_{\sqrt{-\pi}}^{\sqrt{\pi}} \frac{t}{1+\cos t}\,dt$

Answer

The integrand is odd; the integral is zero.

37) $\int_1^3 (2-x)\,dx$ (Hint: Look at the graph of f.)

38) $\int_2^4 (x-3)^3\,dx$ (Hint: Look at the graph of f.)

Answer

The integrand is antisymmetric with respect to $x = 3$. The integral is zero.

In exercises 39 - 44, given that $\int_0^1 x\,dx = \frac{1}{2}$, $\int_0^1 x^2\,dx = \frac{1}{3}$, and $\int_0^1 x^3\,dx = \frac{1}{4}$, compute the integrals.

39) $\int_0^1 (1 + x + x^2 + x^3)\,dx$

40) $\int_0^1 (1 - x + x^2 - x^3)\,dx$

Answer

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$$

41) $\int_0^1 (1-x)^2\,dx$

42) $\int_0^1 (1-2x)^3\,dx$

Answer

$$\int_0^1 (1 - 6x + 12x^2 - 8x^3)\,dx = 1 - 6\left(\frac{1}{2}\right) + 12\left(\frac{1}{3}\right) - 8\left(\frac{1}{4}\right) = 1 - 3 + 4 - 2 = 0$$

43) $\int_0^1 \left(6x - \frac{4}{3}x^2\right)\,dx$

44) $\int_0^1 (7 - 5x^3)\,dx$

Answer

$$7 - \tfrac{5}{4} = \tfrac{23}{4}$$

In exercises 45 - 50, use the comparison theorem.

45) Show that $\int_0^3 (x^2 - 6x + 9)\,dx \geq 0$.

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46) Show that $\int_{-2}^{3}(x-3)(x+2)\,dx \le 0$.

Answer

The integrand is negative over $[-2,3]$.

47) Show that $\int_0^1 \sqrt{1+x^3}\,dx \le \int_0^1 \sqrt{1+x^2}\,dx$.

48) Show that $\int_1^2 \sqrt{1+x}\,dx \le \int_1^2 \sqrt{1+x^2}\,dx$.

Answer

$x \le x^2$ over $[1,2]$, so $\sqrt{1+x} \le \sqrt{1+x^2}$ over $[1,2]$.

49) Show that $\int_0^{\pi/2} \sin t dt \ge \frac{\pi}{4}$ (Hint: $\sin t \ge \frac{2t}{\pi}$ over $[0, \frac{\pi}{2}]$)

50) Show that $\int_{-\pi/4}^{\pi/4} \cos t\,dt \ge \pi\sqrt{2}/4$.

Answer

$\cos(t) \ge \frac{\sqrt{2}}{2}$. Multiply by the length of the interval to get the inequality.

In exercises 51 - 56, find the average value f_{ave} of f between a and b, and find a point c, where $f(c) = f_{ave}$

51) $f(x) = x^2,\ a = -1,\ b = 1$

52) $f(x) = x^5,\ a = -1,\ b = 1$

Answer

$f_{ave} = 0;\ c = 0$

53) $f(x) = \sqrt{4-x^2},\ a = 0,\ b = 2$

54) $f(x) = 3 - |x|,\ a = -3,\ b = 3$

Answer

$\frac{3}{2}$ when $c = \pm\frac{3}{2}$

55) $f(x) = \sin x,\ a = 0,\ b = 2\pi$

56) $f(x) = \cos x,\ a = 0,\ b = 2\pi$

Answer

$f_{ave} = 0;\ c = \frac{\pi}{2}, \frac{3\pi}{2}$

In exercises 57 - 60, approximate the average value using Riemann sums L_{100} and R_{100}. How does your answer compare with the exact given answer?

57) [T] $y = \ln(x)$ over the interval $[1,4]$; the exact solution is $\frac{\ln(256)}{3} - 1$.

58) [T] $y = e^{x/2}$ over the interval $[0,1]$; the exact solution is $2(\sqrt{e}-1)$.

Answer

$L_{100} = 1.294,\ R_{100} = 1.301$; the exact average is between these values.

59) [T] $y = \tan x$ over the interval $[0, \frac{\pi}{4}]$; the exact solution is $\frac{2\ln(2)}{\pi}$.

60) [T] $y = \frac{x+1}{\sqrt{4-x^2}}$ over the interval $[-1,1]$; the exact solution is $\frac{\pi}{6}$.

Answer

$L_{100} \times (\frac{1}{2}) = 0.5178,\ R_{100} \times (\frac{1}{2}) = 0.5294$

In exercises 61 - 64, compute the average value using the left Riemann sums L_N for $N = 1, 10, 100$. How does the accuracy compare with the given exact value?

61) [T] $y = x^2 - 4$ over the interval $[0, 2]$; the exact solution is $-\frac{8}{3}$.

62) [T] $y = xe^{x^2}$ over the interval $[0, 2]$; the exact solution is $\frac{1}{4}(e^4 - 1)$.

Answer

$L_1 = 0,\ L_{10} \times (\frac{1}{2}) = 8.743493,\ L_{100} \times (\frac{1}{2}) = 12.861728$. The exact answer ≈ 26.799, so L_{100} is not accurate.

63) [T] $y = \left(\frac{1}{2}\right)^x$ over the interval $[0, 4]$; the exact solution is $\frac{15}{64\ln(2)}$.

64) [T] $y = x\sin(x^2)$ over the interval $[-\pi, 0]$; the exact solution is $\frac{\cos(\pi^2) - 1}{2\pi.}$

Answer

$L_1 \times (\frac{1}{\pi}) = 1.352, L_{10} \times (\frac{1}{\pi}) = -0.1837, L_{100} \times (1\pi) = -0.2956$. The exact answer ≈ -0.303, so L_{100} is not accurate to the first decimal.

65) Suppose that $A = \int_0^{2\pi} \sin^2 t\, dt$ and $B = \int_0^{2\pi} \cos^2 t\, dt$. Show that $A + B = 2\pi$ and $A = B$.

66) Suppose that $A = \int_{-\pi/4}^{\pi/4} \sec^2 t\, dt = \pi$ and $B = \int_{-\pi/}^{\pi/4} 4\tan^2 t\, dt$. Show that $A - B = \frac{\pi}{2}$.

Answer

Use $\tan^2\theta + 1 = \sec^2\theta$. Then, $B - A = \int_{-\pi/4}^{\pi/4} 1\, dx = \frac{\pi}{2}$.

67) Show that the average value of $\sin^2 t$ over $[0, 2\pi]$ is equal to $1/2$. Without further calculation, determine whether the average value of $\sin^2 t$ over $[0, \pi]$ is also equal to $1/2$.

68) Show that the average value of $\cos^2 t$ over $[0, 2\pi]$ is equal to $1/2$. Without further calculation, determine whether the average value of $\cos^2(t)$ over $[0, \pi]$ is also equal to $1/2$.

Answer

$\int_0^{2\pi} \cos^2 t\, dt = \pi$, so divide by the length 2π of the interval. $\cos^2 t$ has period π, so yes, it is true.

69) Explain why the graphs of a quadratic function (parabola) $p(x)$ and a linear function $\ell(x)$ can intersect in at most two points. Suppose that $p(a) = \ell(a)$ and $p(b) = \ell(b)$, and that $\int_a^b p(t)\, dt > \int_a^b \ell(t)\, dt$. Explain why $\int_c^d p(t)\, dt > \int_c^d \ell(t)\, dt$ whenever $a \le c < d \le b$.

70) Suppose that parabola $p(x) = ax^2 + bx + c$ opens downward $(a < 0)$ and has a vertex of $y = \frac{-b}{2a} > 0$. For which interval $[A, B]$ is $\int_A^B (ax^2 + bx + c)\, dx$ as large as possible?

Answer

The integral is maximized when one uses the largest interval on which p is nonnegative. Thus, $A = \frac{-b-\sqrt{b^2-4ac}}{2a}$ and $B = \frac{-b+\sqrt{b^2-4ac}}{2a}$.

71) Suppose $[a, b]$ can be subdivided into subintervals $a = a_0 < a_1 < a_2 < \cdots < a_N = b$ such that either $f \ge 0$ over $[a_{i-1}, a_i]$ or $f \le 0$ over $[a_{i-1}, a_i]$. Set $A_i = \int_{a_{i-1}}^{a_i} f(t)\, dt$.

a. Explain why $\int_a^b f(t)\, dt = A_1 + A_2 + \cdots + A_N$.

b. Then, explain why $\int_a^b f(t)\, dt \le \int_a^b |f(t)|\, dt$.

72) Suppose f and g are continuous functions such that $\int_c^d f(t)\, dt \le \int_c^d g(t)\, dt$ for every subinterval $[c, d]$ of $[a, b]$. Explain why $f(x) \le g(x)$ for all values of x.

Answer

If $f(t_0) > g(t_0)$ for some $t_0 \in [a, b]$, then since $f - g$ is continuous, there is an interval containing t_0 such that $f(t) > g(t)$ over the interval $[c, d]$, and then $\int_d^d f(t)\, dt > \int_c^d g(t)\, dt$over this interval.

73) Suppose the average value of f over $[a,b]$ is 1 and the average value of f over $[b,c]$ is 1 where $a<c<b$. Show that the average value of f over $[a,c]$ is also 1.

74) Suppose that $[a,b]$ can be partitioned. taking $a=a_0<a_1<\cdots<a_N=b$ such that the average value of f over each subinterval $[a_{i-1},a_i]=1$ is equal to 1 for each $i=1,\ldots,N$. Explain why the average value of f over $[a,b]$ is also equal to 1.

Answer

The integral of f over an interval is the same as the integral of the average of f over that interval. Thus, $\int_a^b f(t)\,dt=\int_{a_0}^{a_1} f(t)\,dt$
$+\int_a^{a_2} 1_f(t)\,dt+\cdots+\int_{a_{N+1}}^{a_N} f(t)\,dt=\int_{a_0}^{a_1} 1\,dt+\int_{a_1}^{a_2} 1\,dt+\cdots+\int_{a_{N+1}}^{a_N} 1\,dt$
$=(a_1-a_0)+(a_2-a_1)+\cdots+(a_N-a_{N-1})=a_N-a_0=b-a.$
Dividing through by $b-a$ gives the desired identity.

75) Suppose that for each i such that $1\le i\le N$ one has $\int_{i-1}^{i} f(t)\,dt=i$. Show that $\int_0^N f(t)\,dt=\dfrac{N(N+1)}{2}$.

76) Suppose that for each i such that $1\le i\le N$ one has $\int_{i-1}^{i} f(t)\,dt=i^2$. Show that $\int_0^N f(t)\,dt=\dfrac{N(N+1)(2N+1)}{6}$.

Answer

$$\int_0^N f(t)\,dt=\sum_{i=1}^{N}\int_{i-1}^{i} f(t)\,dt=\sum_{i=1}^{N} i^2=\frac{N(N+1)(2N+1)}{6}$$

77) [T] Compute the left and right Riemann sums L_{10} and R_{10} and their average $\dfrac{L_{10}+R_{10}}{2}$ for $f(t)=t^2$over $[0,1]$. Given that $\int_0^1 t^2\,dt=1/3$, to how many decimal places is $\dfrac{L_{10}+R_{10}}{2}$ accurate?

78) [T] Compute the left and right Riemann sums, L_10 and R_{10}, and their average $\dfrac{L_{10}+R_{10}}{2}$ for $f(t)=(4-t^2)$ over $[1,2]$. Given that $\int_1^2 (4-t^2)\,dt=1.66$, to how many decimal places is $\dfrac{L_{10}+R_{10}}{2}$ accurate?

Answer

$L_{10}=1.815,\ R_{10}=1.515,\ \frac{L_{10}+R_{10}}{2}=1.665$, so the estimate is accurate to two decimal places.

79) If $\int_1^5 \sqrt{1+t^4}\,dt=41.7133\ldots$, what is $\int_1^5 \sqrt{1+u^4}\,du$?

80) Estimate $\int_0^1 t\,dt$ using the left and right endpoint sums, each with a single rectangle. How does the average of these left and right endpoint sums compare with the actual value $\int_0^1 t\,dt$?

Answer

The average is $1/2$, which is equal to the integral in this case.

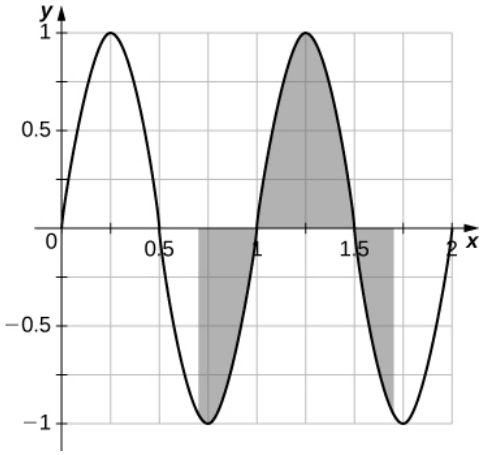
81) Estimate $\int_0^1 t\,dt$ by comparison with the area of a single rectangle with height equal to the value of t at the midpoint $t=\dfrac{1}{2}$. How does this midpoint estimate compare with the actual value $\int_0^1 t\,dt$?

82) From the graph of $\sin(2\pi x)$ shown:

a. Explain why $\int_0^1 \sin(2\pi t)\,dt=0$.

b. Explain why, in general, $\int_a^{a+1} \sin(2\pi t)\,dt=0$ for any value of a.

Answer



a. The graph is antisymmetric with respect to $t = \frac{1}{2}$ over $[0, 1]$, so the average value is zero.
b. For any value of a, the graph between $[a, a + 1]$ is a shift of the graph over $[0, 1]$, so the net areas above and below the axis do not change and the average remains zero.

83) If f is 1-periodic $(f(t + 1) = f(t))$, odd, and integrable over $[0, 1]$, is it always true that $\int_0^1 f(t)\,dt = 0$?

84) If f is 1-periodic and $\int 10f(t)\,dt = A$, is it necessarily true that $\int_a^{1+a} f(t)\,dt = A$ for all A?

Answer

Yes, the integral over any interval of length 1 is the same.

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5.3: The Fundamental Theorem of Calculus

Learning Objectives

- Describe the meaning of the Mean Value Theorem for Integrals.
- State the meaning of the Fundamental Theorem of Calculus, Part 1.
- Use the Fundamental Theorem of Calculus, Part 1, to evaluate derivatives of integrals.
- State the meaning of the Fundamental Theorem of Calculus, Part 2.
- Use the Fundamental Theorem of Calculus, Part 2, to evaluate definite integrals.
- Explain the relationship between differentiation and integration.

In the previous two sections, we looked at the definite integral and its relationship to the area under the curve of a function. Unfortunately, so far, the only tools we have available to calculate the value of a definite integral are geometric area formulas and limits of Riemann sums, and both approaches are extremely cumbersome. In this section we look at some more powerful and useful techniques for evaluating definite integrals.

These new techniques rely on the relationship between differentiation and integration. This relationship was discovered and explored by both Sir Isaac Newton and Gottfried Wilhelm Leibniz (among others) during the late 1600s and early 1700s, and it is codified in what we now call the **Fundamental Theorem of Calculus**, which has two parts that we examine in this section. Its very name indicates how central this theorem is to the entire development of calculus.

Isaac Newton's contributions to mathematics and physics changed the way we look at the world. The relationships he discovered, codified as Newton's laws and the law of universal gravitation, are still taught as foundational material in physics today, and his calculus has spawned entire fields of mathematics.

Before we get to this crucial theorem, however, let's examine another important theorem, the Mean Value Theorem for Integrals, which is needed to prove the Fundamental Theorem of Calculus.

The Mean Value Theorem for Integrals

The Mean Value Theorem for Integrals states that a continuous function on a closed interval takes on its average value at some point in that interval. The theorem guarantees that if $f(x)$ is continuous, a point c exists in an interval $[a, b]$ such that the value of the function at c is equal to the average value of $f(x)$ over $[a, b]$. We state this theorem mathematically with the help of the formula for the average value of a function that we presented at the end of the preceding section.

Theorem 5.3.1: The Mean Value Theorem for Integrals

If $f(x)$ is continuous over an interval $[a, b]$, then there is at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

This formula can also be stated as

$$\int_a^b f(x)\, dx = f(c)(b-a). \tag{5.3.1}$$

Since $f(x)$ is continuous on $[a, b]$, by the extreme value theorem (see section on Maxima and Minima), it assumes minimum and maximum values—m and M, respectively—on $[a, b]$. Then, for all x in $[a, b]$, we have $m \leq f(x) \leq M$. Therefore, by the comparison theorem (see Section on The Definite Integral), we have

$$m(b-a) \leq \int_a^b f(x)\, dx \leq M(b-a).$$

Proof

Since $f(x)$ is continuous on $[a,b]$, by the extreme value theorem (see section on Maxima and Minima), it assumes minimum and maximum values—m and M, respectively—on $[a,b]$. Then, for all x in $[a,b]$, we have $m \le f(x) \le M$. Therefore, by the comparison theorem (see Section on The Definite Integral), we have

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

Dividing by $b-a$ gives us

$$m \le \frac{1}{b-a}\int_a^b f(x)\,dx \le M.$$

Since $\frac{1}{b-a}\int_a^b f(x)\,dx$ is a number between m and M, and since $f(x)$ is continuous and assumes the values m and M over $[a,b]$, by the Intermediate Value Theorem, there is a number c over $[a,b]$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx,$$

and the proof is complete.

□

Example 5.3.1: Finding the Average Value of a Function

Find the average value of the function $f(x) = 8 - 2x$ over the interval $[0,4]$ and find c such that $f(c)$ equals the average value of the function over $[0,4]$.

Solution

The formula states the mean value of $f(x)$ is given by

$$\frac{1}{4-0}\int_0^4 (8-2x)\,dx.$$

We can see in Figure 5.3.1 that the function represents a straight line and forms a right triangle bounded by the x- and y-axes. The area of the triangle is $A = \frac{1}{2}(base)(height)$. We have

$$A = \frac{1}{2}(4)(8) = 16.$$

The average value is found by multiplying the area by $1/(4-0)$. Thus, the average value of the function is

$$\frac{1}{4}(16) = 4$$

Set the average value equal to $f(c)$ and solve for c.

$$\begin{aligned} 8 - 2c &= 4 \\ c &= 2 \end{aligned}$$

At $c = 2$, $f(2) = 4$.

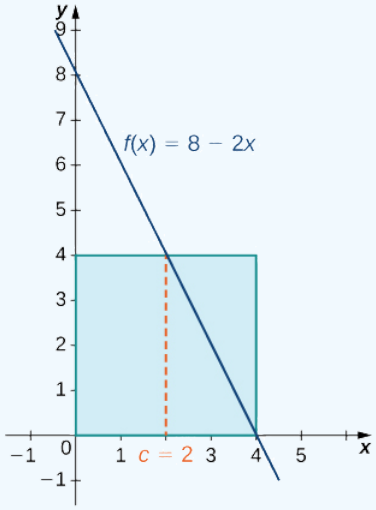


Figure 5.3.1: By the Mean Value Theorem, the continuous function $f(x)$ takes on its average value at c at least once over a closed interval.

Exercise 5.3.1

Find the average value of the function $f(x) = \frac{x}{2}$ over the interval $[0, 6]$ and find c such that $f(c)$ equals the average value of the function over $[0, 6]$.

Hint

Use the procedures from Example 5.3.1 to solve the problem

Answer

The average value is 1.5 and $c = 3$.

Example 5.3.2: Finding the Point Where a Function Takes on Its Average Value

Given $\int_0^3 x^2\,dx = 9$, find c such that $f(c)$ equals the average value of $f(x) = x^2$ over $[0, 3]$.

Solution

We are looking for the value of c such that

$$f(c) = \frac{1}{3-0}\int_0^3 x^2\,dx = \frac{1}{3}(9) = 3.$$

Replacing $f(c)$ with c^2, we have

$$c^2 = 3$$
$$c = \pm\sqrt{3}.$$

Since $-\sqrt{3}$ is outside the interval, take only the positive value. Thus, $c = \sqrt{3}$ (Figure 5.3.2).

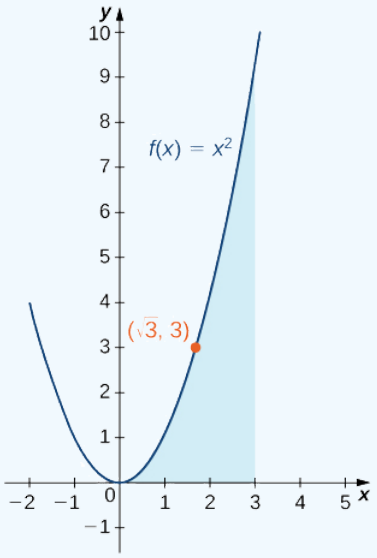


Figure 5.3.2: Over the interval $[0, 3]$, the function $f(x) = x^2$ takes on its average value at $c = \sqrt{3}$.

? Exercise 5.3.2

Given $\int_0^3 (2x^2 - 1)\, dx = 15$, find c such that $f(c)$ equals the average value of $f(x) = 2x^2 - 1$ over $[0, 3]$.

Hint

Use the procedures from Example 5.3.2 to solve the problem.

Answer

$c = \sqrt{3}$

Fundamental Theorem of Calculus Part 1: Integrals and Antiderivatives

As mentioned earlier, the Fundamental Theorem of Calculus is an extremely powerful theorem that establishes the relationship between differentiation and integration, and gives us a way to evaluate definite integrals without using Riemann sums or calculating areas. The theorem is comprised of two parts, the first of which, the **Fundamental Theorem of Calculus, Part 1,** is stated here. Part 1 establishes the relationship between differentiation and integration.

Theorem 5.3.2: The Fundamental Theorem of Calculus, Part 1

If $f(x)$ is continuous over an interval $[a, b]$, and the function $F(x)$ is defined by

$$F(x) = \int_a^x f(t)\, dt,$$

then $F'(x) = f(x)$ over $[a, b]$.

Before we delve into the proof, a couple of subtleties are worth mentioning here. First, a comment on the notation. Note that we have defined a function, $F(x)$, as the definite integral of another function, $f(t)$, from the point a to the point x. At first glance, this is confusing, because we have said several times that a definite integral is a number, and here it looks like it's a function. The key here is to notice that for any particular value of x, the definite integral is a number. So the function $F(x)$ returns a number (the value of the definite integral) for each value of x.

Second, it is worth commenting on some of the key implications of this theorem. There is a reason it is called the *Fundamental Theorem of Calculus*. Not only does it establish a relationship between integration and differentiation, but also it guarantees that any integrable function has an antiderivative. Specifically, it guarantees that any continuous function has an antiderivative.

Proof: Fundamental Theorem of Calculus, Part 1

Applying the definition of the derivative, we have

$$\begin{aligned} F'(x) &= \lim_{h\to 0} \frac{F(x+h)-F(x)}{h} \\ &= \lim_{h\to 0} \frac{1}{h}\left[\int_a^{x+h} f(t)dt - \int_a^x f(t)\,dt\right] \\ &= \lim_{h\to 0} \frac{1}{h}\left[\int_a^{x+h} f(t)\,dt + \int_x^a f(t)\,dt\right] \\ &= \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt. \end{aligned}$$

Looking carefully at this last expression, we see $\frac{1}{h}\int_x^{x+h} f(t)\,dt$ is just the average value of the function $f(x)$ over the interval $[x, x+h]$. Therefore, by Equation 5.3.1, there is some number c in $[x, x+h]$ such that

$$\frac{1}{h}\int_x^{x+h} f(t)\,dt = f(c).$$

In addition, since c is between x and h, c approaches x as h approaches zero. Also, since $f(x)$ is continuous, we have

$$\lim_{h\to 0} f(c) = \lim_{c\to x} f(c) = f(x)$$

Putting all these pieces together, we have

$$F'(x) = \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt = \lim_{h\to 0} f(c) = f(x),$$

and the proof is complete.

□

Example 5.3.3: Finding a Derivative with the Fundamental Theorem of Calculus

Use the Fundamental Theorem of Calculus, Part 1 to find the derivative of

$$g(x) = \int_1^x \frac{1}{t^3+1}\,dt.$$

Solution

According to the Fundamental Theorem of Calculus, the derivative is given by

$$g'(x) = \frac{1}{x^3+1}.$$

Exercise 5.3.3

Use the Fundamental Theorem of Calculus, Part 1 to find the derivative of $g(r) = \int_0^r \sqrt{x^2+4}\,dx$.

Hint

Follow the procedures from Example 5.3.3 to solve the problem.

Answer

$g'(r) = \sqrt{r^2+4}$

Example 5.3.4: Using the Fundamental Theorem and the Chain Rule to Calculate Derivatives

Let $F(x) = \int_1^{\sqrt{x}} \sin t\, dt$. Find $F'(x)$.

Solution

Letting $u(x) = \sqrt{x}$, we have $F(x) = \int_1^{u(x)} \sin t\, dt$.

Thus, by the Fundamental Theorem of Calculus and the chain rule,

$$F'(x) = \sin(u(x))\frac{du}{dx} = \sin(u(x)) \cdot \left(\frac{1}{2}x^{-1/2}\right) = \frac{\sin\sqrt{x}}{2\sqrt{x}}.$$

Exercise 5.3.4

Let $F(x) = \int_1^{x^3} \cos t\, dt$. Find $F'(x)$.

Hint

Use the chain rule to solve the problem.

Answer

$F'(x) = 3x^2 \cos x^3$

Example 5.3.5: Using the Fundamental Theorem of Calculus with Two Variable Limits of Integration

Let $F(x) = \int_x^{2x} t^3\, dt$. Find $F'(x)$.

Solution

We have $F(x) = \int_x^{2x} t^3\, dt$. Both limits of integration are variable, so we need to split this into two integrals. We get

$$\begin{aligned} F(x) &= \int_x^{2x} t^3\, dt = \int_x^0 t^3\, dt + \int_0^{2x} t^3\, dt \\ &= -\int_0^x t^3\, dt + \int_0^{2x} t^3\, dt. \end{aligned}$$

Differentiating the first term, we obtain

$$\frac{d}{dx}\left[-\int_0^x t^3\, dt\right] = -x^3.$$

Differentiating the second term, we first let $u(x) = 2x$. Then,

$$\begin{aligned} \frac{d}{dx}\left[\int_0^{2x} t^3\, dt\right] &= \frac{d}{dx}\left[\int_0^{u(x)} t^3\, dt\right] \\ &= (u(x))^3\frac{du}{dx} \\ &= (2x)^3 \cdot 2 = 16x^3. \end{aligned}$$

Thus,

$$\begin{aligned} F'(x) &= \frac{d}{dx}\left[-\int_0^x t^3\, dt\right] + \frac{d}{dx}\left[\int_0^{2x} t^3\, dt\right] \\ &= -x^3 + 16x^3 = 15x^3 \end{aligned}$$

Exercise 5.3.5

Let $F(x) = \int_x^{x^2} \cos t \, dt$. Find $F'(x)$.

Hint

Use the procedures from Example 5.3.5 to solve the problem

Answer

$F'(x) = 2x \cos x^2 - \cos x$

Fundamental Theorem of Calculus, Part 2: The Evaluation Theorem

The Fundamental Theorem of Calculus, Part 2, is perhaps the most important theorem in calculus. After tireless efforts by mathematicians for approximately 500 years, new techniques emerged that provided scientists with the necessary tools to explain many phenomena. Using calculus, astronomers could finally determine distances in space and map planetary orbits. Everyday financial problems such as calculating marginal costs or predicting total profit could now be handled with simplicity and accuracy. Engineers could calculate the bending strength of materials or the three-dimensional motion of objects. Our view of the world was forever changed with calculus.

After finding approximate areas by adding the areas of n rectangles, the application of this theorem is straightforward by comparison. It almost seems too simple that the area of an entire curved region can be calculated by just evaluating an antiderivative at the first and last endpoints of an interval.

Theorem 5.3.3: The Fundamental Theorem of Calculus, Part 2

If $f(x)$ is continuous over the interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then

$$\int_a^b f(x) \, dx = F(b) - F(a). \tag{5.3.2}$$

We often see the notation $F(x)|_a^b$ to denote the expression $F(b) - F(a)$. We use this vertical bar and associated limits a and b to indicate that we should evaluate the function $F(x)$ at the upper limit (in this case, b), and subtract the value of the function $F(x)$ evaluated at the lower limit (in this case, a).

The **Fundamental Theorem of Calculus, Part 2** (also known as the **evaluation theorem**) states that if we can find an antiderivative for the integrand, then we can evaluate the definite integral by evaluating the antiderivative at the endpoints of the interval and subtracting.

Proof

Let $P = x_i, i = 0, 1, \ldots, n$ be a regular partition of $[a, b]$. Then, we can write

$$\begin{aligned} F(b) - F(a) &= F(x_n) - F(x_0) \\ &= [F(x_n) - F(x_{n-1})] + [F(x_{n-1}) - F(x_{n-2})] + \ldots + [F(x_1) - F(x_0)] \\ &= \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})]. \end{aligned}$$

Now, we know F is an antiderivative of f over $[a, b]$, so by the Mean Value Theorem for derivatives (see The Mean Value Theorem) for $i = 0, 1, \ldots, n$ we can find c_i in $[x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = F'(c_i)(x_i - x_{i-1}) = f(c_i) \, \Delta x.$$

Then, substituting into the previous equation, we have

$$F(b) - F(a) = \sum_{i=1}^{n} f(c_i) \, \Delta x.$$

Taking the limit of both sides as $n \to \infty$, we obtain

$$F(b) - F(a) = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\Delta x = \int_a^b f(x)\,dx.$$

□

Example 5.3.6: Evaluating an Integral with the Fundamental Theorem of Calculus

Use Equation 5.3.2 to evaluate

$$\int_{-2}^{2} (t^2 - 4)\,dt.$$

Solution

Recall the power rule for Antiderivatives:

If $y = x^n$,

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C.$$

Use this rule to find the antiderivative of the function and then apply the theorem. We have

$$\begin{aligned}
\int_{-2}^{2} (t^2 - 4)dt &= \left(\frac{t^3}{3} - 4t\right)\Big|_{-2}^{2} \\
&= \left[\frac{(2)^3}{3} - 4(2)\right] - \left[\frac{(-2)^3}{3} - 4(-2)\right] \\
&= \left[\frac{8}{3} - 8\right] - \left[-\frac{8}{3} + 8\right] \\
&= \frac{8}{3} - 8 + \frac{8}{3} - 8 \\
&= \frac{16}{3} - 16 = -\frac{32}{3}.
\end{aligned}$$

Analysis

Notice that we did not include the "$+C$" term when we wrote the antiderivative. The reason is that, according to the Fundamental Theorem of Calculus, Part 2 (Equation 5.3.2), any antiderivative works. So, for convenience, we chose the antiderivative with $C = 0$. If we had chosen another antiderivative, the constant term would have canceled out. This always happens when evaluating a definite integral.

The region of the area we just calculated is depicted in Figure 5.3.3. Note that the region between the curve and the x-axis is all below the x-axis. Area is always positive, but a definite integral can still produce a negative number (a net signed area). For example, if this were a profit function, a negative number indicates the company is operating at a loss over the given interval.

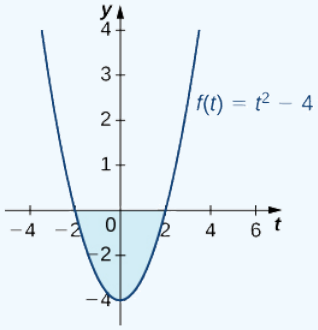


Figure 5.3.3: The evaluation of a definite integral can produce a negative value, even though area is always positive.

Example 5.3.7: Evaluating a Definite Integral Using the Fundamental Theorem of Calculus, Part 2

Evaluate the following integral using the Fundamental Theorem of Calculus, Part 2 (Equation 5.3.2):

$$\int_1^9 \frac{x-1}{\sqrt{x}}dx.$$

Solution

First, eliminate the radical by rewriting the integral using rational exponents. Then, separate the numerator terms by writing each one over the denominator:

$$\int_1^9 \frac{x-1}{x^{1/2}}\,dx = \int_1^9 \left(\frac{x}{x^{1/2}} - \frac{1}{x^{1/2}}\right)\,dx.$$

Use the properties of exponents to simplify:

$$\int_1^9 \left(\frac{x}{x^{1/2}} - \frac{1}{x^{1/2}}\right)\,dx = \int_1^9 (x^{1/2} - x^{-1/2})\,dx.$$

Now, integrate using the power rule:

$$\begin{aligned}\int_1^9 (x^{1/2} - x^{-1/2})\,dx &= \left(\frac{x^{3/2}}{\frac{3}{2}} - \frac{x^{1/2}}{\frac{1}{2}}\right)\Big|_1^9 \\ &= \left[\frac{(9)^{3/2}}{\frac{3}{2}} - \frac{(9)^{1/2}}{\frac{1}{2}}\right] - \left[\frac{(1)^{3/2}}{\frac{3}{2}} - \frac{(1)^{1/2}}{\frac{1}{2}}\right] \\ &= \left[\frac{2}{3}(27) - 2(3)\right] - \left[\frac{2}{3}(1) - 2(1)\right] \\ &= 18 - 6 - \frac{2}{3} + 2 = \frac{40}{3}.\end{aligned}$$

See Figure 5.3.4.

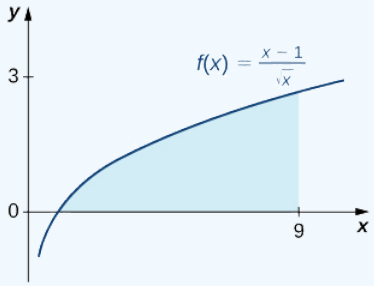


Figure 5.3.4: The area under the curve from $x = 1$ to $x = 9$ can be calculated by evaluating a definite integral.

Exercise 5.3.6

Use Note to evaluate $\int_1^2 x^{-4}\,dx$.

Hint

Use the power rule.

Answer

$\frac{7}{24}$

Example 5.3.8: A Roller-Skating Race

James and Kathy are racing on roller skates. They race along a long, straight track, and whoever has gone the farthest after 5 sec wins a prize. If James can skate at a velocity of $f(t) = 5 + 2t$ ft/sec and Kathy can skate at a velocity of $g(t) = 10 + \cos\left(\frac{\pi}{2}t\right)$ ft/sec, who is going to win the race?

Solution

We need to integrate both functions over the interval $[0, 5]$ and see which value is bigger. For James, we want to calculate

$$\int_0^5 (5 + 2t)\, dt.$$

Using the power rule, we have

$$\begin{aligned}\int_0^5 (5 + 2t)\, dt &= \left(5t + t^2\right)\Big|_0^5 \\ &= (25 + 25) \\ &= 50.\end{aligned}$$

Thus, James has skated 50 ft after 5 sec. Turning now to Kathy, we want to calculate

$$\int_0^5 10 + \cos\left(\frac{\pi}{2}t\right)\, dt.$$

We know $\sin t$ is an antiderivative of $\cos t$, so it is reasonable to expect that an antiderivative of $\cos\left(\frac{\pi}{2}t\right)$ would involve $\sin\left(\frac{\pi}{2}t\right)$. However, when we differentiate $\sin\left(\pi^2 t\right)$, we get $\pi^2 \cos\left(\pi^2 t\right)$ as a result of the chain rule, so we have to account for this additional coefficient when we integrate. We obtain

$$\begin{aligned}\int_0^5 10 + \cos\left(\frac{\pi}{2}t\right)\, dt &= \left(10t + \frac{2}{\pi}\sin\left(\frac{\pi}{2}t\right)\right)\Big|_0^5 \\ &= \left(50 + \frac{2}{\pi}\right) - \left(0 - \frac{2}{\pi}\sin 0\right) \approx 50.6.\end{aligned}$$

Kathy has skated approximately 50.6 ft after 5 sec. Kathy wins, but not by much!

Exercise 5.3.7

Suppose James and Kathy have a rematch, but this time the official stops the contest after only 3 sec. Does this change the outcome?

Hint

Change the limits of integration from those in Example 5.3.7.

Answer

Kathy still wins, but by a much larger margin: James skates 24 ft in 3 sec, but Kathy skates 29.3634 ft in 3 sec.

A Parachutist in Free Fall

Julie is an avid skydiver with more than 300 jumps under her belt and has mastered the art of making adjustments to her body position in the air to control how fast she falls. If she arches her back and points her belly toward the ground, she reaches a terminal velocity of approximately 120 mph (176 ft/sec). If, instead, she orients her body with her head straight down, she falls faster, reaching a terminal velocity of 150 mph (220 ft/sec).



Figure 5.3.5: Skydivers can adjust the velocity of their dive by changing the position of their body during the free fall. (credit: Jeremy T. Lock)

Since Julie will be moving (falling) in a downward direction, we assume the downward direction is positive to simplify our calculations. Julie executes her jumps from an altitude of 12,500 ft. After she exits the aircraft, she immediately starts falling at a velocity given by $v(t) = 32t$.

She continues to accelerate according to this velocity function until she reaches terminal velocity. After she reaches terminal velocity, her speed remains constant until she pulls her ripcord and slows down to land.

On her first jump of the day, Julie orients herself in the slower "belly down" position (terminal velocity is 176 ft/sec). Using this information, answer the following questions.

1. How long after she exits the aircraft does Julie reach terminal velocity?
2. Based on your answer to question 1, set up an expression involving one or more integrals that represents the distance Julie falls after 30 sec.
3. If Julie pulls her ripcord at an altitude of 3000 ft, how long does she spend in a free fall?
4. Julie pulls her ripcord at 3000 ft. It takes 5 sec for her parachute to open completely and for her to slow down, during which time she falls another 400 ft. After her canopy is fully open, her speed is reduced to 16 ft/sec. Find the total time Julie spends in the air, from the time she leaves the airplane until the time her feet touch the ground. On Julie's second jump of the day, she decides she wants to fall a little faster and orients herself in the "head down" position. Her terminal velocity in this position is 220 ft/sec. Answer these questions based on this velocity:
5. How long does it take Julie to reach terminal velocity in this case?
6. Before pulling her ripcord, Julie reorients her body in the "belly down" position so she is not moving quite as fast when her parachute opens. If she begins this maneuver at an altitude of 4000 ft, how long does she spend in a free fall before beginning the reorientation?

Some jumpers wear "**wingsuits**" (Figure 5.3.6). These suits have fabric panels between the arms and legs and allow the wearer to glide around in a free fall, much like a flying squirrel. (Indeed, the suits are sometimes called "flying squirrel suits.") When wearing these suits, terminal velocity can be reduced to about 30 mph (44 ft/sec), allowing the wearers a much longer time in the air. Wingsuit flyers still use parachutes to land; although the vertical velocities are within the margin of safety, horizontal velocities can exceed 70 mph, much too fast to land safely.



Figure 5.3.6: The fabric panels on the arms and legs of a wingsuit work to reduce the vertical velocity of a skydiver's fall. (credit: Richard Schneider)

Answer the following question based on the velocity in a wingsuit.

7. If Julie dons a wingsuit before her third jump of the day, and she pulls her ripcord at an altitude of 3000 ft, how long does she get to spend gliding around in the air

Key Concepts

- The Mean Value Theorem for Integrals states that for a continuous function over a closed interval, there is a value c such that $f(c)$ equals the average value of the function.
- The Fundamental Theorem of Calculus, Part 1 shows the relationship between the derivative and the integral.
- The Fundamental Theorem of Calculus, Part 2 is a formula for evaluating a definite integral in terms of an antiderivative of its integrand. The total area under a curve can be found using this formula.

Key Equations

- **Mean Value Theorem for Integrals**

  If $f(x)$ is continuous over an interval $[a, b]$, then there is at least one point $c \in [a, b]$ such that

  $$f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

- **Fundamental Theorem of Calculus, Part 1**

  If $f(x)$ is continuous over an interval $[a, b]$, and the function $F(x)$ is defined by

  $$F(x) = \int_a^x f(t)\, dt,$$

  then

  $$F'(x) = f(x).$$

- **Fundamental Theorem of Calculus, Part 2**

  If f is continuous over the interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then

  $$\int_a^b f(x)\, dx = F(b) - F(a).$$

Glossary

fundamental theorem of calculus

the theorem, central to the entire development of calculus, that establishes the relationship between differentiation and integration

fundamental theorem of calculus, part 1

uses a definite integral to define an antiderivative of a function

fundamental theorem of calculus, part 2

(also, **evaluation theorem**) we can evaluate a definite integral by evaluating the antiderivative of the integrand at the endpoints of the interval and subtracting

mean value theorem for integrals

guarantees that a point c exists such that $f(c)$ is equal to the average value of the function

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5.3E: EXERCISES FOR SECTION 5.3

1) Consider two athletes running at variable speeds $v_1(t)$ and $v_2(t)$. The runners start and finish a race at exactly the same time. Explain why the two runners must be going the same speed at some point.

2) Two mountain climbers start their climb at base camp, taking two different routes, one steeper than the other, and arrive at the peak at exactly the same time. Is it necessarily true that, at some point, both climbers increased in altitude at the same rate?

Answer

Yes. It is implied by the Mean Value Theorem for Integrals.

3) To get on a certain toll road a driver has to take a card that lists the mile entrance point. The card also has a timestamp. When going to pay the toll at the exit, the driver is surprised to receive a speeding ticket along with the toll. Explain how this can happen.

4) Set $F(x) = \int_1^x (1-t)\,dt$. Find $F'(2)$ and the average value of F' over $[1,2]$.

Answer

$F'(2) = -1$; average value of F' over $[1,2]$ is $-1/2$.

In exercises 5 - 16, use the Fundamental Theorem of Calculus, Part 1, to find each derivative.

5) $\frac{d}{dx}\left[\int_1^x e^{-t^2}\,dt\right]$

6) $\frac{d}{dx}\left[\int_1^x e^{\cos t}\,dt\right]$

Answer

$\frac{d}{dx}\left[\int_1^x e^{\cos t}\,dt\right] = e^{\cos x}$

7) $\frac{d}{dx}\left[\int_3^x \sqrt{9-y^2}\,dy\right]$

8) $\frac{d}{dx}\left[\int_3^x \frac{ds}{\sqrt{16-s^2}}\right]$

Answer

$\frac{d}{dx}\left[\int_3^x \frac{ds}{\sqrt{16-s^2}}\right] = \frac{1}{\sqrt{16-x^2}}$

9) $\frac{d}{dx}\left[\int^{2x} x_t\,dt\right]$

10) $\frac{d}{dx}\left[\int_0^{\sqrt{x}} t\,dt\right]$

Answer

$\frac{d}{dx}\left[\int_0^{\sqrt{x}} t\,dt\right] = \sqrt{x}\frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2}$

11) $\frac{d}{dx}\left[\int_0^{\sin x} \sqrt{1-t^2}\,dt\right]$

12) $\frac{d}{dx}\left[\int_{\cos x}^1 \sqrt{1-t^2}\,dt\right]$

Answer

$\frac{d}{dx}\left[\int_{\cos x}^1 \sqrt{1-t^2}\,dt\right] = -\sqrt{1-\cos^2 x}\frac{d}{dx}\left(\cos x\right) = |\sin x|\sin x$

13) $\frac{d}{dx}\left[\int_1^{\sqrt{x}} \frac{t^2}{1+t^4}\,dt\right]$

14) $\frac{d}{dx}\left[\int_1^{x^2} \frac{\sqrt{t}}{1+t}\,dt\right]$

Answer

$$\frac{d}{dx}\left[\int_1^{x^2}\frac{\sqrt{t}}{1+t}\,dt\right]=2x\frac{|x|}{1+x^2}$$

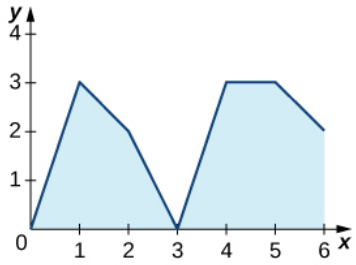
15) $\frac{d}{dx}\left[\int_0^{\ln x} e^t\,dt\right]$

16) $\frac{d}{dx}\left[\int_1^{e^x}\ln u^2\,du\right]$

Answer

$$\frac{d}{dx}\left[\int_1^{e^x}\ln u^2\,du\right]=\ln(e^{2x})\frac{d}{dx}\left(e^x\right)=2xe^x$$

17) The graph of $y=\int_0^x f(t)\,dt$, where f is a piecewise constant function, is shown here.

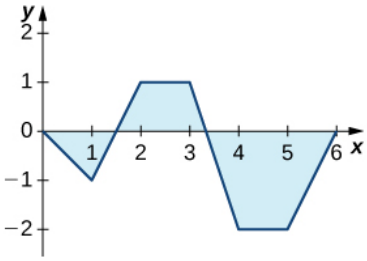


a. Over which intervals is f positive? Over which intervals is it negative? Over which intervals, if any, is it equal to zero?

b. What are the maximum and minimum values of f?

c. What is the average value of f?

18) The graph of $y=\int_0^x f(t)\,dt$, where f is a piecewise constant function, is shown here.



a. Over which intervals is f positive? Over which intervals is it negative? Over which intervals, if any, is it equal to zero?

b. What are the maximum and minimum values of f?

c. What is the average value of f?

Answer

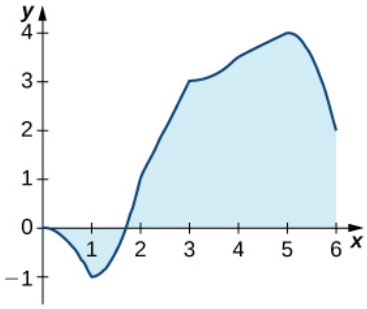
a. f is positive over $[1,2]$ and $[5,6]$, negative over $[0,1]$ and $[3,4]$, and zero over $[2,3]$ and $[4,5]$.

b. The maximum value is 2 and the minimum is -3.

c. The average value is 0.

19) The graph of $y=\int_0^x \ell(t)\,dt$, where ℓ is a piecewise linear function, is shown here.

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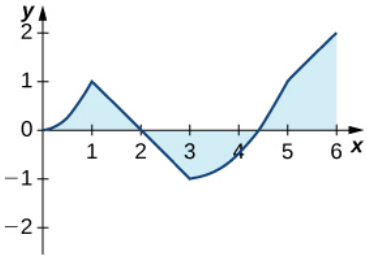


a. Over which intervals is ℓ positive? Over which intervals is it negative? Over which, if any, is it zero?

b. Over which intervals is ℓ increasing? Over which is it decreasing? Over which, if any, is it constant?

c. What is the average value of ℓ?

20) The graph of $y = \int_0^x \ell(t)\,dt$, where ℓ is a piecewise linear function, is shown here.



a. Over which intervals is ℓ positive? Over which intervals is it negative? Over which, if any, is it zero?

b. Over which intervals is ℓ increasing? Over which is it decreasing? Over which intervals, if any, is it constant?

c. What is the average value of ℓ?

Answer

a. ℓ is positive over $[0,1]$ and $[3,6]$, and negative over $[1,3]$.

b. It is increasing over $[0,1]$ and $[3,5]$, and it is constant over $[1,3]$ and $[5,6]$.

c. Its average value is $\frac{1}{3}$.

In exercises 21 - 26, use a calculator to estimate the area under the curve by computing T_{10}, the average of the left- and right-endpoint Riemann sums using $N = 10$ rectangles. Then, using the Fundamental Theorem of Calculus, Part 2, determine the exact area.

21) [T] $y = x^2$ over $[0,4]$

22) [T] $y = x^3 + 6x^2 + x - 5$ over $[-4,2]$

Answer

$T_{10} = 49.08, \quad \int_{-2}^{3} \left(x^3 + 6x^2 + x - 5\right)\,dx = 48$

23) [T] $y = \sqrt{x^3}$ over $[0,6]$

24) [T] $y = \sqrt{x} + x^2$ over $[1,9]$

Answer

$T_{10} = 260.836, \quad \int_1^9 (\sqrt{x} + x^2)\,dx = 260$

25) [T] $\int (\cos x - \sin x)\,dx$ over $[0,\pi]$

26) [T] $\int \frac{4}{x^2}\,dx$ over $[1,4]$

Answer

$T_{10} = 3.058, \quad \int_1^4 \frac{4}{x^2}\,dx = 3$

In exercises 27 - 46, evaluate each definite integral using the Fundamental Theorem of Calculus, Part 2.

27) $\int_{-1}^{2} (x^2 - 3x)\,dx$

28) $\int_{-2}^{3} (x^2 + 3x - 5)\,dx$

Answer

$F(x) = \frac{x^3}{3} + \frac{3x^2}{2} - 5x, \quad F(3) - F(-2) = -\frac{35}{6}$

29) $\int_{-2}^{3} (t+2)(t-3)dt$

30) $\int_{2}^{3} (t^2 - 9)(4 - t^2)dt$

Answer

$F(x) = -\frac{t^5}{5} + \frac{13t^3}{3} - 36t, \quad F(3) - F(2) = \frac{62}{15}$

31) $\int_1^2 x^9\,dx$

32) $\int_0^1 x^{99}\,dx$

Answer

$F(x) = \frac{x^{100}}{100}, \quad F(1) - F(0) = \frac{1}{100}$

33) $\int_4^8 (4t^{5/2} - 3t^{3/2})dt$

34) $\int_{1/4}^{4} \left(x^2 - \frac{1}{x^2}\right)\,dx$

Answer

$F(x) = \frac{x^3}{3} + \frac{1}{x}, \quad F(4) - F\left(\frac{1}{4}\right) = \frac{1125}{64}$

35) $\int_1^2 \frac{2}{x^3}\,dx$

36) $\int_1^4 \frac{1}{2\sqrt{x}}\,dx$

Answer

$F(x) = \sqrt{x}, \quad F(4) - F(1) = 1$

37) $\int_1^4 \frac{2 - \sqrt{t}}{t^2}\,dt$

38) $\int_1^{16} \frac{dt}{t^{1/4}}$

Answer

$F(t) = \frac{4}{3}t^{3/4}, \quad F(16) - F(1) = \frac{28}{3}$

39) $\int_0^{2\pi} \cos\theta\,d\theta$

40) $\int_0^{\pi/2} \sin\theta\,d\theta$

Answer

$F(\theta) = -\cos\theta, \quad F(\frac{\pi}{2}) - F(0) = 1$

41) $\int_0^{\pi/4} \sec^2\theta\, d\theta$

42) $\int_0^{\pi/4} \sec\theta\tan\theta\, d\theta$

Answer

$F(\theta) = \sec\theta, \quad F(\frac{\pi}{4}) - F(0) = \sqrt{2} - 1$

43) $\int_{\pi/3}^{\pi/4} \csc\theta\cot\theta\, d\theta$

44) $\int_{\pi/4}^{\pi/2} \csc^2\theta\, d\theta$

Answer

$F(\theta) = -\cot\theta, \quad F(\frac{\pi}{2}) - F(\frac{\pi}{4}) = 1$

45) $\int_1^2 \left(\frac{1}{t^2} - \frac{1}{t^3}\right) dt$

46) $\int_{-2}^{-1} \left(\frac{1}{t^2} - \frac{1}{t^3}\right) dt$

Answer

$F(t) = -\frac{1}{t} + \frac{1}{2t^2}, \quad F(-1) - F(-2) = \frac{7}{8}$

In exercises 47 - 50, use the evaluation theorem to express the integral as a function $F(x)$.

47) $\int_a^x t^2\, dt$

48) $\int_1^x e^t\, dt$

Answer

$F(x) = e^x - e$

49) $\int_0^x \cos t\, dt$

50) $\int_{-x}^x \sin t\, dt$

Answer

$F(x) = 0$

In exercises 51 - 54, identify the roots of the integrand to remove absolute values, then evaluate using the Fundamental Theorem of Calculus, Part 2.

51) $\int_{-2}^3 |x|\, dx$

52) $\int_{-2}^4 |\, t^2 - 2t - 3\,|\, dt$

Answer

$\int_{-2}^{-1} (t^2 - 2t - 3)\, dt - \int_{-1}^3 (t^2 - 2t - 3)\, dt + \int_3^4 (t^2 - 2t - 3)\, dt = \frac{46}{3}$

53) $\int_0^{\pi} |\cos t|\, dt$

54) $\int_{-\pi/2}^{\pi/2} |\sin t|\, dt$

Answer

$-\int_{-\pi/2}^0 \sin t\, dt + \int_0^{\pi/2} \sin t\, dt = 2$

55) Suppose that the number of hours of daylight on a given day in Seattle is modeled by the function $-3.75\cos(\frac{\pi t}{6}) + 12.25$, with t given in months and $t = 0$ corresponding to the winter solstice.

a. What is the average number of daylight hours in a year?

b. At which times t_1 and t_2, where $0 \le t_1 < t_2 < 12$, do the number of daylight hours equal the average number?

c. Write an integral that expresses the total number of daylight hours in Seattle between t_1 and t_2

d. Compute the mean hours of daylight in Seattle between t_1 and t_2, where $0 \le t_1 < t_2 < 12$, and then between t_2 and t_1, and show that the average of the two is equal to the average day length.

56) Suppose the rate of gasoline consumption in the United States can be modeled by a sinusoidal function of the form $(11.21 - \cos(\frac{\pi t}{6})) \times 10^9$ gal/mo.

a. What is the average monthly consumption, and for which values of t is the rate at time t equal to the average rate?

b. What is the number of gallons of gasoline consumed in the United States in a year?

c. Write an integral that expresses the average monthly U.S. gas consumption during the part of the year between the beginning of April $(t = 3)$ and the end of September $(t = 9)$.

Answer

a. The average is 11.21×10^9 since $\cos(\frac{\pi t}{6})$ has period 12 and integral 0 over any period. Consumption is equal to the average when $\cos(\frac{\pi t}{6}) = 0$, when $t = 3$, and when $t = 9$.

b. Total consumption is the average rate times duration: $11.21 \times 12 \times 10^9 = 1.35 \times 10^{11}$

c. $10^9(11.21 - \frac{1}{6}\int_3^9 \cos(\frac{\pi t}{6})\,dt) = 10^9(11.21 + 2\pi) = 11.84x10^9$

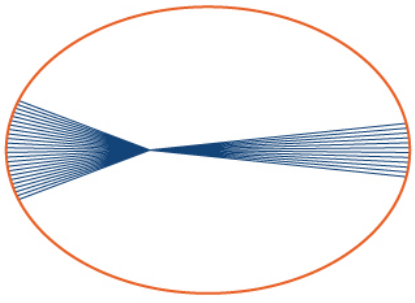
57) Explain why, if f is continuous over $[a, b]$, there is at least one point $c \in [a, b]$ such that $f(c) = \frac{1}{b-a}\int_a^b f(t)\,dt$.

58) Explain why, if f is continuous over $[a, b]$ and is not equal to a constant, there is at least one point $M \in [a, b]$ such that $f(M) = \frac{1}{b-a}\int_a^b f(t)\,dt$ and at least one point $m \in [a, b]$ such that $f(m) < \frac{1}{b-a}\int_a^b f(t)\,dt$.

Answer

If f is not constant, then its average is strictly smaller than the maximum and larger than the minimum, which are attained over $[a, b]$ by the extreme value theorem.

59) Kepler's first law states that the planets move in elliptical orbits with the Sun at one focus. The closest point of a planetary orbit to the Sun is called the **perihelion** (for Earth, it currently occurs around January 3) and the farthest point is called the **aphelion** (for Earth, it currently occurs around July 4). Kepler's second law states that planets sweep out equal areas of their elliptical orbits in equal times. Thus, the two arcs indicated in the following figure are swept out in equal times. At what time of year is Earth moving fastest in its orbit? When is it moving slowest? Kepler's first law states that the planets move in elliptical orbits with the Sun at one focus. The closest point of a planetary orbit to the Sun is called the perihelion (for Earth, it currently occurs around January 3) and the farthest point is called the aphelion (for Earth, it currently occurs around July 4). Kepler's second law states that planets sweep out equal areas of their elliptical orbits in equal times. Thus, the two arcs indicated in the following figure are swept out in equal times. At what time of year is Earth moving fastest in its orbit? When is it moving slowest?



60) A point on an ellipse with major axis length $2a$ and minor axis length $2b$ has the coordinates $(a\cos\theta, b\sin\theta)$, $0 \le \theta \le 2\pi$.

a. Show that the distance from this point to the focus at $(-c, 0)$ is $d(\theta) = a + c\cos\theta$, where $c = \sqrt{a^2 - b^2}$.

Use these coordinates to show that the average distance $\bar{d}$ from a point on the ellipse to the focus at $(-c, 0)$, with respect to angle θ, is a.

Answer

a. $d^2\theta = (a\cos\theta + c)^2 + b^2\sin^2\theta = a^2 + c^2\cos^2\theta + 2ac\cos\theta = (a + c\cos\theta)^2$;

b. $\bar{d} = \dfrac{1}{2\pi}\displaystyle\int_0^{2\pi}(a + 2c\cos\theta)\,d\theta = a$

61) As implied earlier, according to Kepler's laws, Earth's orbit is an ellipse with the Sun at one focus. The perihelion for Earth's orbit around the Sun is 147,098,290 km and the aphelion is 152,098,232 km.

a. By placing the major axis along the x-axis, find the average distance from Earth to the Sun.

b. The classic definition of an astronomical unit (AU) is the distance from Earth to the Sun, and its value was computed as the average of the perihelion and aphelion distances. Is this definition justified?

62) The force of gravitational attraction between the Sun and a planet is $F(\theta) = \dfrac{GmM}{r^2(\theta)}$, where m is the mass of the planet, M is the mass of the Sun, G is a universal constant, and $r(\theta)$ is the distance between the Sun and the planet when the planet is at an angle θ with the major axis of its orbit. Assuming that M, m, and the ellipse parameters a and b (half-lengths of the major and minor axes) are given, set up—but do not evaluate—an integral that expresses in terms of G, m, M, a, b the average gravitational force between the Sun and the planet.

Answer

Mean gravitational force $= \dfrac{GmM}{2}\displaystyle\int_0^{2\pi}\frac{1}{(a + 2\sqrt{a^2 - b^2}\cos\theta)^2}\,d\theta.$

The displacement from rest of a mass attached to a spring satisfies the simple harmonic motion equation $x(t) = A\cos(\omega t - \phi)$, where ϕ is a phase constant, ω is the angular frequency, and A is the amplitude. Find the average velocity, the average speed (magnitude of velocity), the average displacement, and the average distance from rest (magnitude of displacement) of the mass.

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5.4: Integration Formulas and the Net Change Theorem

Learning Objectives

- Apply the basic integration formulas.
- Explain the significance of the net change theorem.
- Use the net change theorem to solve applied problems.
- Apply the integrals of odd and even functions.

In this section, we use some basic integration formulas studied previously to solve some key applied problems. It is important to note that these formulas are presented in terms of indefinite integrals. Although definite and indefinite integrals are closely related, there are some key differences to keep in mind. A definite integral is either a number (when the limits of integration are constants) or a single function (when one or both of the limits of integration are variables). An indefinite integral represents a family of functions, all of which differ by a constant. As you become more familiar with integration, you will get a feel for when to use definite integrals and when to use indefinite integrals. You will naturally select the correct approach for a given problem without thinking too much about it. However, until these concepts are cemented in your mind, think carefully about whether you need a definite integral or an indefinite integral and make sure you are using the proper notation based on your choice.

Basic Integration Formulas

Recall the integration formulas given in the section on Antiderivatives and the properties of definite integrals. Let's look at a few examples of how to apply these formulas and properties.

Example 5.4.1: Integrating a Function Using the Power Rule

Use the power rule to integrate the function $\int_1^4 \sqrt{t}(1+t)\,dt$.

Solution

The first step is to rewrite the function and simplify it so we can apply the power rule:

$$\begin{aligned}\int_1^4 \sqrt{t}(1+t)\,dt &= \int_1^4 t^{1/2}(1+t)\,dt \\ &= \int_1^4 (t^{1/2}+t^{3/2})\,dt.\end{aligned}$$

Now apply the power rule:

$$\begin{aligned}\int_1^4 (t^{1/2}+t^{3/2})\,dt &= \left(\frac{2}{3}t^{3/2}+\frac{2}{5}t^{5/2}\right)\Big|_1^4 \\ &= \left[\frac{2}{3}(4)^{3/2}+\frac{2}{5}(4)^{5/2}\right]-\left[\frac{2}{3}(1)^{3/2}+\frac{2}{5}(1)^{5/2}\right] \\ &= \frac{256}{15}.\end{aligned}$$

Exercise 5.4.1

Find the definite integral of $f(x)=x^2-3x$ over the interval $[1,3]$.

Hint

Follow the process from Example 5.4.1 to solve the problem.

Answer

$$\int_1^3 \left(x^2 - 3x\right)\, dx = -\frac{10}{3}$$

The Net Change Theorem

The **net change theorem** considers the integral of a **rate of change**. It says that when a quantity changes, the new value equals the initial value plus the integral of the rate of change of that quantity. The formula can be expressed in two ways. The second is more familiar; it is simply the definite integral.

Net Change Theorem

The new value of a changing quantity equals the initial value plus the integral of the rate of change:

$$F(b) = F(a) + \int_a^b F'(x)dx \tag{5.4.1}$$

or

$$\int_a^b F'(x)dx = F(b) - F(a). \tag{5.4.2}$$

Subtracting $F(a)$ from both sides of the Equation 5.4.1 yields Equation 5.4.2. Since they are equivalent formulas, which one we use depends on the application.

The significance of the net change theorem lies in the results. Net change can be applied to area, distance, and volume, to name only a few applications. Net change accounts for negative quantities automatically without having to write more than one integral. To illustrate, let's apply the net change theorem to a **velocity** function in which the result is **displacement.**

We looked at a simple example of this in The Definite Integral section. Suppose a car is moving due north (the positive direction) at 40 mph between 2 p.m. and 4 p.m., then the car moves south at 30 mph between 4 p.m. and 5 p.m. We can graph this motion as shown in Figure 5.4.1.

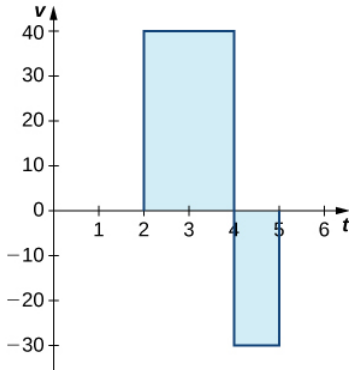


Figure 5.4.1: The graph shows speed versus time for the given motion of a car.

Just as we did before, we can use definite integrals to calculate the net displacement as well as the total distance traveled. The **net displacement** is given by

$$\int_2^5 v(t)\, dt = \int_2^4 40\, dt + \int_4^5 -30\, dt = 80 - 30 = 50.$$

Thus, at 5 p.m. the car is 50 mi north of its starting position. The **total distance** traveled is given by

$$\int_2^5 |v(t)|\, dt = \int_2^4 40\, dt + \int_4^5 30\, dt = 80 + 30 = 110.$$

Therefore, between 2 p.m. and 5 p.m., the car traveled a total of 110 mi.

To summarize, net displacement may include both positive and negative values. In other words, the velocity function accounts for both forward distance and backward distance. To find net displacement, integrate the velocity function over the interval. Total distance traveled, on the other hand, is always positive. To find the total distance traveled by an object, regardless of direction, we need to integrate the absolute value of the velocity function.

Example 5.4.2: Finding Net Displacement

Given a velocity function $v(t) = 3t - 5$ (in meters per second) for a particle in motion from time $t = 0$ to time $t = 3$, find the net displacement of the particle.

Solution

Applying the net change theorem, we have

$$\int_0^3 (3t-5)\,dt = \left(\frac{3t^2}{2} - 5t\right)\Big|_0^3 = \left[\frac{3(3)^2}{2} - 5(3)\right] - 0 = \frac{27}{2} - 15 = \frac{27}{2} - \frac{30}{2} = -\frac{3}{2}.$$

The net displacement is $-\frac{3}{2}$ m (Figure 5.4.2).

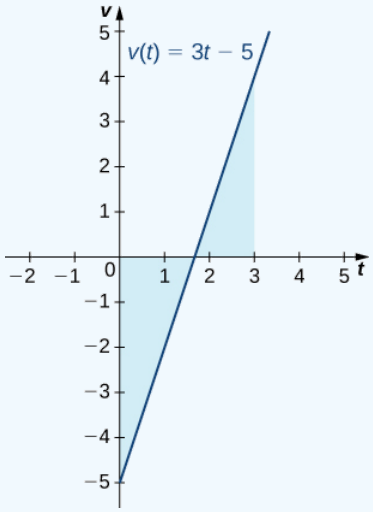


Figure 5.4.2: The graph shows velocity versus time for a particle moving with a linear velocity function.

Example 5.4.3: Finding the Total Distance Traveled

Use Example 5.4.2 to find the total distance traveled by a particle according to the velocity function $v(t) = 3t - 5$ m/sec over a time interval $[0, 3]$.

Solution

The total distance traveled includes both the positive and the negative values. Therefore, we must integrate the absolute value of the velocity function to find the total distance traveled.

To continue with the example, use two integrals to find the total distance. First, find the t-intercept of the function, since that is where the division of the interval occurs. Set the equation equal to zero and solve for t. Thus,

$$\begin{aligned} 3t - 5 &= 0 \\ 3t &= 5 \\ t &= \frac{5}{3}. \end{aligned}$$

The two subintervals are $[0, \frac{5}{3}]$ and $[\frac{5}{3}, 3]$. To find the total distance traveled, integrate the absolute value of the function. Since the function is negative over the interval $[0, \frac{5}{3}]$, we have $|v(t)| = -v(t)$ over that interval. Over $[\frac{5}{3}, 3]$, the function is positive, so $|v(t)| = v(t)$. Thus, we have

$$\begin{aligned}\int_0^3 |v(t)|\,dt &= \int_0^{5/3} -v(t)\,dt + \int_{5/3}^3 v(t)\,dt \\ &= \int_0^{5/3} 5-3t\,dt + \int_{5/3}^3 3t-5\,dt \\ &= \left(5t-\frac{3t^2}{2}\right)\Big|_0^{5/3} + \left(\frac{3t^2}{2}-5t\right)\Big|_{5/3}^3 \\ &= \left[5(\frac{5}{3})-\frac{3(5/3)^2}{2}\right]-0+\left[\frac{27}{2}-15\right]-\left[\frac{3(5/3)^2}{2}-\frac{25}{3}\right] \\ &= \frac{25}{3}-\frac{25}{6}+\frac{27}{2}-15-\frac{25}{6}+\frac{25}{3} \\ &= \frac{41}{6}\end{aligned}$$

So, the total distance traveled is $\frac{41}{6}$ m.

? Exercise 5.4.2

Find the net displacement and total distance traveled in meters given the velocity function $f(t)=\frac{1}{2}e^t-2$ over the interval $[0,2]$.

Hint

Follow the procedures from Examples 5.4.2 and 5.4.3. Note that $f(t)\le 0$ for $t\le \ln 4$ and $f(t)\ge 0$ for $t\ge \ln 4$.

Answer

Net displacement: $\frac{e^2-9}{2}\approx -0.8055$ m; total distance traveled: $4\ln 4-7.5+\frac{e^2}{2}\approx 1.740$ m.

Applying the Net Change Theorem

The net change theorem can be applied to the flow and consumption of fluids, as shown in Example 5.4.4.

✓ Example 5.4.4: How Many Gallons of Gasoline Are Consumed?

If the motor on a motorboat is started at $t=0$ and the boat consumes gasoline at the rate of $5-t^3$ gal/hr, how much gasoline is used in the first 2 hours?

Solution

Express the problem as a definite integral, integrate, and evaluate using the Fundamental Theorem of Calculus. The limits of integration are the endpoints of the interval [0,2]. We have

$$\begin{aligned}\int_0^2 (5-t^3)\,dt &= \left(5t-\frac{t^4}{4}\right)\Big|_0^2 \\ &= \left[5(2)-\frac{(2)^4}{4}\right]-0 \\ &= 10-\frac{16}{4} \\ &= 6.\end{aligned}$$

Thus, the motorboat uses 6 gal of gas in 2 hours.

✓ Example 5.4.5: Chapter Opener: Iceboats

As we saw at the beginning of the chapter, top **iceboat** racers can attain speeds of up to five times the wind speed. Andrew is an intermediate iceboater, though, so he attains speeds equal to only twice the wind speed.



Figure 5.4.3: (credit: modification of work by Carter Brown, Flickr)

Suppose Andrew takes his iceboat out one morning when a light 5-mph breeze has been blowing all morning. As Andrew gets his iceboat set up, though, the wind begins to pick up. During his first half hour of iceboating, the wind speed increases according to the function $v(t) = 20t + 5$. For the second half hour of Andrew's outing, the wind remains steady at 15 mph. In other words, the wind speed is given by

$$v(t) = \begin{cases} 20t + 5, & \text{for } 0 \le t \le \frac{1}{2} \\ 15, & \text{for } \frac{1}{2} \le t \le 1 \end{cases}$$

Recalling that Andrew's iceboat travels at twice the wind speed, and assuming he moves in a straight line away from his starting point, how far is Andrew from his starting point after 1 hour?

Solution

To figure out how far Andrew has traveled, we need to integrate his velocity, which is twice the wind speed. Then

$$\text{Distance} = \int_0^1 2v(t)\,dt.$$

Substituting the expressions we were given for $v(t)$, we get

$$\begin{aligned} \int_0^1 2v(t)\,dt &= \int_0^{1/2} 2v(t)\,dt + \int_{1/2}^1 2v(t)\,dt \\ &= \int_0^{1/2} 2(20t+5)\,dt + \int_{1/3}^1 2(15)\,dt \\ &= \int_0^{1/2} (40t+10)\,dt + \int_{1/2}^1 30\,dt \\ &= \left[20t^2 + 10t\right]\Big|_0^{1/2} + \left[30t\right]\Big|_{1/2}^1 \\ &= \left(\frac{20}{4} + 5\right) - 0 + (30 - 15) \\ &= 25. \end{aligned}$$

Andrew is 25 mi from his starting point after 1 hour.

? Exercise 5.4.3

Suppose that, instead of remaining steady during the second half hour of Andrew's outing, the wind starts to die down according to the function $v(t) = -10t + 15$. In other words, the wind speed is given by

$$v(t) = \begin{cases} 20t + 5, & \text{for } 0 \le t \le \frac{1}{2} \\ -10t + 15, & \text{for } \frac{1}{2} \le t \le 1 \end{cases}.$$

Under these conditions, how far from his starting point is Andrew after 1 hour?

Hint

Don't forget that Andrew's iceboat moves twice as fast as the wind.

Answer

17.5 mi

Integrating Even and Odd Functions

We saw in Functions and Graphs that an **even function** is a function in which $f(-x) = f(x)$ for all x in the domain—that is, the graph of the curve is unchanged when x is replaced with $-x$. The graphs of even functions are symmetric about the y-axis. An **odd function** is one in which $f(-x) = -f(x)$ for all x in the domain, and the graph of the function is symmetric about the origin.

Integrals of even functions, when the limits of integration are from $-a$ to a, involve two equal areas, because they are symmetric about the y-axis. Integrals of odd functions, when the limits of integration are similarly $[-a, a]$, evaluate to zero because the areas above and below the x-axis are equal.

Integrals of Even and Odd Functions

For continuous even functions such that $f(-x) = f(x)$,

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx.$$

For continuous odd functions such that $f(-x) = -f(x)$,

$$\int_{-a}^{a} f(x)\,dx = 0.$$

Example 5.4.6: Integrating an Even Function

Integrate the even function $\int_{-2}^{2}(3x^8 - 2)\,dx$ and verify that the integration formula for even functions holds.

Solution

The symmetry appears in the graphs in Figure 5.4.4. Graph (a) shows the region below the curve and above the x-axis. We have to zoom in to this graph by a huge amount to see the region. Graph (b) shows the region above the curve and below the x-axis. The signed area of this region is negative. Both views illustrate the symmetry about the y-axis of an even function. We have

$$\begin{aligned}\int_{-2}^{2}(3x^8 - 2)\,dx &= \left(\frac{x^9}{3} - 2x\right)\Big|_{-2}^{2} \\ &= \left[\frac{(2)^9}{3} - 2(2)\right] - \left[\frac{(-2)^9}{3} - 2(-2)\right] \\ &= \left(\frac{512}{3} - 4\right) - \left(-\frac{512}{3} + 4\right) \\ &= \frac{1000}{3}.\end{aligned}$$

To verify the integration formula for even functions, we can calculate the integral from 0 to 2 and double it, then check to make sure we get the same answer.

$$\int_{0}^{2}(3x^8 - 2)\,dx = \left(\frac{x^9}{3} - 2x\right)\Big|_{0}^{2} = \frac{512}{3} - 4 = \frac{500}{3}$$

Since $2 \cdot \frac{500}{3} = \frac{1000}{3}$, we have verified the formula for even functions in this particular example.

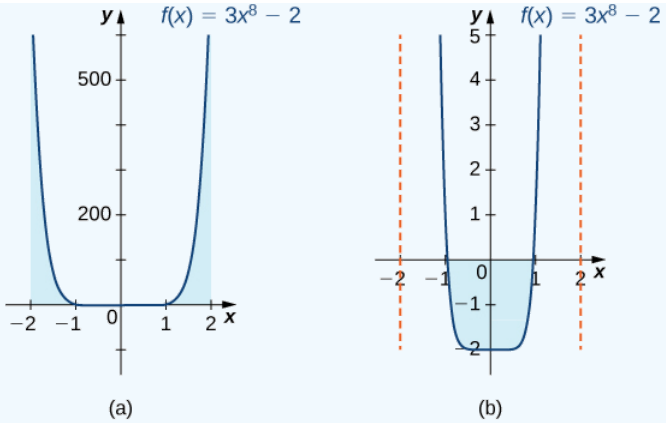


Figure 5.4.4: Graph (a) shows the positive area between the curve and the x-axis, whereas graph (b) shows the negative area between the curve and the x-axis. Both views show the symmetry about the y-axis.

Example 5.4.7: Integrating an Odd Function

Evaluate the definite integral of the odd function $-5\sin x$ over the interval $[-\pi, \pi]$.

Solution

The graph is shown in Figure 5.4.5. We can see the symmetry about the origin by the positive area above the x-axis over $[-\pi, 0]$, and the negative area below the x-axis over $[0, \pi]$. we have

$$\begin{aligned}\int_{-\pi}^{\pi} -5\sin x\, dx &= -5(-\cos x)\Big|_{-\pi}^{\pi} \\ &= 5\cos x\Big|_{-\pi}^{\pi} \\ &= [5\cos\pi] - [5\cos(-\pi)] \\ &= -5 - (-5) = 0.\end{aligned}$$

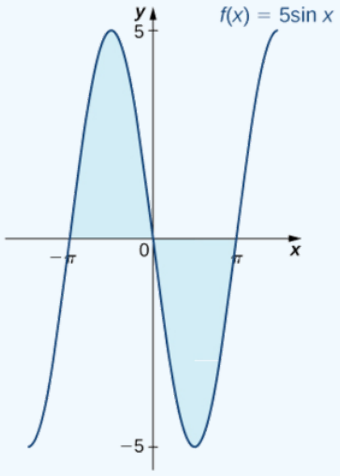


Figure 5.4.5:The graph shows areas between a curve and the x-axis for an odd function.

Exercise 5.4.4

Integrate the function $\int_{-2}^{2} x^4\, dx$.

Hint

Integrate an even function.

Answer

$\frac{64}{5}$

Key Concepts

- The net change theorem states that when a quantity changes, the final value equals the initial value plus the integral of the rate of change. Net change can be a positive number, a negative number, or zero.
- The area under an even function over a symmetric interval can be calculated by doubling the area over the positive x-axis. For an odd function, the integral over a symmetric interval equals zero, because half the area is negative.

Key Equations

- Net Change Theorem

$$F(b) = F(a) + \int_a^b F'(x)\,dx$$

or

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

Glossary

net change theorem

if we know the rate of change of a quantity, the net change theorem says the future quantity is equal to the initial quantity plus the integral of the rate of change of the quantity

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5.4E: EXERCISES FOR SECTION 5.4

In exercises 1 - 6, use basic integration formulas to compute the following antiderivatives.

1) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) dx$

Answer

$$\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) dx = \int x^{1/2}\, dx - \int x^{-1/2}\, dx = \frac{2}{3}x^{3/2} + C_1 - 2x^{1/2} + C_2 = \frac{2}{3}x^{3/2} - 2x^{1/2} + C$$

2) $\int \left(e^{2x} - \frac{1}{2}e^{x/2}\right) dx$

3) $\int \frac{dx}{2x}$

Answer

$$\int \frac{dx}{2x} = \frac{1}{2}\ln|x| + C$$

4) $\int \frac{x-1}{x^2}\, dx$

5) $\int_0^{\pi} (\sin x - \cos x)\, dx$

Answer

$$\int_0^{\pi} \sin x\, dx - \int_0^{\pi} \cos x\, dx = -\cos x\big|_0^{\pi} - (\sin x)\big|_0^{\pi} = (-(-1) + 1) - (0 - 0) = 2$$

6) $\int_0^{\pi/2} (x - \sin x)\, dx$

7) Write an integral that expresses the increase in the perimeter $P(s)$ of a square when its side length s increases from 2 units to 4 units and evaluate the integral.

Answer

$P(s) = 4s$, so $\frac{dP}{ds} = 4$ and $\int_2^4 4\, ds = 8$.

8) Write an integral that quantifies the change in the area $A(s) = s^2$ of a square when the side length doubles from S units to $2S$ units and evaluate the integral.

9) A regular N-gon (an N-sided polygon with sides that have equal length s, such as a pentagon or hexagon) has perimeter Ns. Write an integral that expresses the increase in perimeter of a regular N-gon when the length of each side increases from 1 unit to 2 units and evaluate the integral.

Answer

$$\int_1^2 N\, ds = N$$

10) The area of a regular pentagon with side length $a > 0$ is pa^2 with $p = \frac{1}{4}\sqrt{5 + \sqrt{5 + 2\sqrt{5}}}$. The Pentagon in Washington, DC, has inner sides of length 360 ft and outer sides of length 920 ft. Write an integral to express the area of the roof of the Pentagon according to these dimensions and evaluate this area.

11) A dodecahedron is a Platonic solid with a surface that consists of 12 pentagons, each of equal area. By how much does the surface area of a dodecahedron increase as the side length of each pentagon doubles from 1 unit to 2 units?

Answer

With p as in the previous exercise, each of the 12 pentagons increases in area from $2p$ to $4p$ units so the net increase in the area of the dodecahedron is $36p$ units.

12) An icosahedron is a Platonic solid with a surface that consists of 20 equilateral triangles. By how much does the surface area of an icosahedron increase as the side length of each triangle doubles from a unit to $2a$ units?

13) Write an integral that quantifies the change in the area of the surface of a cube when its side length doubles from s unit to $2s$ units and evaluate the integral.

Answer

$18s^2 = 6\int_s^{2s} 2x\,dx$

14) Write an integral that quantifies the increase in the volume of a cube when the side length doubles from s unit to $2s$ units and evaluate the integral.

15) Write an integral that quantifies the increase in the surface area of a sphere as its radius doubles from R unit to $2R$ units and evaluate the integral.

Answer

$12\pi R^2 = 8\pi\int_R^{2R} r\,dr$

16) Write an integral that quantifies the increase in the volume of a sphere as its radius doubles from R unit to $2R$ units and evaluate the integral.

17) Suppose that a particle moves along a straight line with velocity $v(t) = 4 - 2t$, where $0 \le t \le 2$ (in meters per second). Find the displacement at time t and the total distance traveled up to $t = 2$.

Answer

$d(t) = \int_0^t v(s)\,ds = 4t - t^2$. The total distance is $d(2) = 4$ m.

18) Suppose that a particle moves along a straight line with velocity defined by $v(t) = t^2 - 3t - 18$, where $0 \le t \le 6$ (in meters per second). Find the displacement at time t and the total distance traveled up to $t = 6$.

19) Suppose that a particle moves along a straight line with velocity defined by $v(t) = |2t - 6|$, where $0 \le t \le 6$ (in meters per second). Find the displacement at time t and the total distance traveled up to $t = 6$.

Answer

$d(t) = \int_0^t v(s)\,ds$. For $t < 3$, $d(t) = \int_0^t (6 - 2t)\,dt = 6t - t^2$. For $t > 3$, $d(t) = d(3) + \int_3^t (2t - 6)\,dt = 9 + (t^2 - 6t)$. The total distance is $d(6) = 9$ m.

20) Suppose that a particle moves along a straight line with acceleration defined by $a(t) = t - 3$, where $0 \le t \le 6$ (in meters per second). Find the velocity and displacement at time t and the total distance traveled up to $t = 6$ if $v(0) = 3$ and $d(0) = 0$.

21) A ball is thrown upward from a height of 1.5 m at an initial speed of 40 m/sec. Acceleration resulting from gravity is $-9.8\ \text{m/sec}^2$. Neglecting air resistance, solve for the velocity $v(t)$ and the height $h(t)$ of the ball t seconds after it is thrown and before it returns to the ground.

Answer

$v(t) = 40 - 9.8t;\ h(t) = 1.5 + 40t - 4.9t^2$ m/s

22) A ball is thrown upward from a height of 3 m at an initial speed of 60 m/sec. Acceleration resulting from gravity is $-9.8\ \text{m/sec}^2$. Neglecting air resistance, solve for the velocity $v(t)$ and the height $h(t)$ of the ball t seconds after it is thrown and before it returns to the ground.

23) The area $A(t)$ of a circular shape is growing at a constant rate. If the area increases from 4π units to 9π units between times $t = 2$ and $t = 3$, find the net change in the radius during that time.

Answer

The net increase is 1 unit.

24) A spherical balloon is being inflated at a constant rate. If the volume of the balloon changes from $36\pi\ \text{in}^3$ to $288\pi\ \text{in}^3$ between time $t = 30$ and $t = 60$ seconds, find the net change in the radius of the balloon during that time.

25) Water flows into a conical tank with cross-sectional area πx^2 at height x and volume $\frac{\pi x^3}{3}$ up to height x. If water flows into the tank at a rate of $1\ \text{m}^3/\text{min}$, find the height of water in the tank after 5 min. Find the change in height between 5 min and 10 min.

Answer

At $t = 5$, the height of water is $x = \left(\frac{15}{\pi}\right)^{1/3}$ m. The net change in height from $t = 5$ to $t = 10$ is $\left(\left(\frac{30}{\pi}\right)^{1/3} - \left(\frac{15}{\pi}\right)^{1/3}\right)$ m.

26) A horizontal cylindrical tank has cross-sectional area $A(x) = 4(6x - x^2)\,\text{m}^2$ at height x meters above the bottom when $x \leq 3$.

a. The volume V between heights a and b is $\int_a^b A(x)\,dx$. Find the volume at heights between 2 m and 3 m.

b. Suppose that oil is being pumped into the tank at a rate of 50 L/min. Using the chain rule, $\frac{dx}{dt} = \frac{dx}{dV}\frac{dV}{dt}$, at how many meters per minute is the height of oil in the tank changing, expressed in terms of x, when the height is at x meters?

c. How long does it take to fill the tank to 3 m starting from a fill level of 2 m?

27) The following table lists the electrical power in gigawatts—the rate at which energy is consumed—used in a certain city for different hours of the day, in a typical 24-hour period, with hour 1 corresponding to midnight to 1 a.m.

Hour	Power	Hour	Power
1	28	13	48
2	25	14	49
3	24	15	49
4	23	16	50
5	24	17	50
6	27	18	50
7	29	19	46
8	32	20	43
9	34	21	42
10	39	22	40
11	42	23	37
12	46	24	34

Find the total amount of power in gigawatt-hours (gW-h) consumed by the city in a typical 24-hour period.

Answer

The total daily power consumption is estimated as the sum of the hourly power rates, or 911 gW-h.

28) The average residential electrical power use (in hundreds of watts) per hour is given in the following table.

Hour	Power	Hour	Power
1	8	13	12
2	6	14	13
3	5	15	14
4	4	16	15
5	5	17	17
6	6	18	19
7	7	19	18
8	8	20	17
9	9	21	16
10	10	22	16
11	10	23	13
12	11	24	11

a. Compute the average total energy used in a day in kilowatt-hours (kWh).

b. If a ton of coal generates 1842 kWh, how long does it take for an average residence to burn a ton of coal?

c. Explain why the data might fit a plot of the form $p(t) = 11.5 - 7.5\sin\left(\frac{\pi t}{12}\right)$.

29) The data in the following table are used to estimate the average power output produced by Peter Sagan for each of the last 18 sec of Stage 1 of the 2012 **Tour de France.**

Second	Watts	Second	Watts
1	600	10	1200
2	500	11	1170
3	575	12	1125
4	1050	13	1100
5	925	14	1075
6	950	15	1000
7	1050	16	950
8	950	17	900
9	1100	18	780

Average Power OutputSource: sportsexercisengineering.com

Estimate the net energy used in kilojoules (kJ), noting that 1W = 1 J/s, and the average power output by Sagan during this time interval.

Answer

17 kJ

30) The data in the following table are used to estimate the average power output produced by Peter Sagan for each 15-min interval of Stage 1 of the 2012 Tour de France.

Minutes	Watts	Minutes	Watts
15	200	165	170
30	180	180	220
45	190	195	140
60	230	210	225
75	240	225	170
90	210	240	10
105	210	255	200
1120	220	270	220
135	210	285	250
150	150	300	400

Average Power Output*Source*: sportsexercisengineering.com

Estimate the net energy used in kilojoules, noting that 1W = 1 J/s.

31) The distribution of incomes as of 2012 in the United States in $5000 increments is given in the following table. The kth row denotes the percentage of households with incomes between $\$5000xk$ and $5000xk + 4999$. The row $k = 40$ contains all households with income between $200,000 and $250,000 and $k = 41$ accounts for all households with income exceeding $250,000.

0	3.5	21	1.5
1	4.1	22	1.4
2	5.9	23	1.3
3	5.7	24	1.3
4	5.9	25	1.1
5	5.4	26	1.0
6	5.5	27	0.75
7	5.1	28	0.8
8	4.8	29	1.0
9	4.1	30	0.6
10	4.3	31	0.6
11	3.5	32	.5
12	3.7	33	0.5
13	3.2	34	0.4
14	3.0	35	0.3
15	2.8	36	0.3
16	2.5	37	0.3
17	2.2	38	0.2
18	2.2	39	1.8
19	1.8	40	2.3
20	2.1	41	

Income DistributionsSource: www.census.gov/prod/2013pubs/p60-245.pdf

a. Estimate the percentage of U.S. households in 2012 with incomes less than $55,000.

b. What percentage of households had incomes exceeding $85,000?

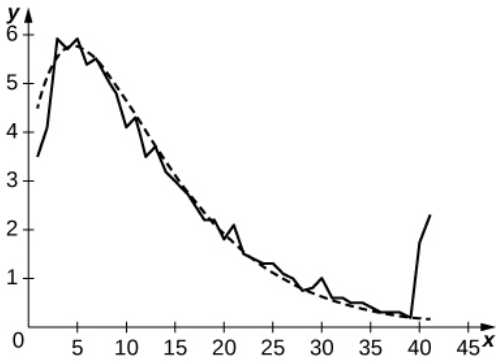
c. Plot the data and try to fit its shape to that of a graph of the form $a(x+c)e^{-b(x+c)}$ for suitable a, b, c.

Answer

a. 54.3%;

b. 27.00%;

c. The curve in the following plot is $2.35(t+3)e^{-0.15(t+3)}$.



32) Newton's law of gravity states that the gravitational force exerted by an object of mass M and one of mass m with centers that are separated by a distance r is $F = G\frac{mM}{r^2}$, with G an empirical constant $G = 6.67x10^{-11}\ \mathrm{m}^3/(\mathrm{kg}\cdot\mathrm{s}^2)$. The work done by a variable force over an interval $[a,b]$ is defined as $W = \int_a^b F(x)\,dx$. If Earth has mass 5.97219×10^{24} and radius 6371 km, compute the amount of work to elevate a polar weather satellite of mass 1400 kg to its orbiting altitude of 850 km above Earth.

33) For a given motor vehicle, the maximum achievable **deceleration** from braking is approximately $7\ \mathrm{m/sec}^2$ on dry concrete. On wet asphalt, it is approximately $2.5\ \mathrm{m/sec}^2$. Given that 1 mph corresponds to 0.447 m/sec, find the total distance that a car travels in meters on

dry concrete after the brakes are applied until it comes to a complete stop if the initial velocity is 67 mph (30 m/sec) or if the initial braking velocity is 56 mph (25 m/sec). Find the corresponding distances if the surface is slippery wet asphalt.

Answer

In dry conditions, with initial velocity $v_0 = 30$ m/s, $D = 64.3$ and, if $v_0 = 25,\ D = 44.64$. In wet conditions, if $v_0 = 30$, and $D = 180$ and if $v_0 = 25,\ D = 125$.

34) John is a 25-year old man who weighs 160 lb. He burns $500 - 50t$ calories/hr while riding his bike for t hours. If an oatmeal cookie has 55 cal and John eats $4t$ cookies during the tth hour, how many net calories has he lost after 3 hours riding his bike?

35) Sandra is a 25-year old woman who weighs 120 lb. She burns $300 - 50t$ cal/hr while walking on her treadmill for t hours. Her caloric intake from drinking Gatorade is $100t$ calories during the tth hour. What is her net decrease in calories after walking for 3 hours?

Answer

225 cal

36) A motor vehicle has a maximum efficiency of 33 mpg at a cruising speed of 40 mph. The efficiency drops at a rate of 0.1 mpg/mph between 40 mph and 50 mph, and at a rate of 0.4 mpg/mph between 50 mph and 80 mph. What is the efficiency in miles per gallon if the car is cruising at 50 mph? What is the efficiency in miles per gallon if the car is cruising at 80 mph? If gasoline costs $3.50/gal, what is the cost of fuel to drive 50 mi at 40 mph, at 50 mph, and at 80 mph?

37) Although some engines are more efficient at given a horsepower than others, on average, fuel efficiency decreases with horsepower at a rate of $1/25$ mpg/horsepower. If a typical 50-horsepower engine has an average fuel efficiency of 32 mpg, what is the average fuel efficiency of an engine with the following horsepower: 150, 300, 450?

Answer

$E(150) = 28,\ E(300) = 22,\ E(450) = 16$

38) [T] The following table lists the 2013 schedule of **federal income tax** versus taxable income.

Taxable Income Range	The Tax Is ...	... Of the Amount Over
$0–$8925	10%	$0
$8925–$36,250	$892.50 + 15%	$8925
$36,250–$87,850	$4,991.25 + 25%	$36,250
$87,850–$183,250	$17,891.25 + 28%	$87,850
$183,250–$398,350	$44,603.25 + 33%	$183,250
$398,350–$400,000	$115,586.25 + 35%	$398,350
> $400,000	$116,163.75 + 39.6%	$400,000

Federal Income Tax Versus Taxable IncomeSource: http://www.irs.gov/pub/irs-prior/i1040tt--2013.pdf.

Suppose that Steve just received a $10,000 raise. How much of this raise is left after federal taxes if Steve's salary before receiving the raise was $40,000? If it was $90,000? If it was $385,000?

39) [T] The following table provides hypothetical data regarding the level of service for a certain highway.

Highway Speed Range (mph)	Vehicles per Hour per Lane	Density Range (vehicles/mi)
>60	<600	<10
60-57	300-1000	10-20
57-54	1000-1500	20-30
57-54	1500-1900	30-45
46-30	1900-2100	48-70
<30	Unstable	70-200

a. Plot vehicles per hour per lane on the x-axis and highway speed on the y-axis.

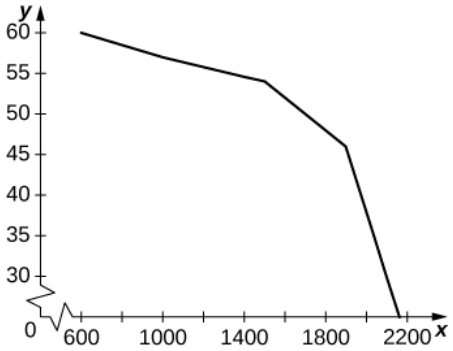
b. Compute the average decrease in speed (in miles per hour) per unit increase in congestion (vehicles per hour per lane) as the latter increases from 600 to 1000, from 1000 to 1500, and from 1500 to 2100. Does the decrease in miles per hour depend linearly on the increase

in vehicles per hour per lane?

c. Plot minutes per mile (60 times the reciprocal of miles per hour) as a function of vehicles per hour per lane. Is this function linear?

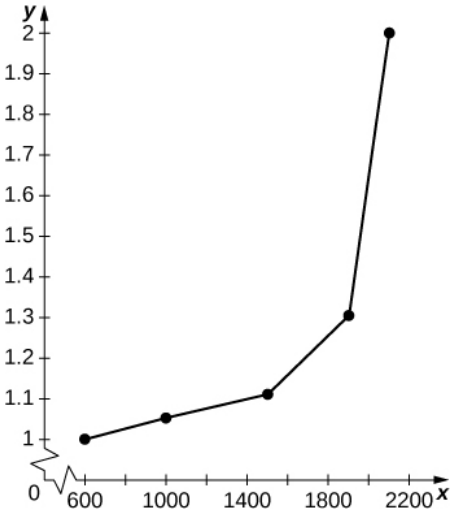
Answer

a.



b. Between 600 and 1000 the average decrease in vehicles per hour per lane is −0.0075. Between 1000 and 1500 it is −0.006 per vehicles per hour per lane, and between 1500 and 2100 it is −0.04 vehicles per hour per lane.

c.



The graph is nonlinear, with minutes per mile increasing dramatically as vehicles per hour per lane reach 2000.

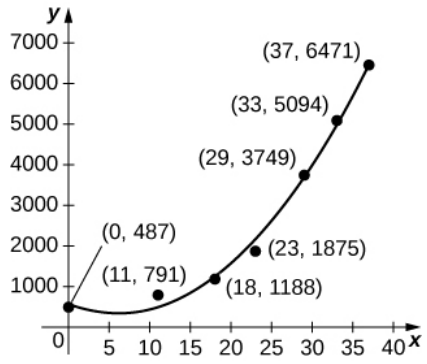
For the next two exercises use the data in the following table, which displays bald eagle populations from 1963 to 2000 in the continental United States.

Year	Population of Breeding Pairs of Bald Eagles
1963	487
1974	791
1981	1188
1986	1875
1992	3749
1996	5094
2000	6471

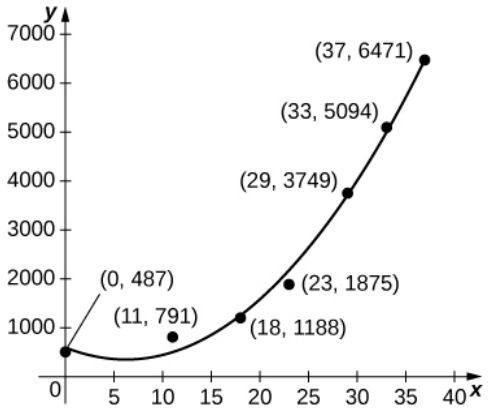
Population of Breeding Bald Eagle Pairs*Source*: www.fws.gov/Midwest/eagle/pop.../chtofprs.html.

40) [T] The graph below plots the quadratic $p(t) = 6.48t^2 - 80.31t + 585.69$ against the data in preceding table, normalized so that $t = 0$ corresponds to 1963. Estimate the average number of bald eagles per year present for the 37 years by computing the average value of p over

$[0, 37]$.



41) [T] The graph below plots the cubic $p(t) = 0.07t^3 + 2.42t^2 - 25.63t + 521.23$ against the data in the preceding table, normalized so that $t = 0$ corresponds to 1963. Estimate the average number of bald eagles per year present for the 37 years by computing the average value of p over $[0, 37]$.



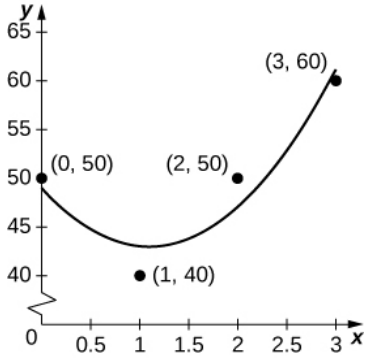
Answer

$$\frac{1}{37}\int_0^{37} p(t)\,dt = \frac{0.07(37)^3}{4} + \frac{2.42(37)^2}{3} - \frac{25.63(37)}{2} + 521.23 \approx 2037$$

Solution:

42) [T] Suppose you go on a road trip and record your speed at every half hour, as compiled in the following table. The best quadratic fit to the data is $q(t) = 5x^2 - 11x + 49$, shown in the accompanying graph. Integrate q to estimate the total distance driven over the 3 hours.

Time (hr)	Speed (m[h)
0 (start)	50
1	40
2	50
3	60

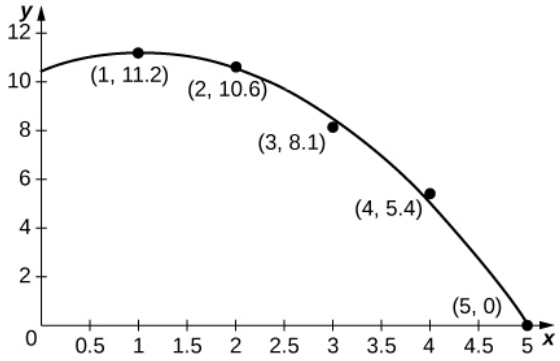


https://math.libretexts.org/@go/page/53520

As a car accelerates, it does not accelerate at a constant rate; rather, the acceleration is variable. For the following exercises, use the following table, which contains the acceleration measured at every second as a driver merges onto a freeway.

Time (sec)	Acceleration (mph/sec)
1	11.2
2	10.6
3	8.1
4	5.4
5	0

43) [T] The accompanying graph plots the best quadratic fit, $a(t) = -0.70t^2 + 1.44t + 10.44$, to the data from the preceding table. Compute the average value of $a(t)$ to estimate the average acceleration between $t = 0$ and $t = 5$.



Answer

Average acceleration is $A = \frac{1}{5}\int_0^5 a(t)\,dt = -\frac{0.7(5^2)}{3} + \frac{1.44(5)}{2} + 10.44 \approx 8.2$ mph/s

44) [T] Using your acceleration equation from the previous exercise, find the corresponding velocity equation. Assuming the final velocity is 0 mph, find the velocity at time $t = 0$.

45) [T] Using your velocity equation from the previous exercise, find the corresponding distance equation, assuming your initial distance is 0 mi. How far did you travel while you accelerated your car? (Hint: You will need to convert time units.)

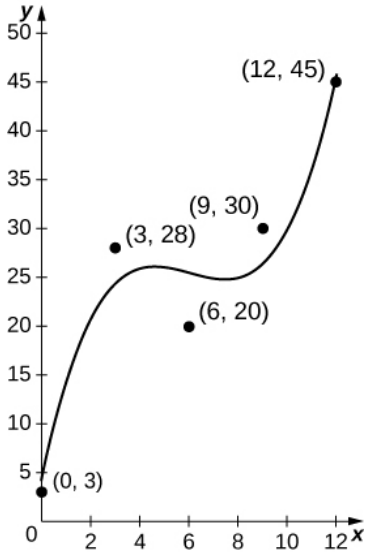
Answer

$d(t) = \int_0^1 |v(t)|\,dt = \int_0^t \left(\frac{7}{30}t^3 - 0.72t^2 - 10.44t + 41.033\right)\,dt = \frac{7}{120}t^4 - 0.24t^3 - 5.22t^3 + 41.033t$. Then, $d(5) \approx 81.12\,\text{mph} \times \text{sec} \approx 119$ feet.

46) [T] The number of hamburgers sold at a restaurant throughout the day is given in the following table, with the accompanying graph plotting the best cubic fit to the data, $b(t) = 0.12t^3 - 2.13t^3 + 12.13t + 3.91$, with $t = 0$ corresponding to 9 a.m. and $t = 12$ corresponding to 9 p.m. Compute the average value of $b(t)$ to estimate the average number of hamburgers sold per hour.

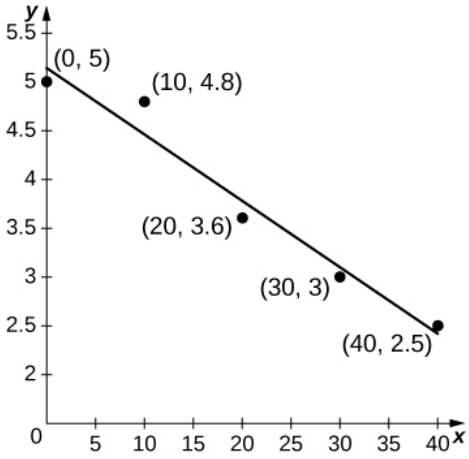
Hours Past Midnight	No. of Burgers Sold
9	3
12	28
15	20
18	30
21	45

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47) [T] An athlete runs by a motion detector, which records her speed, as displayed in the following table. The best linear fit to this data, $\ell(t) = -0.068t + 5.14$, is shown in the accompanying graph. Use the average value of $\ell(t)$ between $t = 0$ and $t = 40$ to estimate the runner's average speed.

Minutes	Speed (m/sec)
0	5
10	4.8
20	3.6
30	3.0
40	2.5



Answer

$$\frac{1}{40}\int_0^{40}(-0.068t + 5.14)\,dt = -\frac{0.068(40)}{2} + 5.14 = 3.78 \text{ m/sec}$$

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5.5: Substitution

Learning Objectives

- Use substitution to evaluate indefinite integrals.
- Use substitution to evaluate definite integrals.

The Fundamental Theorem of Calculus gave us a method to evaluate integrals without using Riemann sums. The drawback of this method, though, is that we must be able to find an antiderivative, and this is not always easy. In this section we examine a technique, called *integration by substitution*, to help us find antiderivatives. Specifically, this method helps us find antiderivatives when the integrand is the result of a chain-rule derivative.

At first, the approach to the substitution procedure may not appear very obvious. However, it is primarily a visual task—that is, the integrand shows you what to do; it is a matter of recognizing the form of the function. So, what are we supposed to see? We are looking for an integrand of the form $f\big[g(x)\big]g'(x)\,dx$. For example, in the integral

$$\int (x^2-3)^3\,2x\,dx. \tag{5.5.1}$$

we have

$$f(x)=x^3$$

and

$$g(x)=x^2-3.$$

Then

$$g'(x)=2x.$$

and

$$f[g(x)]g'(x)=(x^2-3)^3(2x),$$

and we see that our integrand is in the correct form. The method is called substitution because we substitute part of the integrand with the variable u and part of the integrand with du. It is also referred to as *change of variables* because we are changing variables to obtain an expression that is easier to work with for applying the integration rules.

Substitution with Indefinite Integrals

Let $u=g(x)$, where $g'(x)$ is continuous over an interval, let $f(x)$ be continuous over the corresponding range of g, and let $F(x)$ be an antiderivative of $f(x)$. Then,

$$\begin{aligned}\int f[g(x)]g'(x)\,dx &= \int f(u)\,du\\ &= F(u)+C\\ &= F(g(x))+C\end{aligned}$$

Proof

Let f, g, u, and F be as specified in the theorem. Then

$$\frac{d}{dx}\big[F(g(x))\big]=F'(g(x))g'(x)=f[g(x)]g'(x).$$

Integrating both sides with respect to x, we see that

$$\int f[g(x)]g'(x)\,dx=F(g(x))+C.$$

If we now substitute $u = g(x)$, and $du = g'(x)\,dx$, we get

$$\int f[g(x)]g'(x)\,dx = \int f(u)\,du = F(u) + C = F(g(x)) + C.$$

□

Returning to the problem we looked at originally, we let $u = x^2 - 3$ and then $du = 2x\,dx$.

Rewrite the integral (Equation 5.5.1) in terms of u:

$$\int (x^2 - 3)^3 (2x\,dx) = \int u^3\,du.$$

Using the power rule for integrals, we have

$$\int u^3\,du = \frac{u^4}{4} + C.$$

Substitute the original expression for x back into the solution:

$$\frac{u^4}{4} + C = \frac{(x^2 - 3)^4}{4} + C.$$

We can generalize the procedure in the following Problem-Solving Strategy.

Problem-Solving Strategy: Integration by Substitution

1. Look carefully at the integrand and select an expression $g(x)$ within the integrand to set equal to u. Let's select $g(x)$. such that $g'(x)$ is also part of the integrand.
2. Substitute $u = g(x)$ and $du = g'(x)dx$. into the integral.
3. We should now be able to evaluate the integral with respect to u. If the integral can't be evaluated we need to go back and select a different expression to use as u.
4. Evaluate the integral in terms of u.
5. Write the result in terms of x and the expression $g(x)$.

Example 5.5.1: Using Substitution to Find an Antiderivative

Use substitution to find the antiderivative of $\int 6x(3x^2 + 4)^4\,dx$.

Solution

The first step is to choose an expression for u. We choose $u = 3x^2 + 4$ because then $du = 6x\,dx$ and we already have du in the integrand. Write the integral in terms of u:

$$\int 6x(3x^2 + 4)^4\,dx = \int u^4\,du.$$

Remember that du is the derivative of the expression chosen for u, regardless of what is inside the integrand. Now we can evaluate the integral with respect to u:

$$\int u^4\,du = \frac{u^5}{5} + C = \frac{(3x^2 + 4)^5}{5} + C.$$

Analysis

We can check our answer by taking the derivative of the result of integration. We should obtain the integrand. Picking a value for C of 1, we let $y = \frac{1}{5}(3x^2 + 4)^5 + 1$. We have

$$y = \frac{1}{5}(3x^2 + 4)^5 + 1,$$

so

$$y' = \left(\frac{1}{5}\right) 5(3x^2+4)^4 6x$$
$$= 6x(3x^2+4)^4.$$

This is exactly the expression we started with inside the integrand.

? Exercise 5.5.1

Use substitution to find the antiderivative of $\int 3x^2(x^3-3)^2\,dx$.

Hint

Let $u = x^3 - 3$.

Answer

$$\int 3x^2(x^3-3)^2\,dx = \frac{1}{3}(x^3-3)^3 + C$$

Sometimes we need to adjust the constants in our integral if they don't match up exactly with the expressions we are substituting.

✓ Example 5.5.2: Using Substitution with Alteration

Use substitution to find the antiderivative of

$$\int z\sqrt{z^2-5}\,dz.$$

Solution

Rewrite the integral as $\int z(z^2-5)^{1/2}\,dz$. Let $u = z^2 - 5$ and $du = 2z\,dz$. Now we have a problem because $du = 2z\,dz$ and the original expression has only $z\,dz$. We have to alter our expression for du or the integral in u will be twice as large as it should be. If we multiply both sides of the du equation by $\frac{1}{2}$. we can solve this problem. Thus,

$$u = z^2 - 5$$
$$du = 2z\,dz$$
$$\frac{1}{2}du = \frac{1}{2}(2z)\,dz = z\,dz.$$

Write the integral in terms of u, but pull the $\frac{1}{2}$ outside the integration symbol:

$$\int z(z^2-5)^{1/2}\,dz = \frac{1}{2}\int u^{1/2}\,du.$$

Integrate the expression in u:

$$\frac{1}{2}\int u^{1/2}\,du = \left(\frac{1}{2}\right)\frac{u^{3/2}}{\frac{3}{2}} + C$$
$$= \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)u^{3/2} + C$$
$$= \frac{1}{3}u^{3/2} + C$$
$$= \frac{1}{3}(z^2-5)^{3/2} + C$$

? Exercise 5.5.2

Use substitution to find the antiderivative of $\int x^2(x^3+5)^9\,dx$.

Hint

Multiply the du equation by $\frac{1}{3}$.

Answer

$$\int x^2(x^3+5)^9\,dx = \frac{(x^3+5)^{10}}{30} + C$$

✓ Example 5.5.3: Using Substitution with Integrals of Trigonometric Functions

Use substitution to evaluate the integral $\int \frac{\sin t}{\cos^3 t}\,dt$.

Solution

We know the derivative of $\cos t$ is $-\sin t$, so we set $u = \cos t$. Then $du = -\sin t\,dt$.

Substituting into the integral, we have

$$\int \frac{\sin t}{\cos^3 t}\,dt = -\int \frac{du}{u^3}.$$

Evaluating the integral, we get

$$-\int \frac{du}{u^3} = -\int u^{-3}\,du = -\left(-\frac{1}{2}\right)u^{-2} + C.$$

Putting the answer back in terms of t, we get

$$\int \frac{\sin t}{\cos^3 t}\,dt = \frac{1}{2u^2} + C = \frac{1}{2\cos^2 t} + C.$$

? Exercise 5.5.3

Use substitution to evaluate the integral $\int \frac{\cos t}{\sin^2 t}\,dt$.

Hint

Use the process from Example 5.5.3 to solve the problem.

Answer

$$\int \frac{\cos t}{\sin^2 t}\,dt = -\frac{1}{\sin t} + C$$

? Exercise 5.5.4

Use substitution to evaluate the indefinite integral $\int \cos^3 t \sin t\,dt$.

Hint

Use the process from Example 5.5.3 to solve the problem.

Answer

$$\int \cos^3 t \sin t\,dt = -\frac{\cos^4 t}{4} + C$$

Sometimes we need to manipulate an integral in ways that are more complicated than just multiplying or dividing by a constant. We need to eliminate all the expressions within the integrand that are in terms of the original variable. When we are done, u should be the only variable in the integrand. In some cases, this means solving for the original variable in terms of u. This technique should become clear in the next example.

Example 5.5.4: Finding an Antiderivative Using u-Substitution

Use substitution to find the antiderivative of

$$\int \frac{x}{\sqrt{x-1}}\,dx.$$

Solution

If we let $u = x - 1$, then $du = dx$. But this does not account for the x in the numerator of the integrand. We need to express x in terms of u. If $u = x - 1$, then $x = u + 1$. Now we can rewrite the integral in terms of u :

$$\int \frac{x}{\sqrt{x-1}}\,dx = \int \frac{u+1}{\sqrt{u}}\,du = \int \left(\sqrt{u} + \frac{1}{\sqrt{u}}\right)\,du = \int \left(u^{1/2} + u^{-1/2}\right)\,du.$$

Then we integrate in the usual way, replace u with the original expression, and factor and simplify the result. Thus,

$$\begin{aligned}
\int (u^{1/2} + u^{-1/2})\,du &= \frac{2}{3}u^{3/2} + 2u^{1/2} + C \\
&= \frac{2}{3}(x-1)^{3/2} + 2(x-1)^{1/2} + C \\
&= (x-1)^{1/2}\left[\frac{2}{3}(x-1) + 2\right] + C \\
&= (x-1)^{1/2}\left(\frac{2}{3}x - \frac{2}{3} + \frac{6}{3}\right) \\
&= (x-1)^{1/2}\left(\frac{2}{3}x + \frac{4}{3}\right) \\
&= \frac{2}{3}(x-1)^{1/2}(x+2) + C.
\end{aligned}$$

Substitution for Definite Integrals

Substitution can be used with definite integrals, too. However, using substitution to evaluate a definite integral requires a change to the limits of integration. If we change variables in the integrand, the limits of integration change as well.

Substitution with Definite Integrals

Let $u = g(x)$ and let g' be continuous over an interval $[a, b]$, and let f be continuous over the range of $u = g(x)$. Then,

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Although we will not formally prove this theorem, we justify it with some calculations here. From the substitution rule for indefinite integrals, if $F(x)$ is an antiderivative of $f(x)$, we have

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C.$$

Then

$$\begin{aligned}\int_a^b f[g(x)]g'(x)\,dx &= F(g(x))\Big|_{x=a}^{x=b}\\ &= F(g(b)) - F(g(a))\\ &= F(u)\Big|_{u=g(a)}^{u=g(b)}\\ &= \int_{g(a)}^{g(b)} f(u)\,du\end{aligned}$$

and we have the desired result.

Example 5.5.5: Using Substitution to Evaluate a Definite Integral

Use substitution to evaluate

$$\int_0^1 x^2(1+2x^3)^5\,dx.$$

Solution

Let $u = 1 + 2x^3$, so $du = 6x^2\,dx$. Since the original function includes one factor of x^2 and $du = 6x^2\,dx$, multiply both sides of the du equation by $1/6$. Then,

$$\begin{aligned} du &= 6x^2\,dx\\ \text{becomes}\quad \frac{1}{6}du &= x^2\,dx.\end{aligned}$$

To adjust the limits of integration, note that when $x = 0, u = 1 + 2(0) = 1$, and when $x = 1,\ u = 1 + 2(1) = 3$.

Then

$$\int_0^1 x^2(1+2x^3)^5 dx = \frac{1}{6}\int_1^3 u^5\,du.$$

Evaluating this expression, we get

$$\begin{aligned}\frac{1}{6}\int_1^3 u^5\,du &= \left(\frac{1}{6}\right)\left(\frac{u^6}{6}\right)\Big|_1^3\\ &= \frac{1}{36}\left[(3)^6 - (1)^6\right]\\ &= \frac{182}{9}.\end{aligned}$$

Exercise 5.5.5

Use substitution to evaluate the definite integral $\int_{-1}^0 y(2y^2-3)^5\,dy$.

Hint

Use the steps from Example 5.5.5 to solve the problem.

Answer

$$\int_{-1}^0 y(2y^2-3)^5\,dy = \frac{91}{3}$$

Exercise 5.5.6

Use substitution to evaluate $\int_0^1 x^2 \cos\left(\frac{\pi}{2}x^3\right)\,dx$.

Hint

Use the process from Example 5.5.5 to solve the problem.

Answer

$$\int_0^1 x^2 \cos\left(\frac{\pi}{2}x^3\right)\,dx = \frac{2}{3\pi} \approx 0.2122$$

Example 5.5.6: Using Substitution with an Exponential Function

Use substitution to evaluate

$$\int_0^1 xe^{4x^2+3}\,dx.$$

Solution

Let $u = 4x^3 + 3$. Then, $du = 8x\,dx$. To adjust the limits of integration, we note that when $x = 0$, $u = 3$, and when $x = 1$, $u = 7$. So our substitution gives

$$\begin{aligned}\int_0^1 xe^{4x^2+3}\,dx &= \frac{1}{8}\int_3^7 e^u\,du \\ &= \frac{1}{8}e^u\Big|_3^7 \\ &= \frac{e^7 - e^3}{8} \\ &\approx 134.568\end{aligned}$$

Substitution may be only one of the techniques needed to evaluate a definite integral. All of the properties and rules of integration apply independently, and trigonometric functions may need to be rewritten using a trigonometric identity before we can apply substitution. Also, we have the option of replacing the original expression for u after we find the antiderivative, which means that we do not have to change the limits of integration. These two approaches are shown in Example 5.5.7.

Example 5.5.7: Using Substitution to Evaluate a Trigonometric Integral

Use substitution to evaluate

$$\int_0^{\pi/2} \cos^2\theta\,d\theta.$$

Solution

Let us first use a trigonometric identity to rewrite the integral. The trig identity $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ allows us to rewrite the integral as

$$\int_0^{\pi/2} \cos^2\theta\,d\theta = \int_0^{\pi/2} \frac{1+\cos 2\theta}{2}\,d\theta.$$

Then,

$$\begin{aligned}\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right)\,d\theta &= \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right)\,d\theta \\ &= \frac{1}{2}\int_0^{\pi/2} d\theta + \int_0^{\pi/2} \cos 2\theta\,d\theta.\end{aligned}$$

We can evaluate the first integral as it is, but we need to make a substitution to evaluate the second integral. Let $u = 2\theta$. Then, $du = 2\,d\theta$, or $\frac{1}{2}\,du = d\theta$. Also, when $\theta = 0$, $u = 0$, and when $\theta = \pi/2$, $u = \pi$. Expressing the second integral in terms of u, we have

$$\begin{aligned}\frac{1}{2}\int_0^{\pi/2} d\theta + \frac{1}{2}\int_0^{\pi/2} \cos 2\theta\, d\theta &= \frac{1}{2}\int_0^{\pi/2} d\theta + \frac{1}{2}\left(\frac{1}{2}\right)\int_0^{\pi} \cos u\, du \\ &= \frac{\theta}{2}\Big|_{\theta=0}^{\theta=\pi/2} + \frac{1}{4}\sin u\Big|_{u=0}^{u=\pi} \\ &= \left(\frac{\pi}{4} - 0\right) + (0 - 0) = \frac{\pi}{4}\end{aligned}$$

Key Concepts

- Substitution is a technique that simplifies the integration of functions that are the result of a chain-rule derivative. The term 'substitution' refers to changing variables or substituting the variable u and du for appropriate expressions in the integrand.
- When using substitution for a definite integral, we also have to change the limits of integration.

Key Equations

- **Substitution with Indefinite Integrals**

$$\int f[g(x)]g'(x)\,dx = \int f(u)\,du = F(u) + C = F(g(x)) + C$$

- **Substitution with Definite Integrals**

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

Glossary

change of variables

the substitution of a variable, such as u, for an expression in the integrand

integration by substitution

a technique for integration that allows integration of functions that are the result of a chain-rule derivative

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5.5E: EXERCISES FOR SECTION 5.5

1) Why is u-substitution referred to as a *change of variable*?

2) If $f = g \circ h$, when reversing the chain rule, $\frac{d}{dx}(g \circ h)(x) = g'(h(x))h'(x)$, should you take $u = g(x)$ or $u = h(x)$?

Answer

$u = h(x)$

In exercises 3 - 7, verify each identity using differentiation. Then, using the indicated u-substitution, identify f such that the integral takes the form $\int f(u)\,du$.

3) $\int x\sqrt{x+1}\,dx = \frac{2}{15}(x+1)^{3/2}(3x-2)+C; \quad u = x+1$

4) $\int \frac{x^2}{\sqrt{x-1}}\,dx = \frac{2}{15}\sqrt{x-1}(3x^2+4x+8)+C, \quad (x>1); \quad u = x-1$

Answer

$f(u) = \frac{(u+1)^2}{\sqrt{u}}$

5) $\int x\sqrt{4x^2+9}\,dx = \frac{1}{12}(4x^2+9)^{3/2}+C; \quad u = 4x^2+9$

6) $\int \frac{x}{\sqrt{4x^2+9}}\,dx = \frac{1}{4}\sqrt{4x^2+9}+C; \quad u = 4x^2+9$

Answer

$du = 8x\,dx; \quad f(u) = \frac{1}{8\sqrt{u}}$

7) $\int \frac{x}{(4x^2+9)^2}\,dx = -\frac{1}{8(4x^2+9)}+C; \quad u = 4x^2+9$

In exercises 8 - 17, find the antiderivative using the indicated substitution.

8) $\int (x+1)^4\,dx; \quad u = x+1$

Answer

$\int (x+1)^4\,dx = \frac{1}{5}(x+1)^5+C$

9) $\int (x-1)^5\,dx; \quad u = x-1$

10) $\int (2x-3)^{-7}\,dx; \quad u = 2x-3$

Answer

$\int (2x-3)^{-7}\,dx = -\frac{1}{12(2x-3)^6}+C$

11) $\int (3x-2)^{-11}\,dx; \quad u = 3x-2$

12) $\int \frac{x}{\sqrt{x^2+1}}\,dx; \quad u = x^2+1$

Answer

$\int \frac{x}{\sqrt{x^2+1}}\,dx = \sqrt{x^2+1}+C$

13) $\int \frac{x}{\sqrt{1-x^2}}\,dx; \quad u = 1-x^2$

14) $\int (x-1)(x^2-2x)^3\,dx; \quad u = x^2-2x$

Answer

$\int (x-1)(x^2-2x)^3\,dx = \frac{1}{8}(x^2-2x)^4+C$

15) $\int (x^2-2x)(x^3-3x^2)^2\,dx; \quad u=x^3=3x^2$

16) $\int \cos^3\theta\,d\theta; \quad u=\sin\theta$ (Hint: $\cos^2\theta=1-\sin^2\theta$)

Answer

$$\int \cos^3\theta\,d\theta=\sin\theta-\frac{\sin^3\theta}{3}+C$$

17) $\int \sin^3\theta\,d\theta; \quad u=\cos\theta$ (Hint: $\sin^2\theta=1-\cos^2\theta$)

In exercises 18 - 34, use a suitable change of variables to determine the indefinite integral.

18) $\int x(1-x)^{99}\,dx$

Answer

$$\int x(1-x)^{99}\,dx=\frac{(1-x)^{101}}{101}-\frac{(1-x)^{100}}{100}+C$$
$$=-\frac{(1-x)^{100}}{10100}\left[100x+1\right]+C$$

19) $\int t(1-t^2)^{10}dt$

20) $\int (11x-7)^{-3}\,dx$

Answer

$$\int (11x-7)^{-3}\,dx=-\frac{1}{22(11x-7)^2}+C$$

21) $\int (7x-11)^4\,dx$

22) $\int \cos^3\theta\sin\theta\,d\theta$

Answer

$$\int \cos^3\theta\sin\theta\,d\theta=-\frac{\cos^4\theta}{4}+C$$

23) $\int \sin^7\theta\cos\theta\,d\theta$

24) $\int \cos^2(\pi t)\sin(\pi t)\,dt$

Answer

$$\int \cos^2(\pi t)\sin(\pi t)\,dt=-\frac{cos^3(\pi t)}{3\pi}+C$$

25) $\int \sin^2 x\cos^3 x\,dx$ (Hint: $\sin^2 x+\cos^2 x=1$)

26) $\int t\sin(t^2)\cos(t^2)\,dt$

Answer

$$\int t\sin(t^2)\cos(t^2)\,dt=-\frac{1}{4}\cos^2(t^2)+C$$

27) $\int t^2\cos^2(t^3)\sin(t^3)\,dt$

28) $\int \frac{x^2}{(x^3-3)^2}\,dx$

Answer

$$\int \frac{x^2}{(x^3-3)^2}\,dx=-\frac{1}{3(x^3-3)}+C$$

29) $\int \frac{x^3}{\sqrt{1-x^2}}\,dx$

30) $\int \frac{y^5}{(1-y^3)^{3/2}}\,dy$

Answer

$$\int \frac{y^5}{(1-y^3)^{3/2}}\,dy = -\frac{2(y^3-2)}{3\sqrt{1-y^3}} + C$$

31) $\int \cos\theta(1-\cos\theta)^{99}\sin\theta\,d\theta$

32) $\int (1-\cos^3\theta)^{10}\cos^2\theta\sin\theta\,d\theta$

Answer

$$\int (1-\cos^3\theta)^{10}\cos^2\theta\sin\theta\,d\theta = \frac{1}{33}(1-\cos^3\theta)^{11} + C$$

33) $\int (\cos\theta - 1)(\cos^2\theta - 2\cos\theta)^3\sin\theta\,d\theta$

34) $\int (\sin^2\theta - 2\sin\theta)(\sin^3\theta - 3\sin^2\theta)^3\cos\theta\,d\theta$

Answer

$$\int (\sin^2\theta - 2\sin\theta)(\sin^3\theta - 3\sin^2\theta)^3\cos\theta\,d\theta = \frac{1}{12}(\sin^3\theta - 3\sin^2\theta)^4 + C$$

In exercises 35 - 38, use a calculator to estimate the area under the curve using left Riemann sums with 50 terms, then use substitution to solve for the exact answer.

35) [T] $y = 3(1-x)^2$ over $[0, 2]$

36) [T] $y = x(1-x^2)^3$ over $[-1, 2]$

Answer

$L_{50} = -8.5779$. The exact area is $\frac{-81}{8}$ units^2.

37) [T] $y = \sin x(1-\cos x)^2$ over $[0, \pi]$

38) [T] $y = \frac{x}{(x^2+1)^2}$ over $[-1, 1]$

Answer

$L_{50} = -0.006399$. The exact area is 0.

In exercises 39 - 44, use a change of variables to evaluate the definite integral.

39) $\int_0^1 x\sqrt{1-x^2}\,dx$

40) $\int_0^1 \frac{x}{\sqrt{1+x^2}}\,dx$

Answer

$$u = 1+x^2, \quad du = 2x\,dx, \quad \int_0^1 \frac{x}{\sqrt{1+x^2}}\,dx = \frac{1}{2}\int_1^2 u^{-1/2}du = \sqrt{2}-1$$

41) $\int_0^2 \frac{t}{\sqrt{5+t^2}}\,dt$

42) $\int_0^1 \frac{t^2}{\sqrt{1+t^3}}\,dt$

Answer

$$u = 1+t^3, \quad du = 3t^2, \quad \int_0^1 \frac{t^2}{\sqrt{1+t^3}}\,dt = \frac{1}{3}\int_1^2 u^{-1/2}du = \frac{2}{3}(\sqrt{2}-1)$$

43) $\int_0^{\pi/4} \sec^2\theta\tan\theta\,d\theta$

44) $\int_0^{\pi/4} \frac{\sin\theta}{\cos^4\theta}\,d\theta$

Answer

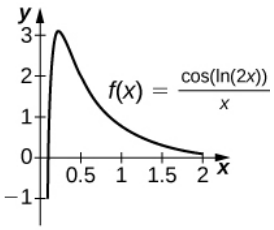
$$u=\cos\theta,\quad du=-\sin\theta\,d\theta,\quad \int_0^{\pi/4}\frac{\sin\theta}{\cos^4\theta}\,d\theta=-\int_1^{\sqrt{2}/2}u^{-4}\,du=\int_{\sqrt{2}/2}^1 u^{-4}\,du=\frac{1}{3}(2\sqrt{2}-1)$$

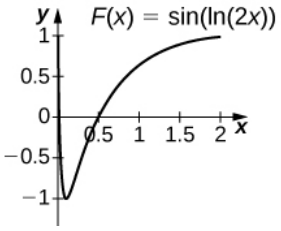
In exercises 45 - 50, evaluate the indefinite integral $\int f(x)\,dx$ with constant $C=0$ using u-substitution. Then, graph the function and the antiderivative over the indicated interval. If possible, estimate a value of C that would need to be added to the antiderivative to make it equal to the definite integral $F(x)=\int_a^x f(t)\,dt$, with a the left endpoint of the given interval.

45) [T] $\int (2x+1)e^{x^2+x-6}\,dx$ over $[-3,2]$

46) [T] $\int \frac{\cos(\ln(2x))}{x}\,dx$ on $[0,2]$

Answer





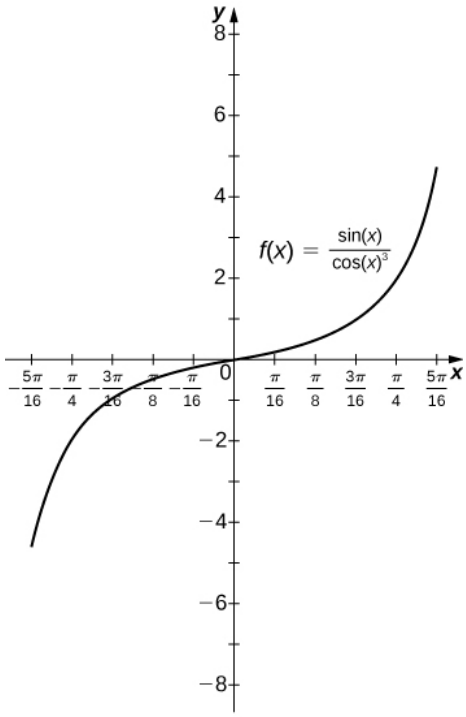
The antiderivative is $y=\sin(\ln(2x))$. Since the antiderivative is not continuous at $x=0$, one cannot find a value of C that would make $y=\sin(\ln(2x))-C$ work as a definite integral.

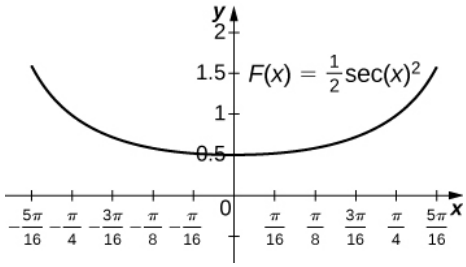
47) [T] $\int \frac{3x^2+2x+1}{\sqrt{x^3+x^2+x+4}}\,dx$ over $[-1,2]$

48) [T] $\int \frac{\sin x}{\cos^3 x}\,dx$ over $[-\frac{\pi}{3},\frac{\pi}{3}]$

Answer

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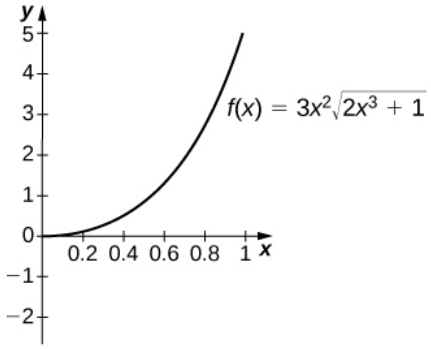
The antiderivative is $y = \frac{1}{2}\sec^2 x$. You should take $C = -2$ so that $F(-\frac{\pi}{3}) = 0$.

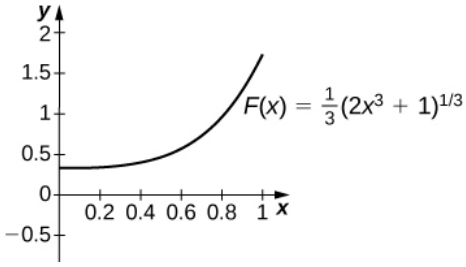
49) [T] $\int (x+2)e^{-x^2-4x+3}\,dx$ over $[-5, 1]$

50) [T] $\int 3x^2\sqrt{2x^3+1}\,dx$ over $[0, 1]$

Answer

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The antiderivative is $y = \frac{1}{3}(2x^3+1)^{3/2}$. One should take $C = -\frac{1}{3}$.

51) If $h(a) = h(b)$ in $\int_a^b g'(h(x))h(x)\,dx$, what can you say about the value of the integral?

52) Is the substitution $u = 1 - x^2$ in the definite integral $\int_0^2 \frac{x}{1-x^2}\,dx$ okay? If not, why not?

Answer

No, because the integrand is discontinuous at $x = 1$.

In exercises 53 - 59, use a change of variables to show that each definite integral is equal to zero.

53) $\int_0^{\pi} \cos^2(2\theta)\sin(2\theta)\,d\theta$

54) $\int_0^{\sqrt{\pi}} t\cos(t^2)\sin(t^2)\,dt$

Answer

$u = \sin(t^2)$; the integral becomes $\frac{1}{2}\int_0^0 u\,du$.

55) $\int_0^1 (1-2t)\,dt$

56) $\int_0^1 \frac{1-2t}{1+(t-\frac{1}{2})^2}\,dt$

Answer

$u = 1 + (t-\frac{1}{2})^2$; the integral becomes $-\int_{5/4}^{5/4} \frac{1}{u}\,du$.

57) $\int_0^{\pi} \sin\left((t-\frac{\pi}{2})^3\right)\cos\left(t-\frac{\pi}{2}\right)\,dt$

58) $\int_0^2 (1-t)\cos(\pi t)\,dt$

Answer

$u = 1 - t$; Since the integrand is odd, the integral becomes

$$\int_1^{-1} u\cos\left(\pi(1-u)\right)\,du = \int_1^{-1} u[\cos\pi\cos u - \sin\pi\sin u]\,du = -\int_1^{-1} u\cos u\,du = \int_{-1}^{1} u\cos u\,du = 0$$

59) $\int_{\pi/4}^{3\pi/4} \sin^2 t \cos t \, dt$

60) Show that the average value of $f(x)$ over an interval $[a, b]$ is the same as the average value of $f(cx)$ over the interval $\left[\frac{a}{c}, \frac{b}{c}\right]$ for $c > 0$.

Answer

Setting $u = cx$ and $du = c\,dx$ gets you $\frac{1}{\frac{b}{c} - \frac{a}{c}} \int_{a/c}^{b/c} f(cx)\,dx = \frac{c}{b-a} \int_{u=a}^{u=b} f(u) \frac{du}{c} = \frac{1}{b-a} \int_a^b f(u)\,du.$

61) Find the area under the graph of $f(t) = \frac{t}{(1+t^2)^a}$ between $t = 0$ and $t = x$ where $a > 0$ and $a \neq 1$ is fixed, and evaluate the limit as $x \to \infty$.

62) Find the area under the graph of $g(t) = \frac{t}{(1-t^2)^a}$ between $t = 0$ and $t = x$, where $0 < x < 1$ and $a > 0$ is fixed. Evaluate the limit as $x \to 1$.

Answer

$\int_0^x g(t)\,dt = \frac{1}{2} \int_{u=1-x^2}^{1} \frac{du}{u^a} = \frac{1}{2(1-a)} u^{1-a} \,|\, 1u = \frac{1}{2(1-a)}(1 - (1-x^2)^{1-a})$ As $x \to 1$ the limit is $\frac{1}{2(1-a)}$ if $a < 1$, and the limit diverges to $+\infty$ if $a > 1$.

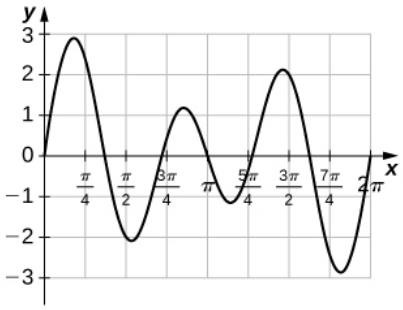
63) The area of a semicircle of radius 1 can be expressed as $\int_{-1}^{1} \sqrt{1-x^2}\,dx$. Use the substitution $x = \cos t$ to express the area of a semicircle as the integral of a trigonometric function. You do not need to compute the integral.

64) The area of the top half of an ellipse with a major axis that is the x-axis from $x = -1$ to a and with a minor axis that is the y-axis from $y = -b$ to $y = b$ can be written as $\int_{-a}^{a} b\sqrt{1 - \frac{x^2}{a^2}}\,dx$. Use the substitution $x = a \cos t$ to express this area in terms of an integral of a trigonometric function. You do not need to compute the integral.

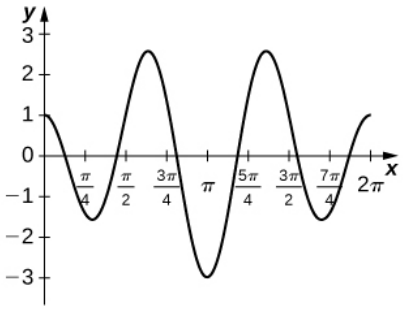
Answer

$\int_{t=\pi}^{t=0} b\sqrt{1-\cos^2 t} \times (-a \sin t)\,dt = \int_{t=0}^{t=\pi} ab \sin^2 t\,dt$

65) [T] The following graph is of a function of the form $f(t) = a \sin(nt) + b \sin(mt)$. Estimate the coefficients a and b and the frequency parameters n and m. Use these estimates to approximate $\int_0^{\pi} f(t)\,dt$.



66) [T] The following graph is of a function of the form $f(x) = a \cos(nt) + b \cos(mt)$. Estimate the coefficients a and b and the frequency parameters n and m. Use these estimates to approximate $\int_0^{\pi} f(t)\,dt$.



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Answer

$$f(t) = 2\cos(3t) - \cos(2t); \quad \int_0^{\pi/2} (2\cos(3t) - \cos(2t))\, dt = -\frac{2}{3}$$

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5.6: Integrals Involving Exponential and Logarithmic Functions

Learning Objectives

- Integrate functions involving exponential functions.
- Integrate functions involving logarithmic functions.

Exponential and logarithmic functions are used to model population growth, cell growth, and financial growth, as well as depreciation, radioactive decay, and resource consumption, to name only a few applications. In this section, we explore integration involving exponential and logarithmic functions.

Integrals of Exponential Functions

The exponential function is perhaps the most efficient function in terms of the operations of calculus. The exponential function, $y = e^x$, is its own derivative and its own integral.

Rule: Integrals of Exponential Functions

Exponential functions can be integrated using the following formulas.

$$\int e^x \, dx = e^x + C \tag{5.6.1}$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C \tag{5.6.2}$$

Example 5.6.1: Finding an Antiderivative of an Exponential Function

Find the antiderivative of the exponential function e^{-x}.

Solution

Use substitution, setting $u = -x$, and then $du = -1 \, dx$. Multiply the du equation by -1, so you now have $-du = dx$. Then,

$$\int e^{-x} \, dx = -\int e^u \, du = -e^u + C = -e^{-x} + C.$$

Exercise 5.6.1

Find the antiderivative of the function using substitution: $x^2 e^{-2x^3}$.

Hint

Let u equal the exponent on e.

Answer

$$\int x^2 e^{-2x^3} \, dx = -\frac{1}{6} e^{-2x^3} + C$$

A common mistake when dealing with exponential expressions is treating the exponent on e the same way we treat exponents in polynomial expressions. We cannot use the power rule for the exponent on e. This can be especially confusing when we have both exponentials and polynomials in the same expression, as in the previous checkpoint. In these cases, we should always double-check to make sure we're using the right rules for the functions we're integrating.

Example 5.6.2: Square Root of an Exponential Function

Find the antiderivative of the exponential function $e^x\sqrt{1+e^x}$.

Solution

First rewrite the problem using a rational exponent:

$$\int e^x\sqrt{1+e^x}\,dx = \int e^x(1+e^x)^{1/2}\,dx.$$

Using substitution, choose $u = 1+e^x$. Then, $du = e^x\,dx$. We have

$$\int e^x(1+e^x)^{1/2}\,dx = \int u^{1/2}\,du.$$

Then

$$\int u^{1/2}\,du = \frac{u^{3/2}}{3/2} + C = \frac{2}{3}u^{3/2} + C = \frac{2}{3}(1+e^x)^{3/2} + C$$

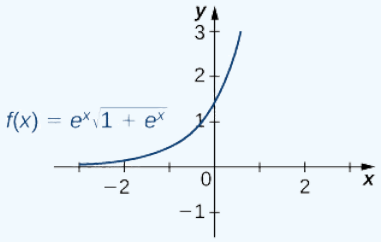


Figure 5.6.1: The graph shows an exponential function times the square root of an exponential function.

Exercise 5.6.2

Find the antiderivative of $e^x(3e^x-2)^2$.

Hint

Let $u = 3e^x - 2$.

Answer

$$\int e^x(3e^x-2)^2\,dx = \frac{1}{9}(3e^x-2)^3 + C$$

Example 5.6.3: Using Substitution with an Exponential Function

Use substitution to evaluate the indefinite integral $\int 3x^2e^{2x^3}\,dx$.

Solution

Here we choose to let u equal the expression in the exponent on e. Let $u = 2x^3$ and $du = 6x^2\,dx$. Again, du is off by a constant multiplier; the original function contains a factor of $3x^2$, not $6x^2$. Multiply both sides of the equation by $\frac{1}{2}$ so that the integrand in u equals the integrand in x. Thus,

$$\int 3x^2e^{2x^3}\,dx = \frac{1}{2}\int e^u\,du.$$

Integrate the expression in u and then substitute the original expression in x back into the u-integral:

$$\frac{1}{2}\int e^u\,du = \frac{1}{2}e^u + C = \frac{1}{2}e^{2x^3} + C.$$

Exercise 5.6.3

Evaluate the indefinite integral $\int 2x^3 e^{x^4}\,dx$.

Hint

Let $u = x^4$.

Answer

$$\int 2x^3 e^{x^4}\,dx = \frac{1}{2}e^{x^4} + C$$

As mentioned at the beginning of this section, exponential functions are used in many real-life applications. The number e is often associated with compounded or accelerating growth, as we have seen in earlier sections about the derivative. Although the derivative represents a rate of change or a growth rate, the integral represents the total change or the total growth. Let's look at an example in which integration of an exponential function solves a common business application.

A price–demand function tells us the relationship between the quantity of a product demanded and the price of the product. In general, price decreases as quantity demanded increases. The marginal price–demand function is the derivative of the price–demand function and it tells us how fast the price changes at a given level of production. These functions are used in business to determine the price–elasticity of demand, and to help companies determine whether changing production levels would be profitable.

Example 5.6.4: Finding a Price–Demand Equation

Find the price–demand equation for a particular brand of toothpaste at a supermarket chain when the demand is 50 tubes per week at \$2.35 per tube, given that the marginal price—demand function, $p'(x)$, for x number of tubes per week, is given as

$$p'(x) = -0.015e^{-0.01x}.$$

If the supermarket chain sells 100 tubes per week, what price should it set?

Solution

To find the price–demand equation, integrate the marginal price–demand function. First find the antiderivative, then look at the particulars. Thus,

$$p(x) = \int -0.015e^{-0.01x}\,dx = -0.015\int e^{-0.01x}\,dx.$$

Using substitution, let $u = -0.01x$ and $du = -0.01\,dx$. Then, divide both sides of the du equation by -0.01. This gives

$$\frac{-0.015}{-0.01}\int e^u\,du = 1.5\int e^u\,du = 1.5e^u + C = 1.5e^{-0.01x} + C.$$

The next step is to solve for C. We know that when the price is \$2.35 per tube, the demand is 50 tubes per week. This means

$$p(50) = 1.5e^{-0.01(50)} + C = 2.35.$$

Now, just solve for C:

$$C = 2.35 - 1.5e^{-0.5} = 2.35 - 0.91 = 1.44.$$

Thus,

$$p(x) = 1.5e^{-0.01x} + 1.44.$$

If the supermarket sells 100 tubes of toothpaste per week, the price would be

$$p(100) = 1.5e^{-0.01(100)} + 1.44 = 1.5e^{-1} + 1.44 \approx 1.99.$$

The supermarket should charge \$1.99 per tube if it is selling 100 tubes per week.

Example 5.6.5: Evaluating a Definite Integral Involving an Exponential Function

Evaluate the definite integral $\int_1^2 e^{1-x}\,dx$.

Solution

Again, substitution is the method to use. Let $u = 1 - x$, so $du = -1\,dx$ or $-du = dx$. Then $\int e^{1-x}\,dx = -\int e^u\,du$.

Next, change the limits of integration. Using the equation $u = 1 - x$, we have:

$$\text{When } x = 1, \quad u = 1 - (1) = 0,$$

$$\text{and when } x = 2, \quad u = 1 - (2) = -1.$$

The integral then becomes

$$\begin{aligned}\int_1^2 e^{1-x}\,dx &= -\int_0^{-1} e^u\,du \\ &= \int_{-1}^0 e^u\,du \\ &= e^u\Big|_{-1}^0 = e^0 - (e^{-1}) \\ &= -e^{-1} + 1.\end{aligned}$$

See Figure 5.6.2.

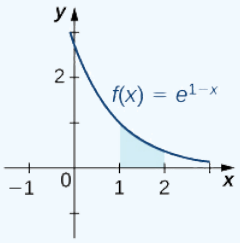


Figure 5.6.2: The indicated area can be calculated by evaluating a definite integral using substitution.

Exercise 5.6.4

Evaluate $\int_0^2 e^{2x}\,dx$.

Hint

Let $u = 2x$.

Answer

$$\frac{1}{2}\int_0^4 e^u\,du = \frac{1}{2}(e^4 - 1)$$

Example 5.6.6: Growth of Bacteria in a Culture

Suppose the rate of growth of bacteria in a Petri dish is given by $q(t) = 3^t$, where t is given in hours and $q(t)$ is given in thousands of bacteria per hour. If a culture starts with $10,000$ bacteria, find a function $Q(t)$ that gives the number of bacteria in the Petri dish at any time t. How many bacteria are in the dish after 2 hours?

Solution

We have

$$Q(t) = \int 3^t dt = \frac{3^t}{\ln 3} + C.$$

Then, at $t = 0$ we have $Q(0) = 10 = \frac{1}{\ln 3} + C$, so $C \approx 9.090$ and we get

$$Q(t) = \frac{3^t}{\ln 3} + 9.090.$$

At time $t = 2$, we have

$$\begin{aligned} Q(2) &= \frac{3^2}{\ln 3} + 9.090 \\ &\approx 17.282. \end{aligned}$$

After 2 hours, there are 17,282 bacteria in the dish.

Exercise 5.6.5

From Example, suppose the bacteria grow at a rate of $q(t) = 2^t$. Assume the culture still starts with $10,000$ bacteria. Find $Q(t)$. How many bacteria are in the dish after 3 hours?

Hint

Use the procedure from Example $5.6.6$ to solve the problem

Answer

$$Q(t) = \frac{2^t}{\ln 2} + 8.557.$$
$$Q(3) \approx 20,099$$

So there are $20,099$ bacteria in the dish after 3 hours.

Example 5.6.7: Fruit Fly Population Growth

Suppose a population of **fruit flies** increases at a rate of $g(t) = 2e^{0.02t}$, in flies per day. If the initial population of fruit flies is 100 flies, how many flies are in the population after 10 days?

Solution

Let $G(t)$ represent the number of flies in the population at time t. Applying the net change theorem, we have

$$\begin{aligned} G(10) = G(0) + \int_0^{10} 2e^{0.02t}\, dt \\ &= 100 + \left[\frac{2}{0.02} e^{0.02t}\right] |_0^{10} \\ &= 100 + \left[100e^{0.02t}\right] |_0^{10} \\ &= 100 + 100e^{0.2} - 100 \\ &\approx 122. \end{aligned}$$

There are 122 flies in the population after 10 days.

Exercise 5.6.6

Suppose the rate of growth of the fly population is given by $g(t) = e^{0.01t}$, and the initial fly population is 100 flies. How many flies are in the population after 15 days?

Hint

Use the process from Example $5.6.7$ to solve the problem.

Answer

There are 116 flies.

Example 5.6.8: Evaluating a Definite Integral Using Substitution

Evaluate the definite integral using substitution:

$$\int_1^2 \frac{e^{1/x}}{x^2}\,dx.$$

Solution

This problem requires some rewriting to simplify applying the properties. First, rewrite the exponent on e as a power of x, then bring the x^2 in the denominator up to the numerator using a negative exponent. We have

$$\int_1^2 \frac{e^{1/x}}{x^2}\,dx = \int_1^2 e^{x^{-1}}x^{-2}\,dx.$$

Let $u = x^{-1}$, the exponent on e. Then

$$du = -x^{-2}\,dx$$

$$-du = x^{-2}\,dx.$$

Bringing the negative sign outside the integral sign, the problem now reads

$$-\int e^u\,du.$$

Next, change the limits of integration:

$$u = (1)^{-1} = 1$$

$$u = (2)^{-1} = \frac{1}{2}.$$

Notice that now the limits begin with the larger number, meaning we can multiply by -1 and interchange the limits. Thus,

$$-\int_1^{1/2} e^u\,du = \int_{1/2}^1 e^u\,du = e^u\big|_{1/2}^1 = e - e^{1/2} = e - \sqrt{e}.$$

Exercise 5.6.7

Evaluate the definite integral using substitution:

$$\int_1^2 \frac{1}{x^3}e^{4x^{-2}}\,dx.$$

Hint

Let $u = 4x^{-2}$.

Answer

$$\int_1^2 \frac{1}{x^3}e^{4x^{-2}}\,dx = \frac{1}{8}[e^4 - e].$$

Integrals Involving Logarithmic Functions

Integrating functions of the form $f(x) = x^{-1}$ result in the absolute value of the natural log function, as shown in the following rule. Integral formulas for other logarithmic functions, such as $f(x) = \ln x$ and $f(x) = \log_a x$, are also included in the rule.

Rule: Integration Formulas Involving Logarithmic Functions

The following formulas can be used to evaluate integrals involving logarithmic functions.

$$\int x^{-1}\,dx = \ln|x| + C$$

$$\int \ln x\;dx = x\ln x - x + C = x(\ln x - 1) + C$$

$$\int \log_a x\,dx = \frac{x}{\ln a}(\ln x - 1) + C$$

Example 5.6.9: Finding an Antiderivative Involving $\ln x$

Find the antiderivative of the function $\dfrac{3}{x-10}$.

Solution

First factor the 3 outside the integral symbol. Then use the u^{-1} rule. Thus,

$$\int \frac{3}{x-10}\,dx = 3\int \frac{1}{x-10}\,dx = 3\int \frac{du}{u} = 3\ln|u| + C = 3\ln|x-10| + C, \quad x \neq 10.$$

See Figure 5.6.3.

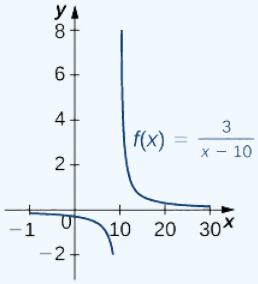


Figure 5.6.3: The domain of this function is $x \neq 10$.

Exercise 5.6.8

Find the antiderivative of $\dfrac{1}{x+2}$.

Hint

Follow the pattern from Example 5.6.9 to solve the problem.

Answer

$$\int \frac{1}{x+2}\,dx = \ln|x+2| + C$$

Example 5.6.10: Finding an Antiderivative of a Rational Function

Find the antiderivative of $\dfrac{2x^3+3x}{x^4+3x^2}$.

Solution

This can be rewritten as $\int (2x^3+3x)(x^4+3x^2)^{-1}\,dx$. Use substitution.

Let $u = x^4+3x^2$, then $du = (4x^3+6x)\,dx$. Alter du by factoring out the 2. Thus,

$$du = (4x^3+6x)\,dx = 2(2x^3+3x)\,dx$$

$$\frac{1}{2}\,du = (2x^3 + 3x)\,dx.$$

Rewrite the integrand in u:

$$\int (2x^3 + 3x)(x^4 + 3x^2)^{-1}\,dx = \frac{1}{2}\int u^{-1}\,du.$$

Then we have

$$\frac{1}{2}\int u^{-1}\,du = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln\,|\,x^4 + 3x^2\,|\,+C.$$

Example 5.6.11: Finding an Antiderivative of a Logarithmic Function

Find the antiderivative of the log function $\log_2 x$.

Solution

Follow the format in the formula listed in the rule on integration formulas involving logarithmic functions. Based on this format, we have

$$\int \log_2 x\,dx = \frac{x}{\ln 2}(\ln x - 1) + C.$$

Exercise 5.6.9

Find the antiderivative of $\log_3 x$.

Hint

Follow Example 5.6.11 and refer to the rule on integration formulas involving logarithmic functions.

Answer

$$\int \log_3 x\,dx = \frac{x}{\ln 3}(\ln x - 1) + C$$

Example 5.6.12 is a definite integral of a trigonometric function. With trigonometric functions, we often have to apply a trigonometric property or an identity before we can move forward. Finding the right form of the integrand is usually the key to a smooth integration.

Example 5.6.12: Evaluating a Definite Integral

Evaluate the definite integral

$$\int_0^{\pi/2} \frac{\sin x}{1 + \cos x}\,dx.$$

Solution

We need substitution to evaluate this problem. Let $u = 1 + \cos x$ so $du = -\sin x\ dx$.

Rewrite the integral in terms of u, changing the limits of integration as well. Thus,

$$u = 1 + \cos(0) = 2$$
$$u = 1 + \cos\left(\frac{\pi}{2}\right) = 1.$$

Then

$$\begin{aligned}\int_0^{\pi/2} \frac{\sin x}{1+\cos x} &= -\int_2^1 u^{-1}\,du \\ &= \int_1^2 u^{-1}\,du \\ &= \ln|u|\Big|_1^2 \\ &= [\ln 2 - \ln 1] = \ln 2\end{aligned}$$

Key Concepts

- Exponential and logarithmic functions arise in many real-world applications, especially those involving growth and decay.
- Substitution is often used to evaluate integrals involving exponential functions or logarithms.

Key Equations

- **Integrals of Exponential Functions**

$$\int e^x\,dx = e^x + C$$

$$\int a^x\,dx = \frac{a^x}{\ln a} + C$$

- **Integration Formulas Involving Logarithmic Functions**

$$\int x^{-1}\,dx = \ln|x| + C$$

$$\int \ln x\,dx = x\ln x - x + C = x(\ln x - 1) + C$$

$$\int \log_a x\,dx = \frac{x}{\ln a}(\ln x - 1) + C$$

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5.6E: EXERCISES FOR SECTION 5.6

For exercises 1 - 8, compute each indefinite integral.

1) $\int e^{2x}\,dx$

2) $\int e^{-3x}\,dx$

Answer

$$\int e^{-3x}\,dx \quad = \quad \frac{-1}{3}e^{-3x}+C$$

3) $\int 2^{x}\,dx$

4) $\int 3^{-x}\,dx$

Answer

$$\int 3^{-x}\,dx \quad = \quad -\frac{3^{-x}}{\ln 3}+C$$

5) $\int \frac{1}{2x}\,dx$

6) $\int \frac{2}{x}\,dx$

Answer

$$\int \frac{2}{x}\,dx \quad = \quad 2\ln|x|+C \quad = \quad \ln(x^2)+C$$

7) $\int \frac{1}{x^2}\,dx$

8) $\int \frac{1}{\sqrt{x}}\,dx$

Answer

$$\int \frac{1}{\sqrt{x}}\,dx \quad = \quad 2\sqrt{x}+C$$

In exercises 9 - 16, find each indefinite integral by using appropriate substitutions.

9) $\int \frac{\ln x}{x}\,dx$

10) $\int \frac{dx}{x(\ln x)^2}$

Answer

$$\int \frac{dx}{x(\ln x)^2} \quad = \quad -\frac{1}{\ln x}+C$$

11) $\int \frac{dx}{x\ln x} \quad (x>1)$

12) $\int \frac{dx}{x\ln x\ln(\ln x)}$

Answer

$$\int \frac{dx}{x\ln x\ln(\ln x)} \quad = \quad \ln(\ln(\ln x))+C$$

13) $\int \tan\theta\,d\theta$

14) $\int \frac{\cos x - x\sin x}{x\cos x}\,dx$

Answer

$$\int \frac{\cos x - x\sin x}{x\cos x}\,dx \quad = \quad \ln(x\cos x)+C$$

15) $\int \frac{\ln(\sin x)}{\tan x}\,dx$

16) $\int \ln(\cos x)\tan x\,dx$

Answer

$$\int \ln(\cos x)\tan x\,dx \quad = \quad -\frac{1}{2}(\ln(\cos(x)))^2 + C$$

17) $\int xe^{-x^2}\,dx$

18) $\int x^2 e^{-x^3}\,dx$

Answer

$$\int x^2 e^{-x^3}\,dx \quad = \quad \frac{-e^{-x^3}}{3} + C$$

19) $\int e^{\sin x}\cos x\,dx$

20) $\int e^{\tan x}\sec^2 x\,dx$

Answer

$$\int e^{\tan x}\sec^2 x\,dx \quad = \quad e^{\tan x} + C$$

21) $\int \frac{e^{\ln x}}{x}\,dx$

22) $\int \frac{e^{\ln(1-t)}}{1-t}\,dt$

Answer

$$\int \frac{e^{\ln(1-t)}}{1-t}\,dt = \int \frac{1-t}{1-t}\,dt = \int 1\,dt \quad = \quad t + C$$

In exercises 23 - 28, verify by differentiation that $\int \ln x\,dx = x(\ln x - 1) + C$, then use appropriate changes of variables to compute the integral.

23) $\int \ln x\,dx$ (Hint: $\int \ln x\,dx = \frac{1}{2}\int x\ln(x^2)\,dx$)

24) $\int x^2 \ln^2 x\,dx$

Answer

$$\int x^2 \ln^2 x\,dx \quad = \quad \frac{1}{9}x^3(\ln(x^3) - 1) + C$$

25) $\int \frac{\ln x}{x^2}\,dx$ (Hint: Set $u = \frac{1}{x}$.)

26) $\int \frac{\ln x}{\sqrt{x}}\,dx$ (Hint: Set $u = \sqrt{x}$.)

Answer

$$\int \frac{\ln x}{\sqrt{x}}\,dx \quad = \quad 2\sqrt{x}(\ln x - 2) + C$$

27) Write an integral to express the area under the graph of $y = \frac{1}{t}$ from $t = 1$ to e^x and evaluate the integral.

28) Write an integral to express the area under the graph of $y = e^t$ between $t = 0$ and $t = \ln x$, and evaluate the integral.

Answer

$$\int_0^{\ln x} e^t\,dt = e^t\Big|_0^{\ln x} = e^{\ln x} - e^0 = x - 1$$

In exercises 29 - 35, use appropriate substitutions to express the trigonometric integrals in terms of compositions with logarithms.

29) $\int \tan(2x)\,dx$

30) $\int \dfrac{\sin(3x)-\cos(3x)}{\sin(3x)+\cos(3x)}\,dx$

Answer

$$\int \frac{\sin(3x)-\cos(3x)}{\sin(3x)+\cos(3x)}\,dx \quad = \quad -\frac{1}{3}\ln|\sin(3x)+\cos(3x)|+C$$

31) $\int \dfrac{x\sin(x^2)}{\cos(x^2)}\,dx$

32) $\int x\csc(x^2)\,dx$

Answer

$$\int x\csc(x^2)\,dx \quad = \quad -\frac{1}{2}\ln|\csc(x^2)+\cot(x^2)|+C$$

33) $\int \ln(\cos x)\tan x\,dx$

34) $\int \ln(\csc x)\cot x\,dx$

Answer

$$\int \ln(\csc x)\cot x\,dx \quad = \quad -\frac{1}{2}(\ln(\csc x))^2+C$$

35) $\int \dfrac{e^x-e^{-x}}{e^x+e^{-x}}\,dx$

In exercises 36 - 40, evaluate the definite integral.

36) $\int_1^2 \dfrac{1+2x+x^2}{3x+3x^2+x^3}\,dx$

Answer

$$\int_1^2 \frac{1+2x+x^2}{3x+3x^2+x^3}\,dx \quad = \quad \frac{1}{3}\ln\left(\tfrac{26}{7}\right)$$

37) $\int_0^{\pi/4} \tan x\,dx$

38) $\int_0^{\pi/3} \dfrac{\sin x-\cos x}{\sin x+\cos x}\,dx$

Answer

$$\int_0^{\pi/3} \frac{\sin x-\cos x}{\sin x+\cos x}\,dx \quad = \quad \ln(\sqrt{3}-1)$$

39) $\int_{\pi/6}^{\pi/2} \csc x\,dx$

40) $\int_{\pi/4}^{\pi/3} \cot x\,dx$

Answer

$$\int_{\pi/4}^{\pi/3} \cot x\,dx \quad = \quad \frac{1}{2}\ln\frac{3}{2}$$

In exercises 41 - 46, integrate using the indicated substitution.

41) $\int \dfrac{x}{x-100}\,dx; \quad u=x-100$

42) $\int \dfrac{y-1}{y+1}\,dy; \quad u=y+1$

Answer

$$\int \frac{y-1}{y+1}\,dy \quad = \quad y-2\ln|y+1|+C$$

43) $\int \frac{1-x^2}{3x-x^3}\,dx; \quad u=3x-x^3$

44) $\int \frac{\sin x+\cos x}{\sin x-\cos x}\,dx; \quad u=\sin x-\cos x$

Answer

$$\int \frac{\sin x+\cos x}{\sin x-\cos x}\,dx \quad = \quad \ln|\sin x-\cos x|+C$$

45) $\int e^{2x}\sqrt{1-e^{2x}}\,dx; \quad u=1-e^{2x}$

46) $\int \ln(x)\frac{\sqrt{1-(\ln x)^2}}{x}\,dx; \quad u=1-(\ln x)^2$

Answer

$$\int \ln(x)\frac{\sqrt{1-(\ln x)^2}}{x}\,dx \quad = \quad -\frac{1}{3}(1-(\ln x)^2)^{3/2}+C$$

47) $\int \frac{\sqrt{x}}{\sqrt{x}+2}\,dx; \quad u=\sqrt{x}+2$

Answer

$$\int \frac{\sqrt{x}}{\sqrt{x}+2}\,dx \quad = \quad (\sqrt{x}+2)^2-8(\sqrt{x}+2)+8\ln(\sqrt{x}+2)+C$$

48) $\int e^x \sec(e^x+1)\tan(e^x+1)\,dx; \quad u=e^x+1$

Answer

$$\int e^x \sec(e^x+1)\tan(e^x+1)\,dx \quad = \quad \sec(e^x+1)+C$$

In exercises 49 - 54, state whether the right-endpoint approximation overestimates or underestimates the exact area. Then calculate the right endpoint estimate R_{50} and solve for the exact area.

49) [T] $y=e^x$ over $[0,1]$

50) [T] $y=e^{-x}$ over $[0,1]$

Answer

Since f is decreasing, the right endpoint estimate underestimates the area.

Exact solution: $\frac{e-1}{e}$, $R_{50}=0.6258$.

51) [T] $y=\ln(x)$ over $[1,2]$

52) [T] $y=\frac{x+1}{x^2+2x+6}$ over $[0,1]$

Answer

Since f is increasing, the right endpoint estimate overestimates the area.

Exact solution: $\frac{2\ln(3)-\ln(6)}{2}$, $R_{50}=0.2033$.

53) [T] $y=2^x$ over $[-1,0]$

54) [T] $y=-2^{-x}$ over $[0,1]$

Answer

Since f is increasing, the right endpoint estimate overestimates the area (the actual area is a larger negative number).

Exact solution: $-\frac{1}{\ln(4)}$, $R_{50}=-0.7164$.

In exercises 55 - 58, $f(x)\geq 0$ for $a\leq x\leq b$. Find the area under the graph of $f(x)$ between the given values a and b by integrating.

55) $f(x)=\frac{\log_{10}(x)}{x}; \quad a=10, b=100$

56) $f(x)=\frac{\log_2(x)}{x}; \quad a=32, b=64$

Answer

$\frac{11}{2}\ln 2$

57) $f(x)=2^{-x}; \quad a=1, b=2$

58) $f(x)=2^{-x}; \quad a=3, b=4$

Answer

$\frac{1}{\ln(65,536)}$

59) Find the area under the graph of the function $f(x)=xe^{-x^2}$ between $x=0$ and $x=5$.

60) Compute the integral of $f(x)=xe^{-x^2}$ and find the smallest value of N such that the area under the graph $f(x)=xe^{-x^2}$ between $x=N$ and $x=N+1$ is, at most, 0.01.

Answer

$\int_N^{N+1} xe^{-x^2}\,dx=\frac{1}{2}(e^{-N^2}-e^{-(N+1)^2})$. The quantity is less than 0.01 when $N=2$.

61) Find the limit, as N tends to infinity, of the area under the graph of $f(x)=xe^{-x^2}$ between $x=0$ and $x=5$.

62) Show that $\int_a^b \frac{dt}{t}=\int_{1/b}^{1/a}\frac{dt}{t}$ when $0<a\le b$.

Answer

$\int_a^b \frac{dx}{x}=\ln(b)-\ln(a)=\ln(\frac{1}{a})-\ln(\frac{1}{b})=\int_{1/b}^{1/a}\frac{dx}{x}$

63) Suppose that $f(x)>0$ for all x and that f and g are differentiable. Use the identity $f^g=e^{g\ln f}$ and the chain rule to find the derivative of f^g.

64) Use the previous exercise to find the antiderivative of $h(x)=x^x(1+\ln x)$ and evaluate $\int_2^3 x^x(1+\ln x)\,dx$.

Answer

23

65) Show that if $c>0$, then the integral of $\frac{1}{x}$ from ac to bc (for $0<a<b$) is the same as the integral of $\frac{1}{x}$ from a to b.

The following exercises are intended to derive the fundamental properties of the natural log starting from the definition $\ln(x)=\int_1^x \frac{dt}{t}$, using properties of the definite integral and making no further assumptions.

66) Use the identity $\ln(x)=\int_1^x \frac{dt}{t}$ to derive the identity $\ln\left(\frac{1}{x}\right)=-\ln x$.

Answer

We may assume that $x>1$,so $\frac{1}{x}<1$. Then, $\int_1^{1/x}\frac{dt}{t}$. Now make the substitution $u=\frac{1}{t}$, so $du=-\frac{dt}{t^2}$ and $\frac{du}{u}=-\frac{dt}{t}$, and change endpoints: $\int_1^{1/x}\frac{dt}{t}=-\int_1^x\frac{du}{u}=-\ln x$.

67) Use a change of variable in the integral $\int_1^{xy}\frac{1}{t}\,dt$ to show that $\ln xy=\ln x+\ln y$ for $x,y>0$.

68) Use the identity $\ln x=\int_1^x\frac{dt}{t}$ to show that $\ln(x)$ is an increasing function of x on $[0,\infty)$, and use the previous exercises to show that the range of $\ln(x)$ is $(-\infty,\infty)$. Without any further assumptions, conclude that $\ln(x)$ has an inverse function defined on $(-\infty,\infty)$.

69) Pretend, for the moment, that we do not know that e^x is the inverse function of $\ln(x)$, but keep in mind that $\ln(x)$ has an inverse function defined on $(-\infty,\infty)$. Call it E. Use the identity $\ln xy=\ln x+\ln y$ to deduce that $E(a+b)=E(a)E(b)$ for any real numbers a, b.

70) Pretend, for the moment, that we do not know that e^x is the inverse function of $\ln x$, but keep in mind that $\ln x$ has an inverse function defined on $(-\infty,\infty)$. Call it E. Show that $E'(t)=E(t)$.

Answer

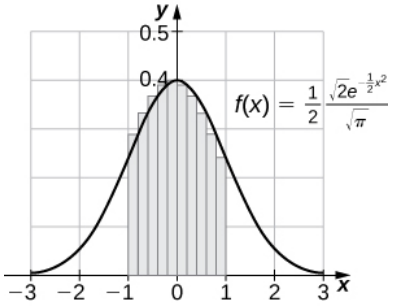
$x = E(\ln(x))$. Then, $1 = \dfrac{E'(\ln x)}{x}$ or $x = E'(\ln x)$. Since any number t can be written $t = \ln x$ for some x, and for such t we have $x = E(t)$, it follows that for any t, $E'(t) = E(t)$.

71) The sine integral, defined as $S(x) = \int_0^x \dfrac{\sin t}{t}\, dt$ is an important quantity in engineering. Although it does not have a simple closed formula, it is possible to estimate its behavior for large x. Show that for $k \geq 1$, $|S(2\pi k) - S(2\pi(k+1))| \leq \dfrac{1}{k(2k+1)\pi}$. (Hint: $\sin(t+\pi) = -\sin t$)

72) [T] The normal distribution in probability is given by $p(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$, where σ is the standard deviation and μ is the average. The standard normal distribution in probability, p_s, corresponds to $\mu = 0$ and $\sigma = 1$. Compute the left endpoint estimates R_{10} and R_{100} of $\int_{-1}^{1} \dfrac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$.

Answer

$R_{10} = 0.6811, \quad R_{100} = 0.6827$



73) [T] Compute the right endpoint estimates R_{50} and R_{100} of $\int_{-3}^{5} \dfrac{1}{2\sqrt{2\pi}} e^{-(x-1)^2/8}$.

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- Paul Seeburger (Monroe Community College) added problems 47-48 to Section 5.6 exercises.

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5.7: Integrals Resulting in Inverse Trigonometric Functions

Learning Objectives

- Integrate functions resulting in inverse trigonometric functions

In this section we focus on integrals that result in inverse trigonometric functions. We have worked with these functions before. Recall, that trigonometric functions are not one-to-one unless the domains are restricted. When working with inverses of trigonometric functions, we always need to be careful to take these restrictions into account. Also, we previously developed formulas for derivatives of inverse trigonometric functions. The formulas developed there give rise directly to integration formulas involving inverse trigonometric functions.

Integrals that Result in Inverse Trigonometric Functions

Let us begin this last section of the chapter with the three formulas. Along with these formulas, we use substitution to evaluate the integrals. We prove the formula for the inverse sine integral.

Rule: Integration Formulas Resulting in Inverse Trigonometric Functions

The following integration formulas yield inverse trigonometric functions:

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C \tag{5.7.1}$$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C \tag{5.7.2}$$

$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{|u|}{a}\right) + C \tag{5.7.3}$$

Proof of the first formula

Let $y = \sin^{-1}\frac{x}{a}$. Then $a\sin y = x$. Now using implicit differentiation, we obtain

$$\frac{d}{dx}(a\sin y) = \frac{d}{dx}(x)$$

$$a\cos y\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{a\cos y}.$$

For $-\frac{\pi}{2} \le y \le \frac{\pi}{2}, \cos y \ge 0$. Thus, applying the Pythagorean identity $\sin^2 y + \cos^2 y = 1$, we have $\cos y = \sqrt{1-\sin^2 y}$. This gives

$$\frac{1}{a\cos y} = \frac{1}{a\sqrt{1-\sin^2 y}} \tag{5.7.4}$$

$$= \frac{1}{\sqrt{a^2-a^2\sin^2 y}} \tag{5.7.5}$$

$$= \frac{1}{\sqrt{a^2-x^2}}. \tag{5.7.6}$$

Then for $-a \le x \le a$, we have

$$\int \frac{1}{\sqrt{a^2-u^2}}\,du = \sin^{-1}\left(\frac{u}{a}\right) + C.$$

□

Example 5.7.1: Evaluating a Definite Integral Using Inverse Trigonometric Functions

Evaluate the definite integral

$$\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}}.$$

Solution

We can go directly to the formula for the antiderivative in the rule on integration formulas resulting in inverse trigonometric functions, and then evaluate the definite integral. We have

$$\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x \Big|_0^{1/2} = \sin^{-1} \tfrac{1}{2} - \sin^{-1} 0 = \frac{\pi}{6} - 0 = \frac{\pi}{6}.$$

Note that since the integrand is simply the derivative of $\sin^{-1} x$, we are really just using this fact to find the antiderivative here.

Exercise 5.7.1

Find the indefinite integral using an inverse trigonometric function and substitution for $\int \frac{dx}{\sqrt{9-x^2}}$.

Hint

Use the formula in the rule on integration formulas resulting in inverse trigonometric functions.

Answer

$$\int \frac{dx}{\sqrt{9-x^2}} = \sin^{-1}\left(\frac{x}{3}\right) + C$$

In many integrals that result in inverse trigonometric functions in the antiderivative, we may need to use substitution to see how to use the integration formulas provided above.

Example 5.7.2: Finding an Antiderivative Involving an Inverse Trigonometric Function using substitution

Evaluate the integral

$$\int \frac{dx}{\sqrt{4-9x^2}}.$$

Solution

Substitute $u = 3x$. Then $du = 3\,dx$ and we have

$$\int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3}\int \frac{du}{\sqrt{4-u^2}}.$$

Applying the formula with $a = 2$, we obtain

$$\int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3}\int \frac{du}{\sqrt{4-u^2}} = \frac{1}{3}\sin^{-1}\left(\frac{u}{2}\right) + C = \frac{1}{3}\sin^{-1}\left(\frac{3x}{2}\right) + C.$$

Exercise 5.7.2

Find the antiderivative of $\int \frac{dx}{\sqrt{1-16x^2}}$.

Hint

Substitute $u = 4x$.

Answer

$$\int \frac{dx}{\sqrt{1-16x^2}} = \frac{1}{4}\sin^{-1}(4x) + C$$

Example 5.7.3: Evaluating a Definite Integral

Evaluate the definite integral

$$\int_0^{\sqrt{3}/2} \frac{du}{\sqrt{1-u^2}}.$$

Solution

The format of the problem matches the inverse sine formula. Thus,

$$\int_0^{\sqrt{3}/2} \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u \Big|_0^{\sqrt{3}/2} = [\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)] - [\sin^{-1}(0)] = \frac{\pi}{3}.$$

Integrals Resulting in Other Inverse Trigonometric Functions

There are six inverse trigonometric functions. However, only three integration formulas are noted in the rule on integration formulas resulting in inverse trigonometric functions because the remaining three are negative versions of the ones we use. The only difference is whether the integrand is positive or negative. Rather than memorizing three more formulas, if the integrand is negative, simply factor out −1 and evaluate the integral using one of the formulas already provided. To close this section, we examine one more formula: the integral resulting in the inverse tangent function.

Example 5.7.4: Finding an Antiderivative Involving the Inverse Tangent Function

Find the antiderivative of $\int \frac{1}{9+x^2}\,dx$.

Solution

Apply the formula with $a = 3$. Then,

$$\int \frac{dx}{9+x^2} = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C.$$

Exercise 5.7.3

Find the antiderivative of $\int \frac{dx}{16+x^2}$.

Hint

Follow the steps in Example 5.7.4.

Answer

$$\int \frac{dx}{16+x^2} = \frac{1}{4}\tan^{-1}\left(\frac{x}{4}\right) + C$$

Example 5.7.5: Applying the Integration Formulas WITH SUBSTITUTION

Find an antiderivative of $\int \frac{1}{1+4x^2}\,dx$.

Solution

Comparing this problem with the formulas stated in the rule on integration formulas resulting in inverse trigonometric functions, the integrand looks similar to the formula for $\tan^{-1} u + C$. So we use substitution, letting $u = 2x$, then $du = 2\,dx$ and $\frac{1}{2}\,du = dx$.Then, we have

$$\frac{1}{2}\int \frac{1}{1+u^2}\,du = \frac{1}{2}\tan^{-1}u + C = \frac{1}{2}\tan^{-1}(2x) + C.$$

Exercise 5.7.4

Use substitution to find the antiderivative of $\int \frac{dx}{25+4x^2}$.

Hint

Use the solving strategy from Example 5.7.5 and the rule on integration formulas resulting in inverse trigonometric functions.

Answer

$$\int \frac{dx}{25+4x^2} = \frac{1}{10}\tan^{-1}\left(\frac{2x}{5}\right) + C$$

Example 5.7.6: Evaluating a Definite Integral

Evaluate the definite integral $\int_{\sqrt{3}/3}^{\sqrt{3}} \frac{dx}{1+x^2}$.

Solution

Use the formula for the inverse tangent. We have

$$\int_{\sqrt{3}/3}^{\sqrt{3}} \frac{dx}{1+x^2} = \tan^{-1}x\Big|_{\sqrt{3}/3}^{\sqrt{3}} = [\tan^{-1}(\sqrt{3})] - [\tan^{-1}\left(\frac{\sqrt{3}}{3}\right)] = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}.$$

Exercise 5.7.5

Evaluate the definite integral $\int_0^2 \frac{dx}{4+x^2}$.

Hint

Follow the procedures from Example 5.7.6 to solve the problem.

Answer

$$\int_0^2 \frac{dx}{4+x^2} = \frac{\pi}{8}$$

Key Concepts

- Formulas for derivatives of inverse trigonometric functions developed in Derivatives of Exponential and Logarithmic Functions lead directly to integration formulas involving inverse trigonometric functions.
- Use the formulas listed in the rule on integration formulas resulting in inverse trigonometric functions to match up the correct format and make alterations as necessary to solve the problem.
- Substitution is often required to put the integrand in the correct form.

Key Equations

- **Integrals That Produce Inverse Trigonometric Functions**

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$$
$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$$
$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{|u|}{a}\right) + C$$

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- Includes some added textual clarifications and edits by Paul Seeburger (Monroe Community College)

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5.7E: EXERCISES FOR SECTION 5.7

In exercises 1 - 6, evaluate each integral in terms of an inverse trigonometric function.

1) $\int_0^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}}$

Answer

$$\int_0^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x\Big|_0^{\sqrt{3}/2} = \frac{\pi}{3}$$

2) $\int_{-1/2}^{1/2} \frac{dx}{\sqrt{1-x^2}}$

3) $\int_{\sqrt{3}}^{1} \frac{dx}{\sqrt{1+x^2}}$

Answer

$$\int_{\sqrt{3}}^{1} \frac{dx}{\sqrt{1+x^2}} = \tan^{-1} x\Big|_{\sqrt{3}}^{1} = -\frac{\pi}{12}$$

4) $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dx}{1+x^2}$

5) $\int_1^{\sqrt{2}} \frac{dx}{|x|\sqrt{x^2-1}}$

Answer

$$\int_1^{\sqrt{2}} \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1} x\Big|_1^{\sqrt{2}} = \frac{\pi}{4}$$

6) $\int_1^{\frac{2}{\sqrt{3}}} \frac{dx}{|x|\sqrt{x^2-1}}$

In exercises 7 - 12, find each indefinite integral, using appropriate substitutions.

7) $\int \frac{dx}{\sqrt{9-x^2}}$

Answer

$$\int \frac{dx}{\sqrt{9-x^2}} = \sin^{-1}\left(\frac{x}{3}\right) + C$$

8) $\int \frac{dx}{\sqrt{1-16x^2}}$

9) $\int \frac{dx}{9+x^2}$

Answer

$$\int \frac{dx}{9+x^2} = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C$$

10) $\int \frac{dx}{25+16x^2}$

11) $\int \frac{dx}{x\sqrt{x^2-9}}$

Answer

$$\int \frac{dx}{x\sqrt{x^2-9}} = \frac{1}{3}\sec^{-1}\left(\frac{|x|}{3}\right) + C$$

12) $\int \frac{dx}{x\sqrt{4x^2-16}}$

13) Explain the relationship $-\cos^{-1} t + C = \int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + C$. Is it true, in general, that $\cos^{-1} t = -\sin^{-1} t$?

Answer

$\cos(\frac{\pi}{2} - \theta) = \sin\theta$. So, $\sin^{-1} t = \frac{\pi}{2} - \cos^{-1} t$. They differ by a constant.

14) Explain the relationship $\sec^{-1} t + C = \int \frac{dt}{|t|\sqrt{t^2-1}} = -\csc^{-1} t + C$. Is it true, in general, that $\sec^{-1} t = -\csc^{-1} t$?

15) Explain what is wrong with the following integral: $\int_1^2 \frac{dt}{\sqrt{1-t^2}}$.

Answer

$\sqrt{1-t^2}$ is not defined as a real number when $t > 1$.

16) Explain what is wrong with the following integral: $\int_{-1}^1 \frac{dt}{|t|\sqrt{t^2-1}}$.

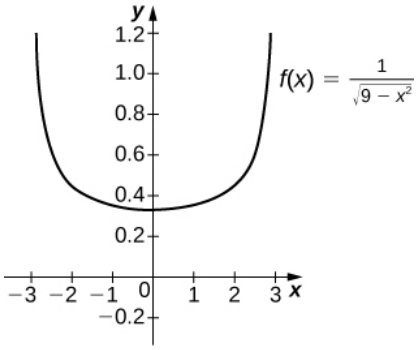
Answer

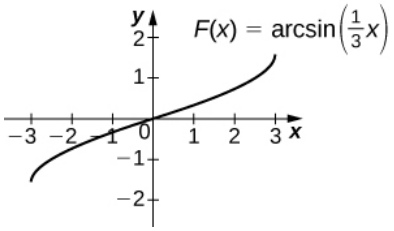
$\sqrt{t^2-1}$ is not defined as a real number when $-1 < t < 1$, and the integrand is undefined when $t = -1$ or $t = 1$.

In exercises 17 - 20, solve for the antiderivative of f with $C = 0$, then use a calculator to graph f and the antiderivative over the given interval $[a, b]$. Identify a value of C such that adding C to the antiderivative recovers the definite integral $F(x) = \int_a^x f(t)\,dt$.

17) [T] $\int \frac{1}{\sqrt{9-x^2}}\,dx$ over $[-3, 3]$

Answer



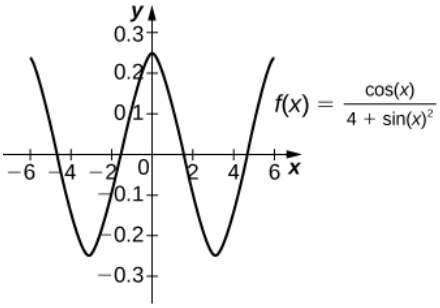


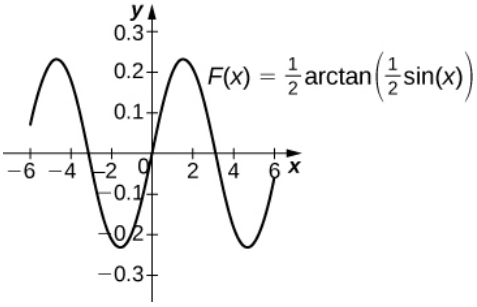
The antiderivative is $\sin^{-1}(\frac{x}{3}) + C$. Taking $C = \frac{\pi}{2}$ recovers the definite integral.

18) [T] $\int \frac{9}{9+x^2}\,dx$ over $[-6, 6]$

19) [T] $\int \frac{\cos x}{4+\sin^2 x}\,dx$ over $[-6, 6]$

Answer





The antiderivative is $\frac{1}{2}\tan^{-1}(\frac{\sin x}{2})+C$. Taking $C=\frac{1}{2}\tan^{-1}(\frac{\sin(6)}{2})$ recovers the definite integral.

20) [T] $\int \frac{e^x}{1+e^{2x}}\,dx$ over $[-6,6]$

In exercises 21 - 26, compute the antiderivative using appropriate substitutions.

21) $\int \frac{\sin^{-1} t}{\sqrt{1-t^2}}\,dt$

Answer

$$\int \frac{\sin^{-1} t\,dt}{\sqrt{1-t^2}} \quad = \quad \tfrac{1}{2}(\sin^{-1} t)^2+C$$

22) $\int \frac{dt}{\sin^{-1} t\sqrt{1-t^2}}$

23) $\int \frac{\tan^{-1}(2t)}{1+4t^2}\,dt$

Answer

$$\int \frac{\tan^{-1}(2t)}{1+4t^2}\,dt \quad = \quad \frac{1}{4}(\tan^{-1}(2t))^2+C$$

24) $\int \frac{t\tan^{-1}(t^2)}{1+t^4}\,dt$

25) $\int \frac{\sec^{-1}(\frac{t}{2})}{|t|\sqrt{t^2-4}}\,dt$

Answer

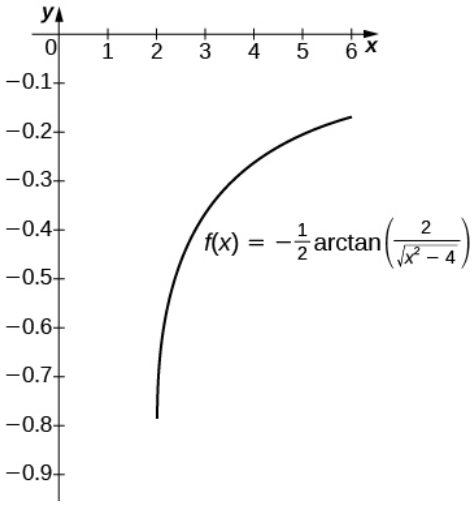
$$\int \frac{\sec^{-1}(\frac{t}{2})}{|t|\sqrt{t^2-4}}\,dt \quad = \quad \tfrac{1}{4}(\sec^{-1}(\tfrac{t}{2}))^2+C$$

26) $\int \frac{t\sec^{-1}(t^2)}{t^2\sqrt{t^4-1}}\,dt$

In exercises 27 - 32, use a calculator to graph the antiderivative of f with $C=0$ over the given interval $[a,b]$. Approximate a value of C, if possible, such that adding C to the antiderivative gives the same value as the definite integral $F(x)=\int_a^x f(t)\,dt$.

27) [T] $\int \frac{1}{x\sqrt{x^2-4}}\,dx$ over $[2,6]$

Answer

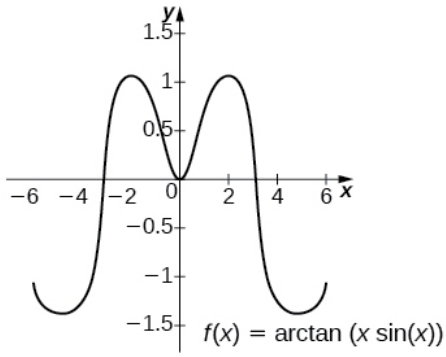


The antiderivative is $\frac{1}{2}\sec^{-1}\left(\frac{x}{2}\right)+C$. Taking $C=0$ recovers the definite integral over $[2,6]$.

28) [T] $\int \frac{1}{(2x+2)\sqrt{x}}\,dx$ over $[0,6]$

29) [T] $\int \frac{(\sin x + x\cos x)}{1+x^2\sin^2 x\,dx}$ over $[-6,6]$

Answer

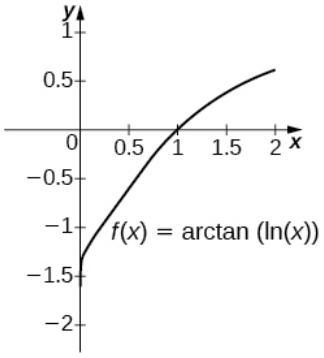


The general antiderivative is $\tan^{-1}(x\sin x)+C$. Taking $C=-\tan^{-1}(6\sin(6))$ recovers the definite integral.

30) [T] $\int \frac{2e^{-2x}}{\sqrt{1-e^{-4x}}}\,dx$ over $[0,2]$

31) [T] $\int \frac{1}{x+x\ln 2x}$ over $[0,2]$

Answer



The general antiderivative is $\tan^{-1}(\ln x)+C$. Taking $C=\frac{\pi}{2}=\lim_{t\to\infty}\tan^{-1}t$ recovers the definite integral.

32) [T] $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}}$ over $[-1,1]$

In exercises 33 - 38, compute each integral using appropriate substitutions.

33) $\int \frac{e^t}{\sqrt{1-e^{2t}}}\,dt$

Answer

$$\int \frac{e^t}{\sqrt{1-e^{2t}}}\,dt = \sin^{-1}(e^t)+C$$

34) $\int \frac{e^t}{1+e^{2t}}\,dt$

35) $\int \frac{dt}{t\sqrt{1-\ln^2 t}}$

Answer

$$\int \frac{dt}{t\sqrt{1-\ln^2 t}} = \sin^{-1}(\ln t)+C$$

36) $\int \frac{dt}{t(1+\ln^2 t)}$

37) $\int \frac{\cos^{-1}(2t)}{\sqrt{1-4t^2}}\,dt$

Answer

$$\int \frac{\cos^{-1}(2t)}{\sqrt{1-4t^2}}\,dt = -\frac{1}{2}(\cos^{-1}(2t))^2+C$$

38) $\int \frac{e^t\cos^{-1}(e^t)}{\sqrt{1-e^{2t}}}\,dt$

In exercises 39 - 42, compute each definite integral.

39) $\int_0^{1/2} \frac{\tan(\sin^{-1} t)}{\sqrt{1-t^2}}\,dt$

Answer

$$\int_0^{1/2} \frac{\tan(\sin^{-1} t)}{\sqrt{1-t^2}}\,dt = \frac{1}{2}\ln\left(\frac{4}{3}\right)$$

40) $\int_{1/4}^{1/2} \frac{\tan(\cos^{-1} t)}{\sqrt{1-t^2}}\,dt$

41) $\int_0^{1/2} \frac{\sin(\tan^{-1} t)}{1+t^2}\,dt$

Answer

$$\int_0^{1/2} \frac{\sin(\tan^{-1} t)}{1+t^2}\,dt = 1-\frac{2}{\sqrt{5}}$$

42) $\int_0^{1/2} \frac{\cos(\tan^{-1} t)}{1+t^2}\,dt$

43) For $A>0$, compute $I(A)=\int_{-A}^{A}\frac{dt}{1+t^2}$ and evaluate $\lim_{a\to\infty} I(A)$, the area under the graph of $\frac{1}{1+t^2}$ on $[-\infty,\infty]$.

Answer

$2\tan^{-1}(A)\to\pi$ as $A\to\infty$

44) For $1<B<\infty$, compute $I(B)=\int_1^B \frac{dt}{t\sqrt{t^2-1}}$ and evaluate $\lim_{B\to\infty} I(B)$, the area under the graph of $\frac{1}{t\sqrt{t^2-1}}$ over $[1,\infty)$.

45) Use the substitution $u=\sqrt{2}\cot x$ and the identity $1+\cot^2 x=\csc^2 x$ to evaluate $\int \frac{dx}{1+\cos^2 x}$. (Hint: Multiply the top and bottom of the integrand by $\csc^2 x$.)

Answer

Using the hint, one has $\int \frac{\csc^2 x}{\csc^2 x + \cot^2 x}\,dx = \int \frac{\csc^2 x}{1 + 2\cot^2 x}\,dx$. Set $u = \sqrt{2}\cot x$. Then, $du = -\sqrt{2}\csc^2 x$ and the integral is $-\frac{1}{\sqrt{2}}\int \frac{du}{1+u^2} = -\frac{\sqrt{2}}{2}\tan^{-1} u + C = \frac{\sqrt{2}}{2}\tan^{-1}(\sqrt{2}\cot x) + C$. If one uses the identity $\tan^{-1} s + \tan^{-1}(\frac{1}{s}) = \frac{\pi}{2}$, then this can also be written $\frac{\sqrt{2}}{2}\tan^{-1}(\frac{\tan x}{\sqrt{2}}) + C$.

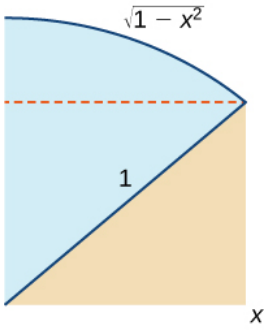
46) [T] Approximate the points at which the graphs of $f(x) = 2x^2 - 1$ and $g(x) = (1 + 4x^2)^{-3/2}$ intersect, and approximate the area between their graphs accurate to three decimal places.

47) [T] Approximate the points at which the graphs of $f(x) = x^2 - 1$ and $f(x) = x^2 - 1$ intersect, and approximate the area between their graphs accurate to three decimal places.

Answer

$x \approx \pm 1.13525$. The left endpoint estimate with $N = 100$ is 2.796 and these decimals persist for $N = 500$.

48) Use the following graph to prove that $\int_0^x \sqrt{1 - t^2}\,dt = \frac{1}{2}x\sqrt{1 - x^2} + \frac{1}{2}\sin^{-1} x$.



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5.7: Integrals Resulting in Inverse Trigonometric Functions

Learning Objectives

- Integrate functions resulting in inverse trigonometric functions

In this section we focus on integrals that result in inverse trigonometric functions. We have worked with these functions before. Recall, that trigonometric functions are not one-to-one unless the domains are restricted. When working with inverses of trigonometric functions, we always need to be careful to take these restrictions into account. Also, we previously developed formulas for derivatives of inverse trigonometric functions. The formulas developed there give rise directly to integration formulas involving inverse trigonometric functions.

Integrals that Result in Inverse Trigonometric Functions

Let us begin this last section of the chapter with the three formulas. Along with these formulas, we use substitution to evaluate the integrals. We prove the formula for the inverse sine integral.

Rule: Integration Formulas Resulting in Inverse Trigonometric Functions

The following integration formulas yield inverse trigonometric functions:

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C \tag{5.7.1}$$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C \tag{5.7.2}$$

$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{|u|}{a}\right) + C \tag{5.7.3}$$

Proof of the first formula

Let $y = \sin^{-1}\frac{x}{a}$. Then $a\sin y = x$. Now using implicit differentiation, we obtain

$$\frac{d}{dx}(a\sin y) = \frac{d}{dx}(x)$$

$$a\cos y\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{a\cos y}.$$

For $-\frac{\pi}{2} \le y \le \frac{\pi}{2}, \cos y \ge 0$. Thus, applying the Pythagorean identity $\sin^2 y + \cos^2 y = 1$, we have $\cos y = \sqrt{1-\sin^2 y}$. This gives

$$\frac{1}{a\cos y} = \frac{1}{a\sqrt{1-\sin^2 y}} \tag{5.7.4}$$

$$= \frac{1}{\sqrt{a^2-a^2\sin^2 y}} \tag{5.7.5}$$

$$= \frac{1}{\sqrt{a^2-x^2}}. \tag{5.7.6}$$

Then for $-a \le x \le a$, we have

$$\int \frac{1}{\sqrt{a^2-u^2}}\,du = \sin^{-1}\left(\frac{u}{a}\right) + C.$$

□

Example 5.7.1: Evaluating a Definite Integral Using Inverse Trigonometric Functions

Evaluate the definite integral

$$\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}}.$$

Solution

We can go directly to the formula for the antiderivative in the rule on integration formulas resulting in inverse trigonometric functions, and then evaluate the definite integral. We have

$$\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x \Big|_0^{1/2} = \sin^{-1} \tfrac{1}{2} - \sin^{-1} 0 = \frac{\pi}{6} - 0 = \frac{\pi}{6}.$$

Note that since the integrand is simply the derivative of $\sin^{-1} x$, we are really just using this fact to find the antiderivative here.

Exercise 5.7.1

Find the indefinite integral using an inverse trigonometric function and substitution for $\int \frac{dx}{\sqrt{9-x^2}}$.

Hint

Use the formula in the rule on integration formulas resulting in inverse trigonometric functions.

Answer

$$\int \frac{dx}{\sqrt{9-x^2}} = \sin^{-1}\left(\frac{x}{3}\right) + C$$

In many integrals that result in inverse trigonometric functions in the antiderivative, we may need to use substitution to see how to use the integration formulas provided above.

Example 5.7.2: Finding an Antiderivative Involving an Inverse Trigonometric Function using substitution

Evaluate the integral

$$\int \frac{dx}{\sqrt{4-9x^2}}.$$

Solution

Substitute $u = 3x$. Then $du = 3\,dx$ and we have

$$\int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3}\int \frac{du}{\sqrt{4-u^2}}.$$

Applying the formula with $a = 2$, we obtain

$$\int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3}\int \frac{du}{\sqrt{4-u^2}} = \frac{1}{3}\sin^{-1}\left(\frac{u}{2}\right) + C = \frac{1}{3}\sin^{-1}\left(\frac{3x}{2}\right) + C.$$

Exercise 5.7.2

Find the antiderivative of $\int \frac{dx}{\sqrt{1-16x^2}}$.

Hint

Substitute $u = 4x$.

Answer

$$\int \frac{dx}{\sqrt{1-16x^2}} = \frac{1}{4}\sin^{-1}(4x) + C$$

Example 5.7.3: Evaluating a Definite Integral

Evaluate the definite integral

$$\int_0^{\sqrt{3}/2} \frac{du}{\sqrt{1-u^2}}.$$

Solution

The format of the problem matches the inverse sine formula. Thus,

$$\int_0^{\sqrt{3}/2} \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u \Big|_0^{\sqrt{3}/2} = [\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)] - [\sin^{-1}(0)] = \frac{\pi}{3}.$$

Integrals Resulting in Other Inverse Trigonometric Functions

There are six inverse trigonometric functions. However, only three integration formulas are noted in the rule on integration formulas resulting in inverse trigonometric functions because the remaining three are negative versions of the ones we use. The only difference is whether the integrand is positive or negative. Rather than memorizing three more formulas, if the integrand is negative, simply factor out −1 and evaluate the integral using one of the formulas already provided. To close this section, we examine one more formula: the integral resulting in the inverse tangent function.

Example 5.7.4: Finding an Antiderivative Involving the Inverse Tangent Function

Find the antiderivative of $\int \frac{1}{9+x^2}\,dx$.

Solution

Apply the formula with $a = 3$. Then,

$$\int \frac{dx}{9+x^2} = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C.$$

Exercise 5.7.3

Find the antiderivative of $\int \frac{dx}{16+x^2}$.

Hint

Follow the steps in Example 5.7.4.

Answer

$$\int \frac{dx}{16+x^2} = \frac{1}{4}\tan^{-1}\left(\frac{x}{4}\right) + C$$

Example 5.7.5: Applying the Integration Formulas WITH SUBSTITUTION

Find an antiderivative of $\int \frac{1}{1+4x^2}\,dx$.

Solution

Comparing this problem with the formulas stated in the rule on integration formulas resulting in inverse trigonometric functions, the integrand looks similar to the formula for $\tan^{-1} u + C$. So we use substitution, letting $u = 2x$, then $du = 2\,dx$ and $\frac{1}{2}\,du = dx$.Then, we have

$$\frac{1}{2}\int \frac{1}{1+u^2}\,du = \frac{1}{2}\tan^{-1}u + C = \frac{1}{2}\tan^{-1}(2x) + C.$$

Exercise 5.7.4

Use substitution to find the antiderivative of $\int \frac{dx}{25+4x^2}$.

Hint

Use the solving strategy from Example 5.7.5 and the rule on integration formulas resulting in inverse trigonometric functions.

Answer

$$\int \frac{dx}{25+4x^2} = \frac{1}{10}\tan^{-1}\left(\frac{2x}{5}\right) + C$$

Example 5.7.6: Evaluating a Definite Integral

Evaluate the definite integral $\int_{\sqrt{3}/3}^{\sqrt{3}} \frac{dx}{1+x^2}$.

Solution

Use the formula for the inverse tangent. We have

$$\int_{\sqrt{3}/3}^{\sqrt{3}} \frac{dx}{1+x^2} = \tan^{-1}x\Big|_{\sqrt{3}/3}^{\sqrt{3}} = [\tan^{-1}(\sqrt{3})] - [\tan^{-1}\left(\frac{\sqrt{3}}{3}\right)] = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}.$$

Exercise 5.7.5

Evaluate the definite integral $\int_0^2 \frac{dx}{4+x^2}$.

Hint

Follow the procedures from Example 5.7.6 to solve the problem.

Answer

$$\int_0^2 \frac{dx}{4+x^2} = \frac{\pi}{8}$$

Key Concepts

- Formulas for derivatives of inverse trigonometric functions developed in Derivatives of Exponential and Logarithmic Functions lead directly to integration formulas involving inverse trigonometric functions.
- Use the formulas listed in the rule on integration formulas resulting in inverse trigonometric functions to match up the correct format and make alterations as necessary to solve the problem.
- Substitution is often required to put the integrand in the correct form.

Key Equations

- **Integrals That Produce Inverse Trigonometric Functions**

$$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$$
$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$$
$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{|u|}{a}\right) + C$$

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- Includes some added textual clarifications and edits by Paul Seeburger (Monroe Community College)

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5.R: CHAPTER 5 REVIEW EXERCISES

In exercises 1 - 4, answer *True or False.* Justify your answer with a proof or a counterexample. Assume all functions f and g are continuous over their domains.

1) If $f(x) > 0,\ f'(x) > 0$ for all x, then the right-hand rule underestimates the integral $\int_a^b f(x)\,dx$. Use a graph to justify your answer.

Answer

False

2) $\int_a^b f(x)^2\,dx = \int_a^b f(x)\,dx$

3) If $f(x) \le g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$.

Answer

True

4) All continuous functions have an antiderivative.

In exercises 5 - 8, evaluate the Riemann sums L_4 and R_4 for the given functions over the specified interval. Compare your answer with the exact answer, when possible, or use a calculator to determine the answer.

5) $y = 3x^2 - 2x + 1)$ over $[-1, 1]$

Answer

$L_4 = 5.25,\ R_4 = 3.25$, exact answer: 4

6) $y = \ln(x^2 + 1)$ over $[0, e]$

7) $y = x^2 \sin x$ over $[0, \pi]$

Answer

$L_4 = 5.364,\ R_4 = 5.364$, exact answer: 5.870

8) $y = \sqrt{x} + \frac{1}{x}$ over $[1, 4]$

In exercises 9 - 12, evaluate the integrals.

9) $\int_{-1}^{1} (x^3 - 2x^2 + 4x)\,dx$

Answer

$-\frac{4}{3}$

10) $\int_0^4 \frac{3t}{\sqrt{1 + 6t^2}}\,dt$

11) $\int_{\pi/3}^{\pi/2} 2\sec(2\theta)\tan(2\theta)\,d\theta$

Answer

1

12) $\int_0^{\pi/4} e^{\cos^2 x} \sin x \cos x\,dx$

In exercises 13 - 16, find the antiderivative.

13) $\int \frac{dx}{(x+4)^3}$

Answer

$-\frac{1}{2(x+4)^2} + C$

14) $\int x \ln(x^2)\, dx$

15) $\int \frac{4x^2}{\sqrt{1-x^6}}\, dx$

Answer

$\frac{4}{3}\sin^{-1}(x^3)+C$

16) $\int \frac{e^{2x}}{1+e^{4x}}\, dx$

In exercises 17 - 20, find the derivative.

17) $\frac{d}{dt}\int_0^t \frac{\sin x}{\sqrt{1+x^2}}\, dx$

Answer

$\frac{\sin t}{\sqrt{1+t^2}}$

18) $\frac{d}{dx}\int_1^{x^3} \sqrt{4-t^2}\, dt$

19) $\frac{d}{dx}\int_1^{\ln(x)} (4t+e^t)\, dt$

Answer

$4\frac{\ln x}{x}+1$

20) $\frac{d}{dx}\int_0^{\cos x} e^{t^2}\, dt$

In exercises 21 - 23, consider the historic average cost per gigabyte of RAM on a computer.

Year	5-Year Change ($)
1980	0
1985	−5,468,750
1990	−755,495
1995	−73,005
2000	−29,768
2005	−918
2010	−177

21) If the average cost per gigabyte of RAM in 2010 is $12, find the average cost per gigabyte of RAM in 1980.

Answer

$6,328,113

Solution: $6,328,113

22) The average cost per gigabyte of RAM can be approximated by the function $C(t) = 8,500,000(0.65)^t$, where t is measured in years since 1980, and C is cost in US dollars. Find the average cost per gigabyte of RAM for the period from 1980 to 2010.

23) Find the average cost of 1 GB RAM from 2005 to 2010.

Answer

$73.36

24) The velocity of a bullet from a rifle can be approximated by $v(t) = 6400t^2 - 6505t + 2686$, where t is seconds after the shot and v is the velocity measured in feet per second. This equation only models the velocity for the first half-second after the shot: $0 \le t \le 0.5$. What is the total distance the bullet travels in 0.5 sec?

25) What is the average velocity of the bullet for the first half-second?

Answer

$\frac{19117}{12}$ ft/sec, or about 1593 ft/sec

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CHAPTER OVERVIEW

6: Applications of Integration

In this chapter, we use definite integrals to calculate the force exerted on the dam when the reservoir is full and we examine how changing water levels affect that force. Hydrostatic force is only one of the many applications of definite integrals we explore in this chapter. From geometric applications such as surface area and volume, to physical applications such as mass and work, to growth and decay models, definite integrals are a powerful tool to help us understand and model the world around us.

6.0: Prelude to Applications of Integration
6.1: Areas between Curves
6.1E: Exercises for Section 6.1
6.2: Determining Volumes by Slicing
6.2E: Exercises for Section 6.2
6.3: Volumes of Revolution - Cylindrical Shells
6.3E: Exercises for Section 6.3
6.4: Arc Length of a Curve and Surface Area
6.4E: Exercises for Section 6.4
6.5: Physical Applications of Integration
6.5E: Exercises for Section 6.5
6.6: Moments and Centers of Mass
6.6E: Exercises for Section 6.6
6.7: Integrals, Exponential Functions, and Logarithms
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6.9E: Exercises for Section 6.9
6.R: Chapter 6 Review Exercises

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6.0: Prelude to Applications of Integration

The Hoover Dam is an engineering marvel. When Lake Mead, the reservoir behind the dam, is full, the dam withstands a great deal of force. However, water levels in the lake vary considerably as a result of droughts and varying water demands. Later in this chapter, we use definite integrals to calculate the force exerted on the dam when the reservoir is full and we examine how changing water levels affect that force. Hydrostatic force is only one of the many applications of definite integrals we explore in this chapter. From geometric applications such as surface area and volume, to physical applications such as mass and work, to growth and decay models, definite integrals are a powerful tool to help us understand and model the world around us.



Figure 6.0.1: Hoover Dam is one of the United States' iconic landmarks, and provides irrigation and hydroelectric power for millions of people in the southwest United States. (credit: modification of work by Lynn Betts, Wikimedia).

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6.1: Areas between Curves

Learning Objectives

- Determine the area of a region between two curves by integrating with respect to the independent variable.
- Find the area of a compound region.
- Determine the area of a region between two curves by integrating with respect to the dependent variable.

In Introduction to Integration, we developed the concept of the definite integral to calculate the area below a curve on a given interval. In this section, we expand that idea to calculate the area of more complex regions. We start by finding the area between two curves that are functions of x, beginning with the simple case in which one function value is always greater than the other. We then look at cases when the graphs of the functions cross. Last, we consider how to calculate the area between two curves that are functions of y.

Area of a Region between Two Curves

Let $f(x)$ and $g(x)$ be continuous functions over an interval $[a, b]$ such that $f(x) \geq g(x)$ on $[a, b]$. We want to find the area between the graphs of the functions, as shown in Figure 6.1.1.

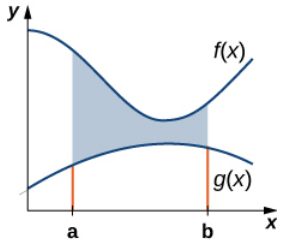


Figure 6.1.1: The area between the graphs of two functions, $f(x)$ and $g(x)$, on the interval $[a, b]$

As we did before, we are going to partition the interval on the **x-axis** and approximate the area between the graphs of the functions with rectangles. So, for $i = 0, 1, 2, \ldots, n$, let $P = x_i$ be a regular partition of $[a, b]$. Then, for $i = 1, 2, \ldots, n$, choose a point $x_i^* \in [x_{i-1}, x_i]$, and on each interval $[x_{i-1}, x_i]$ construct a rectangle that extends vertically from $g(x_i^*)$ to $f(x_i^*)$. Figure 6.1.2a shows the rectangles when x_i^* is selected to be the left endpoint of the interval and $n = 10$. Figure 6.1.2b shows a representative rectangle in detail.

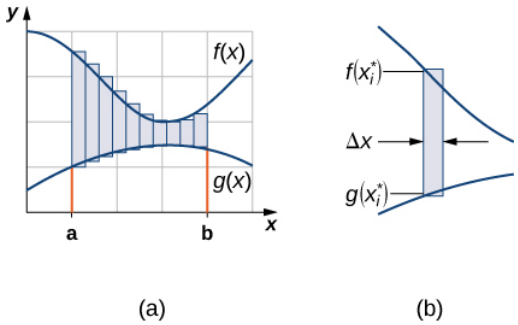


Figure 6.1.2: (a)We can approximate the area between the graphs of two functions, $f(x)$ and $g(x)$, with rectangles. (b) The area of a typical rectangle goes from one curve to the other.

The height of each individual rectangle is $f(x_i^*) - g(x_i^*)$ and the width of each rectangle is Δx. Adding the areas of all the rectangles, we see that the area between the curves is approximated by

$$A \approx \sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)] \Delta x.$$

This is a Riemann sum, so we take the limit as $n \to \infty$ and we get

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)]\Delta x = \int_a^b [f(x) - g(x)]dx.$$

These findings are summarized in the following theorem.

Finding the Area between Two Curves

Let $f(x)$ and $g(x)$ be continuous functions such that $f(x) \geq g(x)$ over an interval $[a, b]$. Let R denote the region bounded above by the graph of $f(x)$, below by the graph of $g(x)$, and on the left and right by the lines $x = a$ and $x = b$, respectively. Then, the area of $\mathbf{R}$ is given by

$$A = \int_a^b [f(x) - g(x)]dx.$$

We apply this theorem in the following example.

Example 6.1.1: Finding the Area of a Region between Two Curves I

If $\mathbf{R}$ is the region bounded above by the graph of the function $f(x) = x + 4$ and below by the graph of the function $g(x) = 3 - \frac{x}{2}$ over the interval $[1, 4]$, find the area of region $\mathbf{R}$.

Solution

The region is depicted in the following figure.

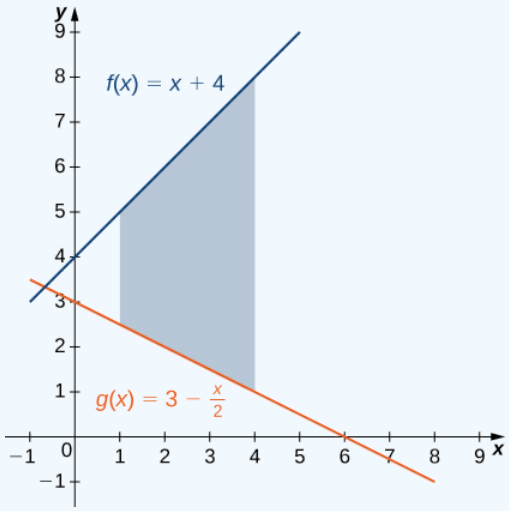


Figure 6.1.3: A region between two curves is shown where one curve is always greater than the other.

We have

$$A = \int_a^b [f(x) - g(x)]\, dx$$
$$= \int_1^4 [(x+4) - (3 - \frac{x}{2})]\, dx = \int_1^4 \left[\frac{3x}{2} + 1\right]\, dx$$
$$= [\frac{3x^2}{4} + x]\Big|_1^4 = (16 - \frac{7}{4}) = \frac{57}{4}.$$

The area of the region is $\frac{57}{4} units^2$.

Exercise 6.1.1

If $\mathbf{R}$ is the region bounded by the graphs of the functions $f(x) = \frac{x}{2} + 5$ and $g(x) = x + \frac{1}{2}$ over the interval $[1, 5]$, find the area of region $\mathbf{R}$.

Hint

Graph the functions to determine which function's graph forms the upper bound and which forms the lower bound, then follow the process used in Example.

Answer

12 units^2

In Example 6.1.1, we defined the interval of interest as part of the problem statement. Quite often, though, we want to define our interval of interest based on where the graphs of the two functions intersect. This is illustrated in the following example.

Example 6.1.2: Finding the Area of a Region between Two Curves II

If $\mathbf{R}$ is the region bounded above by the graph of the function $f(x) = 9 - (x/2)^2$ and below by the graph of the function $g(x) = 6 - x$, find the area of region $\mathbf{R}$.

Solution

The region is depicted in the following figure.

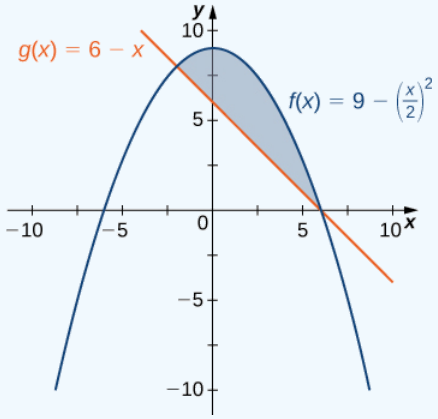


Figure 6.1.4: This graph shows the region below the graph of $f(x)$ and above the graph of $g(x)$.

We first need to compute where the graphs of the functions intersect. Setting $f(x) = g(x)$, we get

$$\begin{aligned} f(x) &= g(x) \\ 9 - (\frac{x}{2})^2 &= 6 - x \\ 9 - \frac{x^2}{4} &= 6 - x \\ 36 - x^2 &= 24 - 4x \\ x^2 - 4x - 12 &= 0 \\ (x - 6)(x + 2) &= 0. \end{aligned}$$

The graphs of the functions intersect when $x = 6$ or $x = -2$, so we want to integrate from -2 to 6. Since $f(x) \geq g(x)$ for $-2 \leq x \leq 6$, we obtain

$$A = \int_a^b [f(x) - g(x)]\,dx$$
$$= \int_{-2}^{6} \left[9 - (\frac{x}{2})^2 - (6 - x)\right]\,dx$$
$$= \int_{-2}^{6} \left[3 - \frac{x^2}{4} + x\right]\,dx$$
$$= \left[3x - \frac{x^3}{12} + \frac{x^2}{2}\right]\Big|_{-2}^{6} = \frac{64}{3}.$$

The area of the region is $64/3$ units^2.

? Exercise 6.1.2

If $\mathbf{R}$ is the region bounded above by the graph of the function $f(x) = x$ and below by the graph of the function $g(x) = x^4$, find the area of region $\mathbf{R}$.

Hint

Use the process from Example 6.1.2.

Answer

$\frac{3}{10}$ unit^2

Areas of Compound Regions

So far, we have required $f(x) \geq g(x)$ over the entire interval of interest, but what if we want to look at regions bounded by the graphs of functions that cross one another? In that case, we modify the process we just developed by using the absolute value function.

Finding the Area of a Region between Curves That Cross

Let $f(x)$ and $g(x)$ be continuous functions over an interval $[a, b]$. Let $\mathbf{R}$ denote the region between the graphs of $f(x)$ and $g(x)$, and be bounded on the left and right by the lines $x = a$ and $x = b$, respectively. Then, the area of $\mathbf{R}$ is given by

$$A = \int_a^b |f(x) - g(x)|dx.$$

In practice, applying this theorem requires us to break up the interval $[a, b]$ and evaluate several integrals, depending on which of the function values is greater over a given part of the interval. We study this process in the following example.

✓ Example 6.1.3: Finding the Area of a Region Bounded by Functions That Cross

If $\mathbf{R}$ is the region between the graphs of the functions $f(x) = \sin x$ and $g(x) = \cos x$ over the interval $[0, \pi]$, find the area of region $\mathbf{R}$.

Solution

The region is depicted in the following figure.

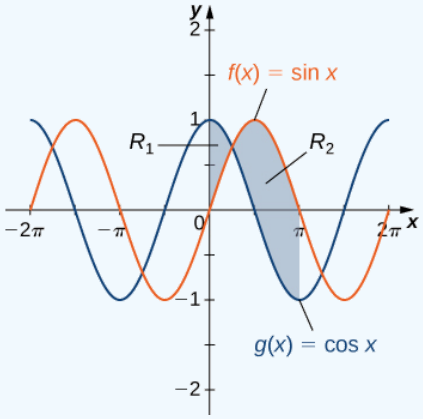


Figure 6.1.5: The region between two curves can be broken into two sub-regions.

The graphs of the functions intersect at $x = \pi/4$. For $x \in [0, \pi/4], \cos x \geq \sin x$, so

$$|f(x) - g(x)| = |\sin x - \cos x| = \cos x - \sin x.$$

On the other hand, for $x \in [\pi/4, \pi], \sin x \geq \cos x$, so

$$|f(x) - g(x)| = |\sin x - \cos x| = \sin x - \cos x.$$

Then

$$\begin{aligned} A &= \int_a^b |f(x) - g(x)|dx \\ &= \int_0^\pi |\sin x - \cos x|dx = \int_0^{\pi/4} (\cos x - \sin x)dx + \int_{\pi/4}^\pi (\sin x - \cos x)dx \\ &= [\sin x + \cos x]|_0^{\pi/4} + [-\cos x - \sin x]|_{\pi/4}^\pi \\ &= (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2}. \end{aligned}$$

The area of the region is $2\sqrt{2}$ units^2.

Exercise 6.1.3

If $\mathbf{R}$ is the region between the graphs of the functions $f(x) = \sin x$ and $g(x) = \cos x$ over the interval $[\pi/2, 2\pi]$, find the area of region $\mathbf{R}$.

Hint

The two curves intersect at $x = (5\pi)/4$.

Answer

$2 + 2\sqrt{2}$ units^2

Example 6.1.4: Finding the Area of a Complex Region

Consider the region depicted in Figure 6.1.6. Find the area of $\mathbf{R}$.

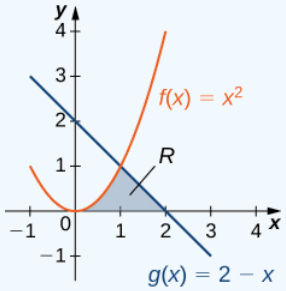


Figure 6.1.6: Two integrals are required to calculate the area of this region.

Solution

As with Example 6.1.3, we need to divide the interval into two pieces. The graphs of the functions intersect at $x = 1$ (set $f(x) = g(x)$ and solve for x), so we evaluate two separate integrals: one over the interval $[0, 1]$ and one over the interval $[1, 2]$.

Over the interval $[0, 1]$, the region is bounded above by $f(x) = x^2$ and below by the x-axis, so we have

$$A_1 = \int_0^1 x^2 dx = \frac{x^3}{3} \,|_0^1 = \frac{1}{3}.$$

Over the interval $[1, 2]$, the region is bounded above by $g(x) = 2 - x$ and below by the **x-axis**, so we have

$$A_2 = \int_1^2 (2 - x)dx = [2x - \frac{x^2}{2}] \,|_1^2 = \frac{1}{2}.$$

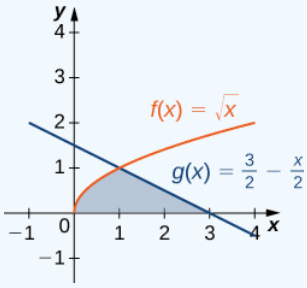
Adding these areas together, we obtain

$$A = A_1 + A_2 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

The area of the region is $5/6$ units^2.

? Exercise 6.1.4

Consider the region depicted in the following figure. Find the area of **R**.



Hint

The two curves intersect at **x=1**

Answer

$\frac{5}{3}$ units^2

Regions Defined with Respect to y

In Example 6.1.4, we had to evaluate two separate integrals to calculate the area of the region. However, there is another approach that requires only one integral. What if we treat the curves as functions of y, instead of as functions of x? Review Figure. Note that the left graph, shown in red, is represented by the function $y = f(x) = x^2$. We could just as easily solve this for x and represent the curve by the function $x = v(y) = \sqrt{y}$. (Note that $x = -\sqrt{y}$ is also a valid representation of the function $y = f(x) = x^2$ as a function of y. However, based on the graph, it is clear we are interested in the positive square root.) Similarly, the right graph is represented by

the function $y = g(x) = 2 - x$, but could just as easily be represented by the function $x = u(y) = 2 - y$. When the graphs are represented as functions of y, we see the region is bounded on the left by the graph of one function and on the right by the graph of the other function. Therefore, if we integrate with respect to y, we need to evaluate one integral only. Let's develop a formula for this type of integration.

Let $u(y)$ and $v(y)$ be continuous functions over an interval $[c, d]$ such that $u(y) \geq v(y)$ for all $y \in [c, d]$. We want to find the area between the graphs of the functions, as shown in Figure 6.1.7.

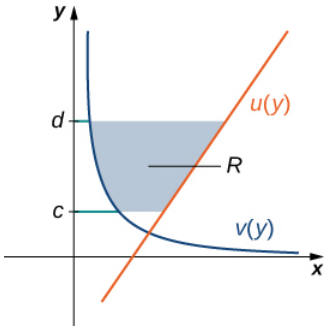


Figure 6.1.7: We can find the area between the graphs of two functions, $u(y)$ and $v(y)$.

This time, we are going to partition the interval on the **y-axis** and use horizontal rectangles to approximate the area between the functions. So, for $i = 0, 1, 2, \ldots, n$, let $Q = y_i$ be a regular partition of $[c, d]$. Then, for $i = 1, 2, \ldots, n$, choose a point $y_i^* \in [y_{i-1}, y_i]$, then over each interval $[y_{i-1}, y_i]$ construct a rectangle that extends horizontally from $v(y^0 *_i)$ to $u(y_i^*)$. Figure 6.1.8$a$shows the rectangles when y_i^* is selected to be the lower endpoint of the interval and $n = 10$. Figure 6.1.8b shows a representative rectangle in detail.

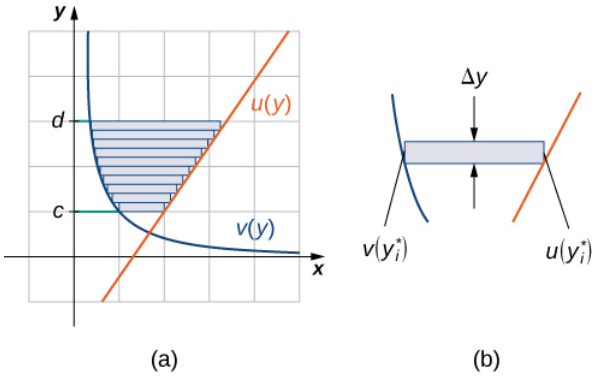


Figure 6.1.8: (a) Approximating the area between the graphs of two functions, $u(y)$ and $v(y)$, with rectangles. (b) The area of a typical rectangle.

The height of each individual rectangle is Δy and the width of each rectangle is $u(y_i^*) - v(y_i^*)$. Therefore, the area between the curves is approximately

$$A \approx \sum_{i=1}^{n} [u(y_i^*) - v(y_i^*)] \Delta y.$$

This is a Riemann sum, so we take the limit as $n \to \infty$, obtaining

$$\begin{aligned} A &= \lim_{n \to \infty} \sum_{i=1}^{n} [u(y_i^*) - v(y_i^*)] \Delta y \\ &= \int_c^d [u(y) - v(y)] dy. \end{aligned}$$

These findings are summarized in the following theorem.

Finding the Area between Two Curves, Integrating along the y-axis

Let $u(y)$ and $v(y)$ be continuous functions such that $u(y) \geq v(y)$for all $y \in [c, d]$. Let $\mathbf{R}$ denote the region bounded on the right by the graph of $u(y)$, on the left by the graph of $v(y)$, and above and below by the lines $y = d$ and $y = c$, respectively. Then, the area of $\mathbf{R}$ is given by

$$A = \int_c^d [u(y) - v(y)]dy.$$

Example 6.1.5: Integrating with Respect to y

Let's revisit Example 6.1.4, only this time let's integrate with respect to y. Let $\mathbf{R}$ be the region depicted in Figure 6.1.9. Find the area of $\mathbf{R}$ by integrating with respect to y.

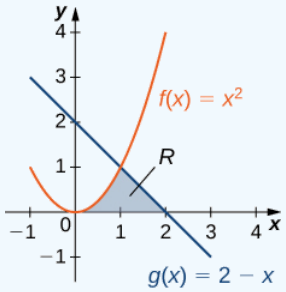


Figure 6.1.9: The area of region $\mathbf{R}$ can be calculated using one integral only when the curves are treated as functions of y.

Solution

We must first express the graphs as functions of y. As we saw at the beginning of this section, the curve on the left can be represented by the function $x = v(y) = \sqrt{y}$, and the curve on the right can be represented by the function $x = u(y) = 2 - y$.

Now we have to determine the limits of integration. The region is bounded below by the x-axis, so the lower limit of integration is $y = 0$. The upper limit of integration is determined by the point where the two graphs intersect, which is the point $(1, 1)$, so the upper limit of integration is $y = 1$. Thus, we have $[c, d] = [0, 1]$.

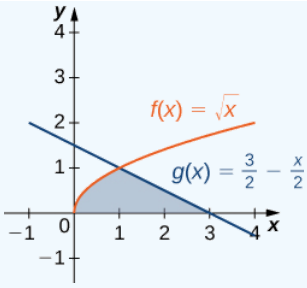
Calculating the area of the region, we get

$$\begin{aligned} A &= \int_c^d [u(y) - v(y)]dy \\ &= \int_0^1 [(2 - y) - \sqrt{y}]dy \\ &= [2y - \frac{y^2}{2} - \frac{2}{3}y^{3/2}] \,|_0^1 \\ &= \frac{5}{6}. \end{aligned}$$

The area of the region is $5/6$ units^2.

Exercise 6.1.5

Let's revisit the checkpoint associated with Example 6.1.4, only this time, let's integrate with respect to y. Let $\mathbf{R}$ be the region depicted in the following figure. Find the area of $\mathbf{R}$ by integrating with respect to y.



Hint

Follow the process from the previous example.

Answer

$\frac{5}{3}$ units^2

Key Concepts

- Just as definite integrals can be used to find the area under a curve, they can also be used to find the area between two curves.
- To find the area between two curves defined by functions, integrate the difference of the functions.
- If the graphs of the functions cross, or if the region is complex, use the absolute value of the difference of the functions. In this case, it may be necessary to evaluate two or more integrals and add the results to find the area of the region.
- Sometimes it can be easier to integrate with respect to y to find the area. The principles are the same regardless of which variable is used as the variable of integration.

Key Equations

- **Area between two curves, integrating on the x-axis**

$$A = \int_a^b [f(x) - g(x)]dx$$

- **Area between two curves, integrating on the y-axis**

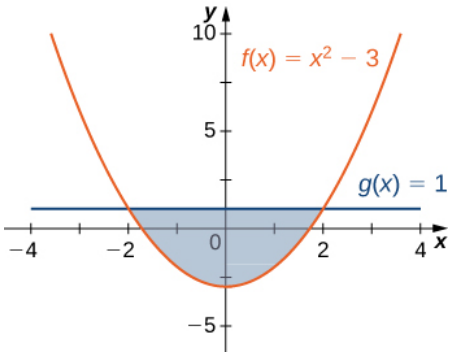
$$A = \int_c^d [u(y) - v(y)]dy$$

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6.1E: EXERCISES FOR SECTION 6.1

For exercises 1 - 2, determine the area of the region between the two curves in the given figure by integrating over the x-axis.

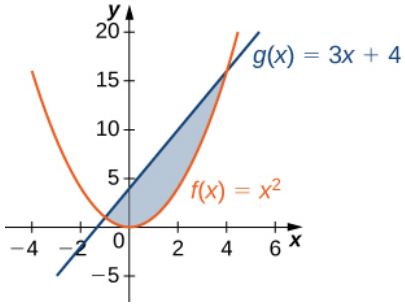
1) $y = x^2 - 3$ and $y = 1$



Answer

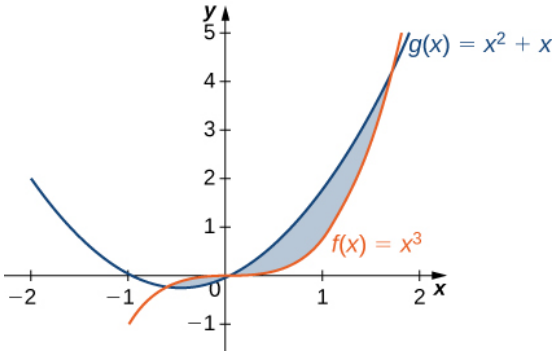
$\frac{32}{3}$ units^2

2) $y = x^2$ and $y = 3x + 4$



For exercises 3 - 4, split the region between the two curves into two smaller regions, then determine the area by integrating over the x-axis. Note that you will have two integrals to solve.

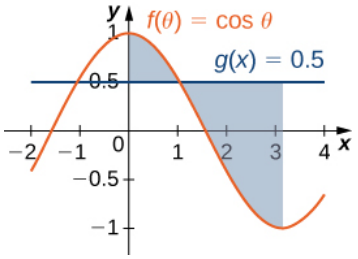
3) $y = x^3$ and $y = x^2 + x$



Answer

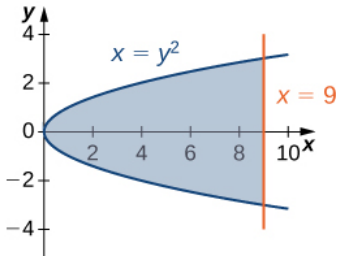
$\frac{13}{12}$ units^2

4) $y = \cos\theta$ and $y = 0.5$, for $0 \le \theta \le \pi$



For exercises 5-6, determine the area of the region between the two curves by integrating over the y-axis.

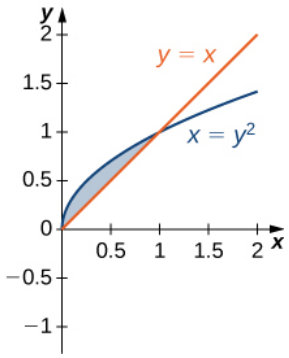
5) $x = y^2$ and $x = 9$



Answer

36 units^2

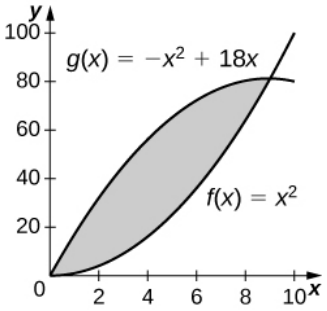
6) $y = x$ and $x = y^2$



For exercises 7 - 13, graph the equations and shade the area of the region between the curves. Determine its area by integrating over the x-axis.

7) $y = x^2$ and $y = -x^2 + 18x$

Answer

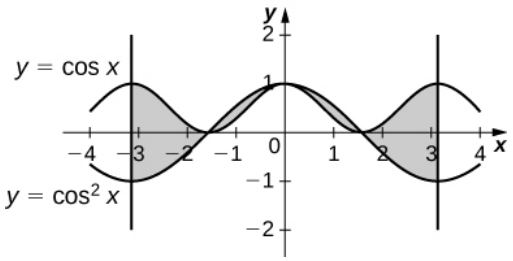


243 square units

8) $y = \frac{1}{x}$, $y = \frac{1}{x^2}$, and $x = 3$

9) $y = \cos x$ and $y = \cos^2 x$ on $x \in [-\pi, \pi]$

Answer

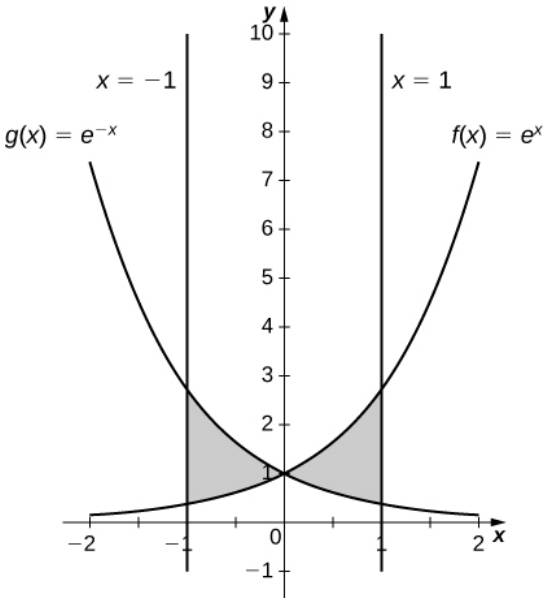


4 square units

10) $y = e^x$, $y = e^{2x-1}$, and $x = 0$

11) $y = e^x$, $y = e^{-x}$, $x = -1$ and $x = 1$

Answer

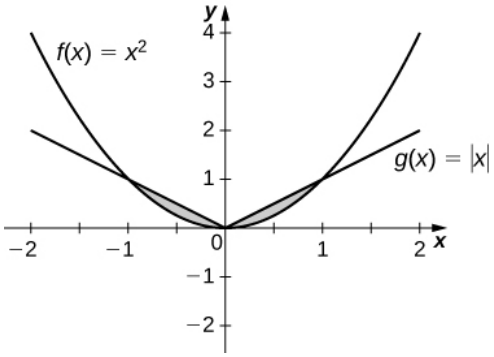


$\frac{2(e-1)^2}{e}$ units^2

12) $y = e$, $y = e^x$, and $y = e^{-x}$

13) $y = |x|$ and $y = x^2$

Answer



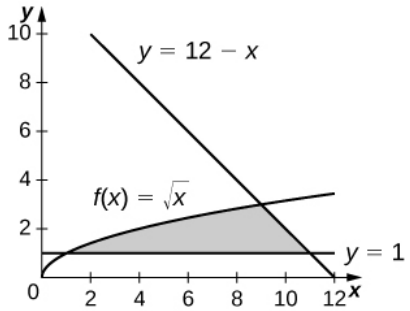
$\frac{1}{3}$ units^2

For exercises 14 - 19, graph the equations and shade the area of the region between the curves. If necessary, break the region into sub-regions to determine its entire area.

14) $y = \sin(\pi x)$, $y = 2x$, and $x > 0$

15) $y = 12 - x$, $y = \sqrt{x}$, and $y = 1$

Answer

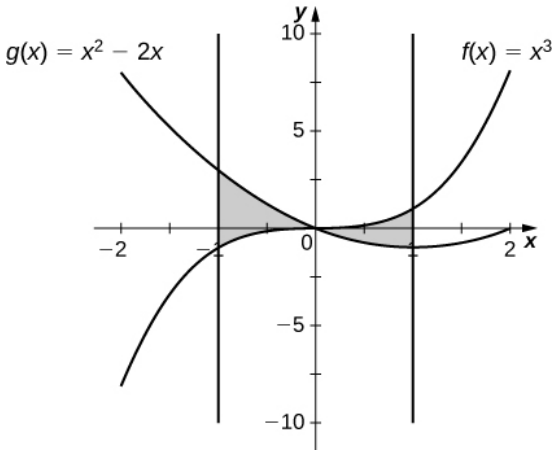


$\frac{34}{3}$ units^2

16) $y = \sin x$ and $y = \cos x$ over $x \in [-\pi, \pi]$

17) $y = x^3$ and $y = x^2 - 2x$ over $x \in [-1, 1]$

Answer



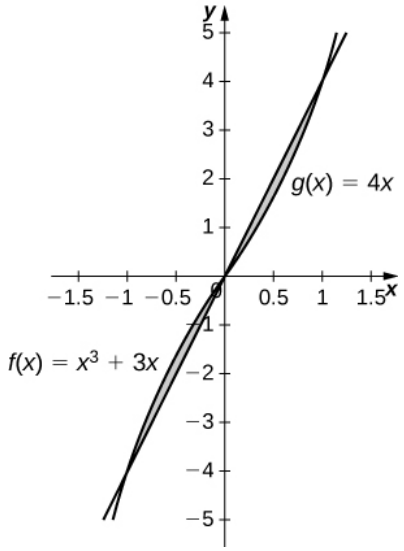
$\frac{5}{2}$ units^2

18) $y = x^2 + 9$ and $y = 10 + 2x$ over $x \in [-1, 3]$

19) $y = x^3 + 3x$ and $y = 4x$

Answer

https://math.libretexts.org/@go/page/69808



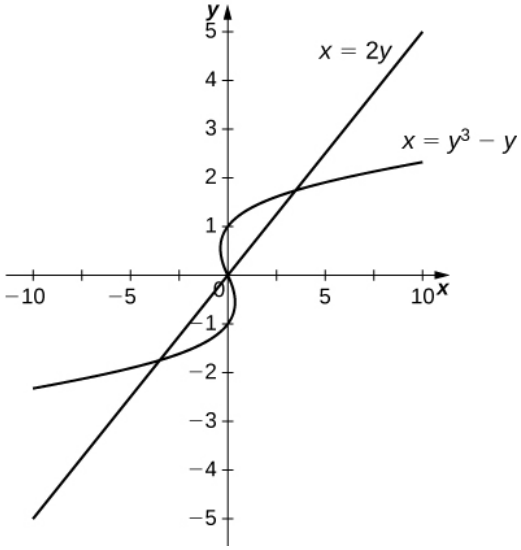
$\frac{1}{2}$ units^2

For exercises 20 -25, graph the equations and shade the area of the region between the curves. Determine its area by integrating over the y-axis.

20) $x = y^3$ and $x = 3y - 2$

21) $x = 2y$ and $x = y^3 - y$

Answer



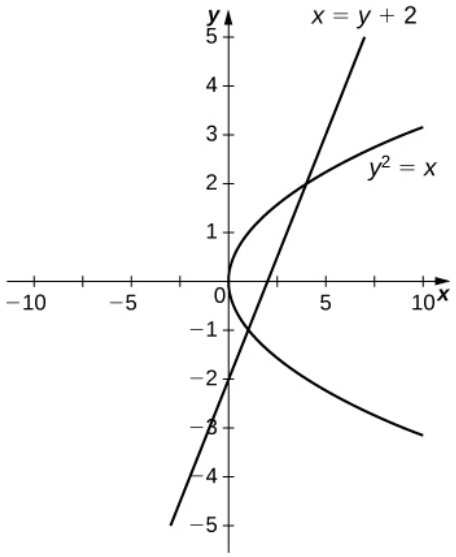
$\frac{9}{2}$ units^2

22) $x = -3 + y^2$ and $x = y - y^2$

23) $y^2 = x$ and $x = y + 2$

Answer

https://math.libretexts.org/@go/page/69808

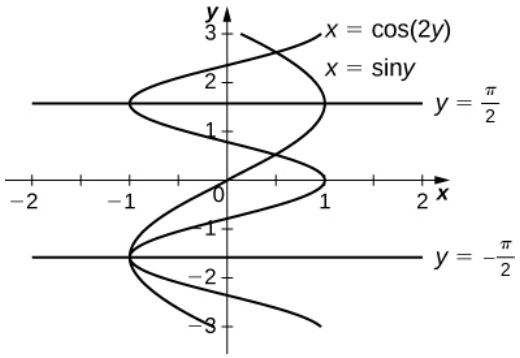


$\frac{9}{2}$ units^2

24) $x = |y|$ and $2x = -y^2 + 2$

25) $x = \sin y, \quad x = \cos(2y), \quad y = \pi/2$, and $y = -\pi/2$

Answer



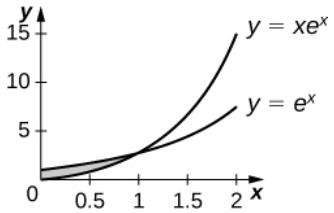
$\frac{3\sqrt{3}}{2}$ units^2

For exercises 26 - 37, graph the equations and shade the area of the region between the curves. Determine its area by integrating over the x-axis or y-axis, whichever seems more convenient.

26) $x = y^4$ and $x = y^5$

27) $y = xe^x, \quad y = e^x, \quad x = 0$, and $x = 1$.

Answer

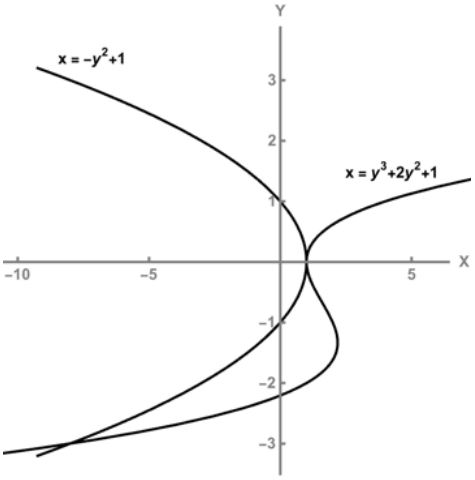


$(e - 2)$ units^2

28) $y = x^6$ and $y = x^4$

29) $x = y^3 + 2y^2 + 1$ and $x = -y^2 + 1$

Answer

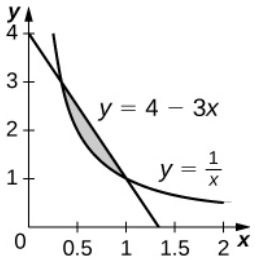


$\frac{27}{4}$ units^2

30) $y = |x|$ and $y = x^2 - 1$

31) $y = 4 - 3x$ and $y = \frac{1}{x}$

Answer

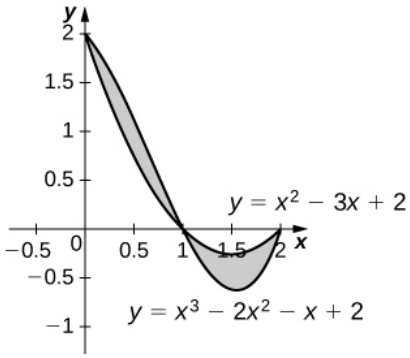


$\left(\frac{4}{3} - \ln(3)\right)$ units^2

32) $y = \sin x, \quad x = -\pi/6, \quad x = \pi/6$, and $y = \cos^3 x$

33) $y = x^2 - 3x + 2$ and $y = x^3 - 2x^2 - x + 2$

Answer

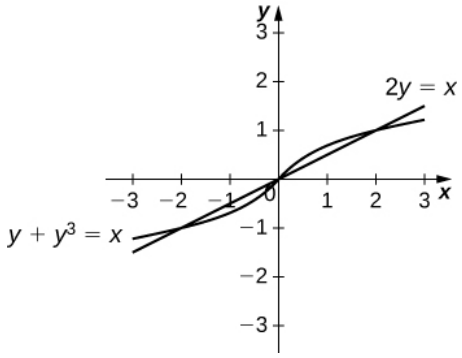


$\frac{1}{2}$ square units

34) $y = 2\cos^3(3x), \quad y = -1, \quad x = \frac{\pi}{4}$, and $x = -\frac{\pi}{4}$

35) $y + y^3 = x$ and $2y = x$

Answer

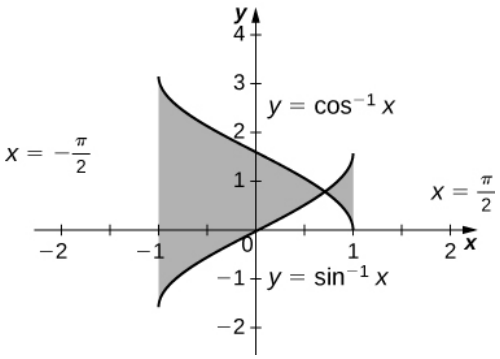


$\frac{1}{2}$ square units

36) $y = \sqrt{1 - x^2}$ and $y = x^2 - 1$

37) $y = \cos^{-1} x, \quad y = \sin^{-1} x, \quad x = -1,$ and $x = 1$

Answer



$-2(\sqrt{2} - \pi)$ square units

For exercises 38 - 47, find the exact area of the region bounded by the given equations if possible. If you are unable to determine the intersection points analytically, use a calculator to approximate the intersection points with three decimal places and determine the approximate area of the region.

38) [T] $x = e^y$ and $y = x - 2$

39) [T] $y = x^2$ and $y = \sqrt{1 - x^2}$

Answer

1.067 square units

40) [T] $y = 3x^2 + 8x + 9$ and $3y = x + 24$

41) [T] $x = \sqrt{4 - y^2}$ and $y^2 = 1 + x^2$

Answer

0.852 square units

42) [T] $x^2 = y^3$ and $x = 3y$

43) [T] $y = \sin^3 x + 2, \quad y = \tan x, \quad x = -1.5,$ and $x = 1.5$

Answer

7.523 square units

44) [T] $y = \sqrt{1 - x^2}$ and $y^2 = x^2$

45) [T] $y = \sqrt{1 - x^2}$ and $y = x^2 + 2x + 1$

Answer

$\frac{3\pi - 4}{12}$ square units

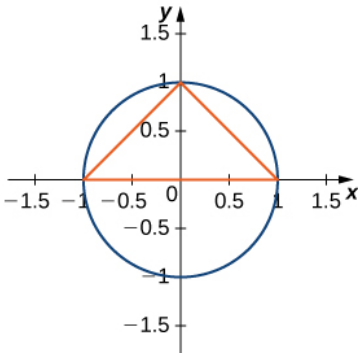
46) [T] $x = 4 - y^2$ and $x = 1 + 3y + y^2$

47) [T] $y = \cos x, \quad y = e^x, \quad x = -\pi, \quad$ and $\quad x = 0$

Answer

1.429 square units

48) The largest triangle with a base on the x-axis that fits inside the upper half of the unit circle $y^2 + x^2 = 1$ is given by $y = 1 + x$ and $y = 1 - x$. See the following figure. What is the area inside the semicircle but outside the triangle?



49) A factory selling cell phones has a marginal cost function $C(x) = 0.01x^2 - 3x + 229$, where x represents the number of cell phones, and a marginal revenue function given by $R(x) = 429 - 2x$. Find the area between the graphs of these curves and $x = 0$. What does this area represent?

Answer

$\$33{,}333.33$ total profit for 200 cell phones sold

50) An amusement park has a marginal cost function $C(x) = 1000e - x + 5$, where x represents the number of tickets sold, and a marginal revenue function given by $R(x) = 60 - 0.1x$. Find the total profit generated when selling 550 tickets. Use a calculator to determine intersection points, if necessary, to two decimal places.

51) The tortoise versus the hare: The speed of the hare is given by the sinusoidal function $H(t) = 1 - \cos((\pi t)/2)$ whereas the speed of the tortoise is $T(t) = (1/2)\tan^{-1}(t/4)$, where t is time measured in hours and the speed is measured in miles per hour. Find the area between the curves from time $t = 0$ to the first time after one hour when the tortoise and hare are traveling at the same speed. What does it represent? Use a calculator to determine the intersection points, if necessary, accurate to three decimal places.

Answer

3.263 mi represents how far ahead the hare is from the tortoise

52) The tortoise versus the hare: The speed of the hare is given by the sinusoidal function $H(t) = (1/2) - (1/2)\cos(2\pi t)$ whereas the speed of the tortoise is $T(t) = \sqrt{t}$, where t is time measured in hours and speed is measured in kilometers per hour. If the race is over in 1 hour, who won the race and by how much? Use a calculator to determine the intersection points, if necessary, accurate to three decimal places.

For exercises 53 - 55, find the area between the curves by integrating with respect to x and then with respect to y. Is one method easier than the other? Do you obtain the same answer?

53) $y = x^2 + 2x + 1$ and $y = -x^2 - 3x + 4$

Answer

$\frac{343}{24}$ square units

54) $y = x^4$ and $x = y^5$

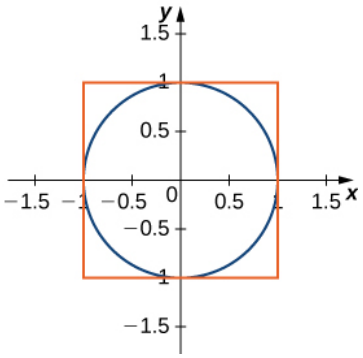
55) $x = y^2 - 2$ and $x = 2y$

Answer

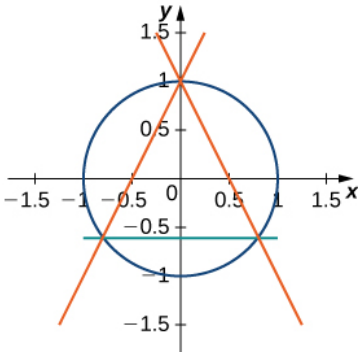
$4\sqrt{3}$ square units

For exercises 56 - 57, solve using calculus, then check your answer with geometry.

56) Determine the equations for the sides of the square that touches the unit circle on all four sides, as seen in the following figure. Find the area between the perimeter of this square and the unit circle. Is there another way to solve this without using calculus?



57) Find the area between the perimeter of the unit circle and the triangle created from $y = 2x + 1$, $y = 1 - 2x$ and $y = -\frac{3}{5}$, as seen in the following figure. Is there a way to solve this without using calculus?



Answer

$\left(\pi - \frac{32}{25}\right)$ square units

CONTRIBUTORS

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6.2: Determining Volumes by Slicing

Learning Objectives

- Determine the volume of a solid by integrating a cross-section (the slicing method).
- Find the volume of a solid of revolution using the disk method.
- Find the volume of a solid of revolution with a cavity using the washer method.

In the preceding section, we used definite integrals to find the area between two curves. In this section, we use definite integrals to find volumes of three-dimensional solids. We consider three approaches—slicing, disks, and washers—for finding these volumes, depending on the characteristics of the solid.

Volume and the Slicing Method

Just as area is the numerical measure of a two-dimensional region, volume is the numerical measure of a three-dimensional solid. Most of us have computed volumes of solids by using basic geometric formulas. The volume of a rectangular solid, for example, can be computed by multiplying length, width, and height: $V = lwh.$ The formulas for the volumes of:

- a sphere

$$V_{sphere} = \frac{4}{3}\pi r^3,$$

- a cone

$$V_{cone} = \frac{1}{3}\pi r^2 h$$

- and a pyramid

$$V_{pyramid} = \frac{1}{3}Ah$$

have also been introduced. Although some of these formulas were derived using geometry alone, all these formulas can be obtained by using integration.

We can also calculate the volume of a cylinder. Although most of us think of a cylinder as having a circular base, such as a soup can or a metal rod, in mathematics the word cylinder has a more general meaning. To discuss cylinders in this more general context, we first need to define some vocabulary.

We define the *cross-section* of a solid to be the intersection of a plane with the solid. A cylinder is defined as any solid that can be generated by translating a plane region along a line perpendicular to the region, called the axis of the cylinder. Thus, all cross-sections perpendicular to the axis of a cylinder are identical. The solid shown in Figure $6.2.1$ is an example of a cylinder with a noncircular base. To calculate the volume of a cylinder, then, we simply multiply the area of the cross-section by the height of the cylinder: $V = A \cdot h.$ In the case of a right circular cylinder (soup can), this becomes $V = \pi r^2 h.$

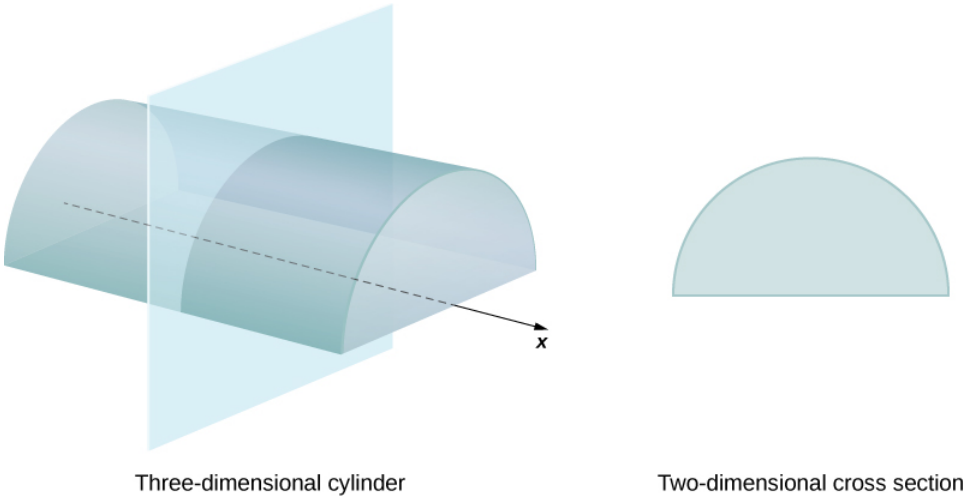


Figure 6.2.1: *Each cross-section of a particular cylinder is identical to the others.*

If a solid does not have a constant cross-section (and it is not one of the other basic solids), we may not have a formula for its volume. In this case, we can use a definite integral to calculate the volume of the solid. We do this by slicing the solid into pieces, estimating the volume of each slice, and then adding those estimated volumes together. The slices should all be parallel to one another, and when we put all the slices together, we should get the whole solid. Consider, for example, the solid S shown in Figure 6.2.2, extending along the x**-axis.**

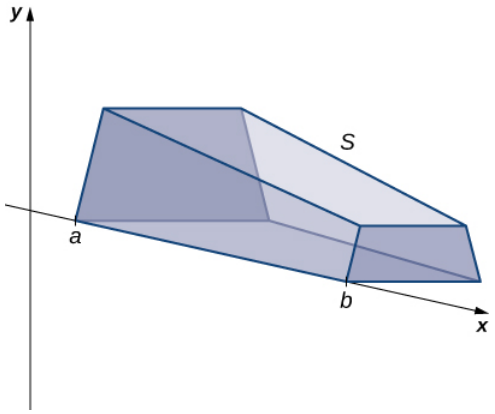


Figure 6.2.2: A solid with a varying cross-section.

We want to divide S into slices perpendicular to the x**-axis**. As we see later in the chapter, there may be times when we want to slice the solid in some other direction—say, with slices perpendicular to the y-axis. The decision of which way to slice the solid is very important. If we make the wrong choice, the computations can get quite messy. Later in the chapter, we examine some of these situations in detail and look at how to decide which way to slice the solid. For the purposes of this section, however, we use slices perpendicular to the x**-axis.**

Because the cross-sectional area is not constant, we let $A(x)$ represent the area of the cross-section at point x. Now let $P = x_0, x_1 \ldots, X_n$ be a regular partition of $[a, b]$, and for $i = 1, 2, \ldots n$, let S_i represent the slice of S stretching from x_{i-1} to x_i. The following figure shows the sliced solid with $n = 3$.

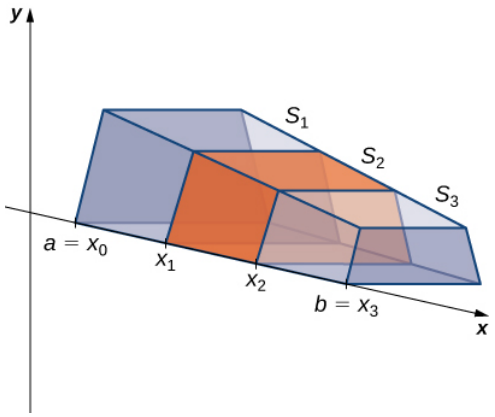


Figure 6.2.3: The solid S has been divided into three slices perpendicular to the x-**axis.**

Finally, for $i = 1, 2, \ldots n$, let x_i^* be an arbitrary point in $[x_{i-1}, x_i]$. Then the volume of slice S_i can be estimated by $V(S_i) \approx A(x_i^*)\,\Delta x$. Adding these approximations together, we see the volume of the entire solid S can be approximated by

$$V(S) \approx \sum_{i=1}^{n} A(x_i^*)\,\Delta x.$$

By now, we can recognize this as a Riemann sum, and our next step is to take the limit as $n \to \infty$. Then we have

$$V(S) = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx.$$

The technique we have just described is called the slicing method. To apply it, we use the following strategy.

Problem-Solving Strategy: Finding Volumes by the Slicing Method

1. Examine the solid and determine the shape of a cross-section of the solid. It is often helpful to draw a picture if one is not provided.
2. Determine a formula for the area of the cross-section.
3. Integrate the area formula over the appropriate interval to get the volume.

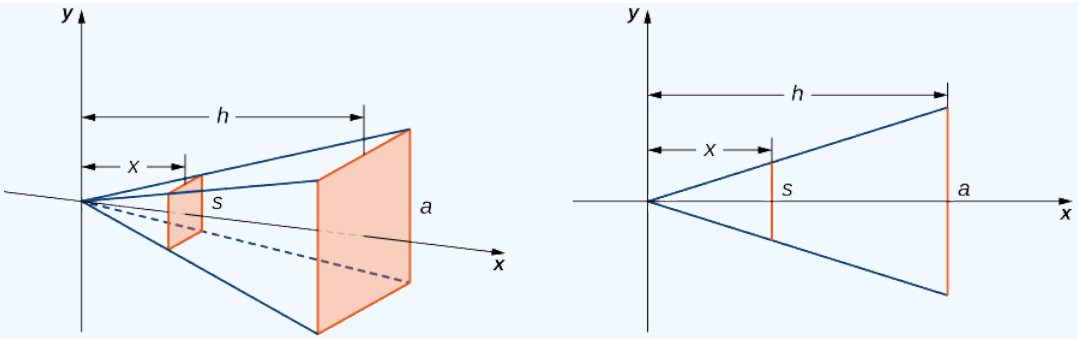
Recall that in this section, we assume the slices are perpendicular to the x-**axis**. Therefore, the area formula is in terms of x and the limits of integration lie on the x-**axis.** However, the problem-solving strategy shown here is valid regardless of how we choose to slice the solid.

Example 6.2.1: Deriving the Formula for the Volume of a Pyramid

We know from geometry that the formula for the volume of a pyramid is $V = \frac{1}{3}Ah$. If the pyramid has a square base, this becomes $V = \frac{1}{3}a^2h$, where a denotes the length of one side of the base. We are going to use the slicing method to derive this formula.

Solution

We want to apply the slicing method to a pyramid with a square base. To set up the integral, consider the pyramid shown in Figure 6.2.4, oriented along the x-**axis.**



(a) (b)

Figure 6.2.4: (a) A pyramid with a square base is oriented along the x-axis. (b) A two-dimensional view of the pyramid is seen from the side.

We first want to determine the shape of a cross-section of the pyramid. We know the base is a square, so the cross-sections are squares as well (step 1). Now we want to determine a formula for the area of one of these cross-sectional squares. Looking at Figure 6.2.4 (b), and using a proportion, since these are similar triangles, we have

$$\frac{s}{a} = \frac{x}{h}$$

or

$$s = \frac{ax}{h}.$$

Therefore, the area of one of the cross-sectional squares is

$$A(x) = s^2 = \left(\frac{ax}{h}\right)^2 \qquad \text{(step 2)}$$

Then we find the volume of the pyramid by integrating from 0 to h (step 3):

$$V = \int_0^h A(x)\,dx = \int_0^h \left(\frac{ax}{h}\right)^2 dx = \frac{a^2}{h^2}\int_0^h x^2\,dx = \left[\frac{a^2}{h^2}\left(\frac{1}{3}x^3\right)\right]\Bigg|_0^h = \frac{1}{3}a^2h.$$

This is the formula we were looking for.

? Exercise 6.2.1

Use the slicing method to derive the formula

$$V = \frac{1}{3}\pi r^2 h$$

for the volume of a circular cone.

Hint

Use similar triangles, as in Example 6.2.1.

Solids of Revolution

If a region in a plane is revolved around a line in that plane, the resulting solid is called a **solid of revolution**, as shown in the following figure.

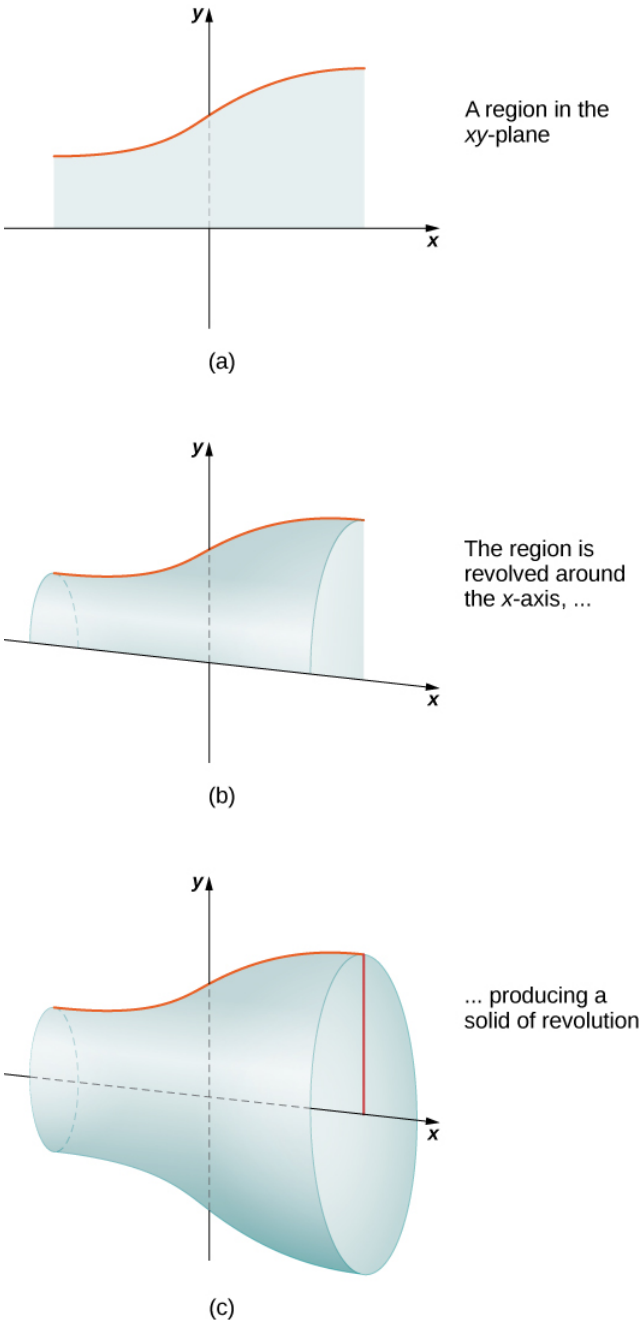


Figure 6.2.5: (a) This is the region that is revolved around the x-axis. (b) As the region begins to revolve around the axis, it sweeps out a solid of revolution. (c) This is the solid that results when the revolution is complete.

Solids of revolution are common in mechanical applications, such as machine parts produced by a lathe. We spend the rest of this section looking at solids of this type. The next example uses the slicing method to calculate the volume of a solid of revolution.

Example 6.2.2: Using the Slicing Method to find the Volume of a Solid of Revolution

Use the slicing method to find the volume of the solid of revolution bounded by the graphs of $f(x) = x^2 - 4x + 5, x = 1$,and $x = 4$, and rotated about the x-axis.

Solution

Using the problem-solving strategy, we first sketch the graph of the quadratic function over the interval $[1, 4]$ as shown in the following figure.

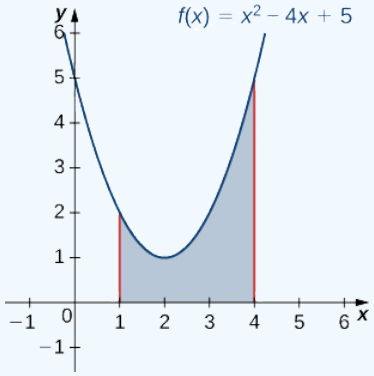
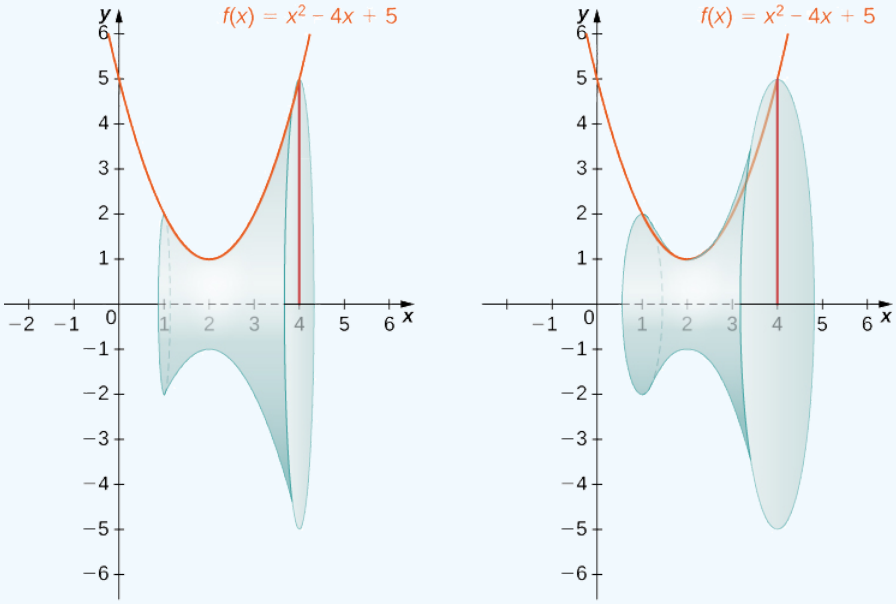


Figure 6.2.6: A region used to produce a solid of revolution.

Next, revolve the region around the x-axis, as shown in the following figure.



(a) (b)

Figure 6.2.7: Two views, (a) and (b), of the solid of revolution produced by revolving the region in Figure 6.2.6 about the x-**axis.**

Since the solid was formed by revolving the region around the x-**axis,** the cross-sections are circles (step 1). The area of the cross-section, then, is the area of a circle, and the radius of the circle is given by $f(x)$. Use the formula for the area of the circle:

$$A(x) = \pi r^2 = \pi[f(x)]^2 = \pi(x^2 - 4x + 5)^2 \qquad \text{(step 2)}.$$

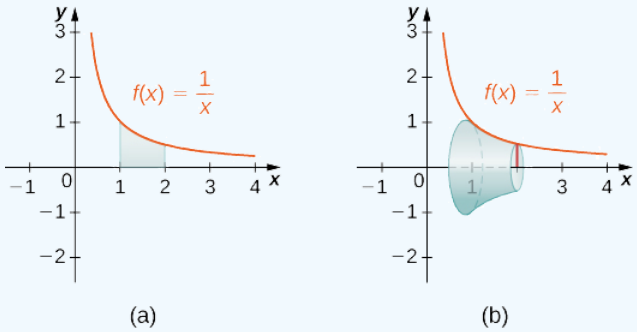
The volume, then, is (step 3)

$$\begin{aligned} V &= \int_a^b A(x)\,dx \\ &= \int_1^4 \pi(x^2 - 4x + 5)^2\,dx \\ &= \pi \int_1^4 (x^4 - 8x^3 + 26x^2 - 40x + 25)\,dx \\ &= \pi \left(\frac{x^5}{5} - 2x^4 + \frac{26x^3}{3} - 20x^2 + 25x \right) \Big|_1^4 \\ &= \frac{78}{5}\pi \end{aligned}$$

The volume is $78\pi/5$ units^3.

Exercise 6.2.2

Use the method of slicing to find the volume of the solid of revolution formed by revolving the region between the graph of the function $f(x) = 1/x$ and the x-**axis** over the interval $[1, 2]$ around the x-**axis.** See the following figure.



Hint

Use the problem-solving strategy presented earlier and follow Example 6.2.2 to help with step 2.

Answer

$\frac{\pi}{2}$ units^3

The Disk Method

When we use the slicing method with solids of revolution, it is often called the disk method because, for solids of revolution, the slices used to over approximate the volume of the solid are disks. To see this, consider the solid of revolution generated by revolving the region between the graph of the function $f(x) = (x - 1)^2 + 1$ and the x-**axis** over the interval $[-1, 3]$ around the x-**axis**. The graph of the function and a representative disk are shown in Figure 6.2.8 (a) and (b). The region of revolution and the resulting solid are shown in Figure 6.2.8 (c) and (d).

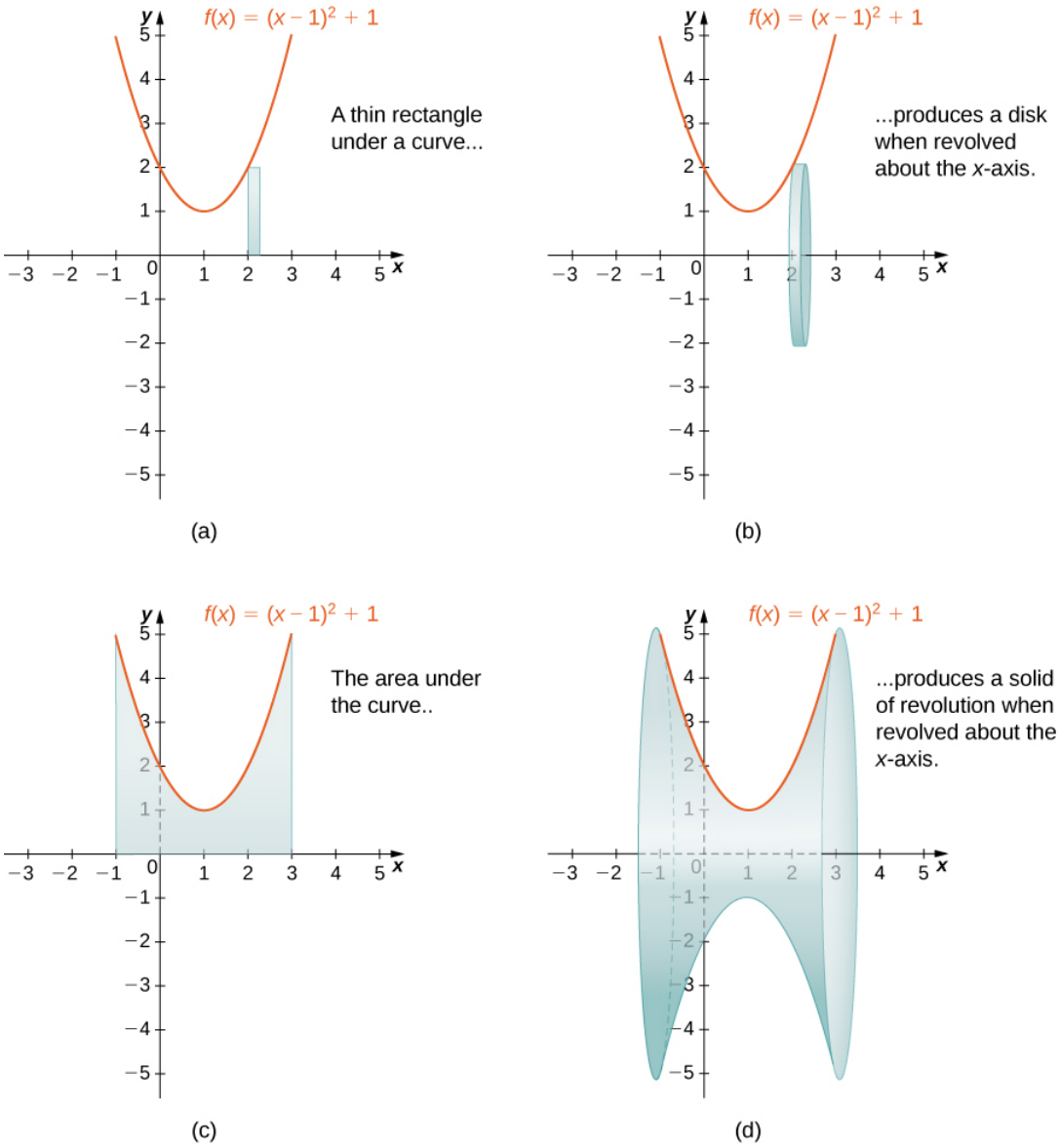


Figure 6.2.8: (a) A thin rectangle for approximating the area under a curve. (b) A representative disk formed by revolving the rectangle about the x**-axis.** (c) The region under the curve is revolved about the x**-axis,** resulting in (d) the solid of revolution.

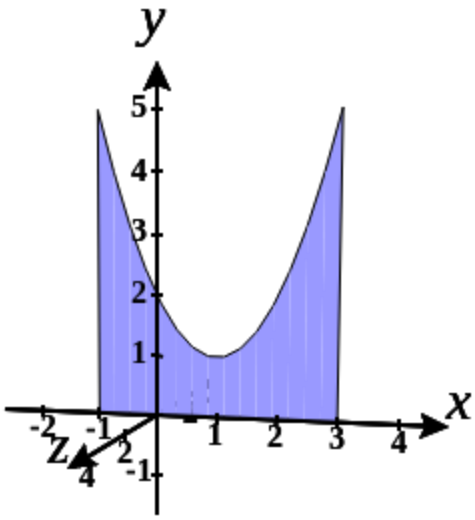


Figure 6.2.8: (e) A dynamic version of this solid of revolution generated using CalcPlot3D.

We already used the formal Riemann sum development of the volume formula when we developed the slicing method. We know that

$$\int_a^b A(x)\,dx.$$

The only difference with the disk method is that we know the formula for the cross-sectional area ahead of time; it is the area of a circle. This gives the following rule.

The Disk Method

Let $f(x)$ be continuous and nonnegative. Define R as the region bounded above by the graph of $f(x)$, below by the x-**axis**, on the left by the line $x = a$, and on the right by the line $x = b$. Then, the volume of the solid of revolution formed by revolving R around the x-**axis** is given by

$$V = \int_a^b \pi[f(x)]^2\,dx.$$

The volume of the solid we have been studying (Figure 6.2.8) is given by

$$
\begin{aligned}
V &= \int_a^b \pi [f(x)]^2 \, dx \\
&= \int_{-1}^3 \pi \left[(x-1)^2+1\right]^2 dx = \pi \int_{-1}^3 \left[(x-1)^4 + 2(x-1)^2 + 1\right]^2 dx \\
&= \pi \left[\frac{1}{5}(x-1)^5 + \frac{2}{3}(x-1)^3 + x\right]\Big|_{-1}^3 \\
&= \pi \left[\left(\frac{32}{5} + \frac{16}{3} + 3\right) - \left(-\frac{32}{5} - \frac{16}{3} - 1\right)\right] \\
&= \frac{412\pi}{15} \text{ units}^3.
\end{aligned}
$$

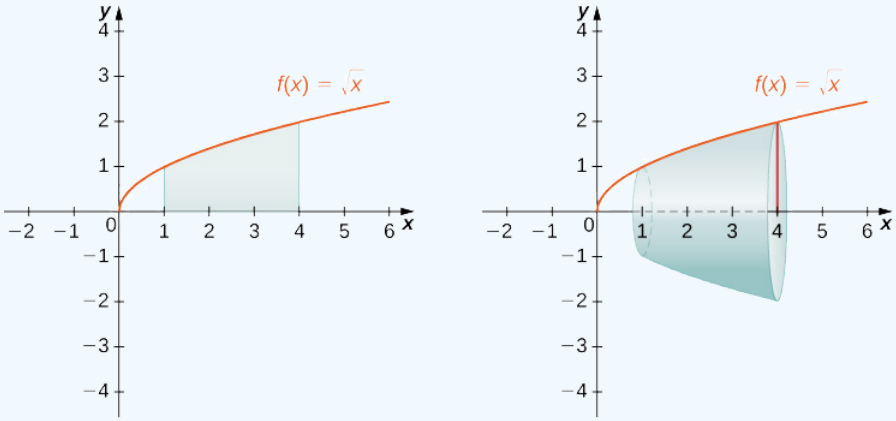
Let's look at some examples.

Example 6.2.3: Using the Disk Method to Find the Volume of a Solid of Revolution 1

Use the disk method to find the volume of the solid of revolution generated by rotating the region between the graph of $f(x) = \sqrt{x}$ and the x-**axis** over the interval $[1, 4]$ around the x-**axis.**

Solution

The graphs of the function and the solid of revolution are shown in the following figure.



(a) (b)

Figure 6.2.9: (a) The function $f(x) = \sqrt{x}$ over the interval $[1, 4]$. (b) The solid of revolution obtained by revolving the region under the graph of $f(x)$ about the x-**axis**.

We have

$$
\begin{aligned}
V &= \int_a^b \pi \left[f(x)\right]^2 dx \\
&= \int_1^4 \pi \left[\sqrt{x}\right]^2 dx = \pi \int_1^4 x \, dx \\
&= \frac{\pi}{2} x^2 \Big|_1^4 = \frac{15\pi}{2}
\end{aligned}
$$

The volume is $(15\pi)/2 \text{ units}^3$.

Exercise 6.2.3

Use the disk method to find the volume of the solid of revolution generated by rotating the region between the graph of $f(x) = \sqrt{4 - x}$ and the x-**axis** over the interval $[0, 4]$ around the x-**axis.**

Hint

Use the procedure from Example 6.2.3.

Answer

8π units^3

So far, our examples have all concerned regions revolved around the x-**axis**, but we can generate a solid of revolution by revolving a plane region around any horizontal or vertical line. In the next example, we look at a solid of revolution that has been generated by revolving a region around the y-**axis**. The mechanics of the disk method are nearly the same as when the x-**axis** is the axis of revolution, but we express the function in terms of y and we integrate with respect to y as well. This is summarized in the following rule.

Rule: The Disk Method for Solids of Revolution around the y-axis

Let $g(y)$ be continuous and nonnegative. Define Q as the region bounded on the right by the graph of $g(y)$, on the left by the y-**axis**, below by the line $y = c$, and above by the line $y = d$. Then, the volume of the solid of revolution formed by revolving Q around the y-**axis** is given by

$$V = \int_c^d \pi \left[g(y)\right]^2 dy.$$

The next example shows how this rule works in practice.

Example 6.2.4: Using the Disk Method to Find the Volume of a Solid of Revolution 2

Let R be the region bounded by the graph of $g(y) = \sqrt{4 - y}$ and the y-**axis** over the y-**axis** interval $[0, 4]$. Use the disk method to find the volume of the solid of revolution generated by rotating R around the y-**axis.**

Solution

Figure 6.2.10 shows the function and a representative disk that can be used to estimate the volume. Notice that since we are revolving the function around the y-**axis,** the disks are horizontal, rather than vertical.

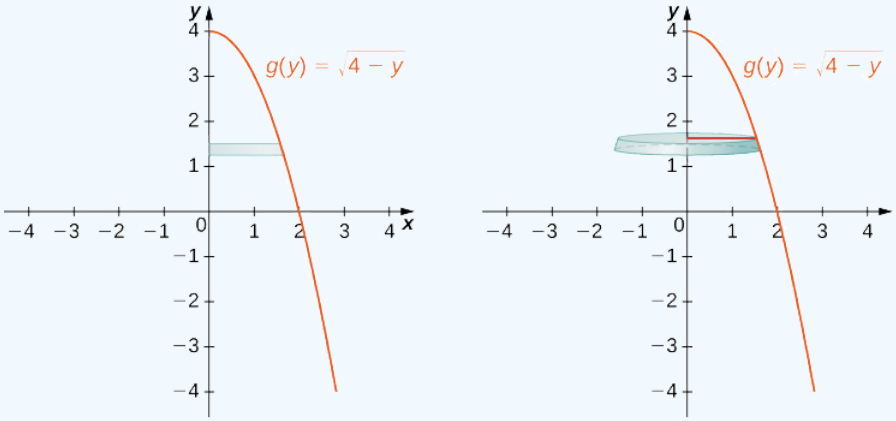


Figure 6.2.10: (a) Shown is a thin rectangle between the curve of the function $g(y) = \sqrt{4 - y}$ and the y-**axis.** (b) The rectangle forms a representative disk after revolution around the y-**axis.**

The region to be revolved and the full solid of revolution are depicted in the following figure.

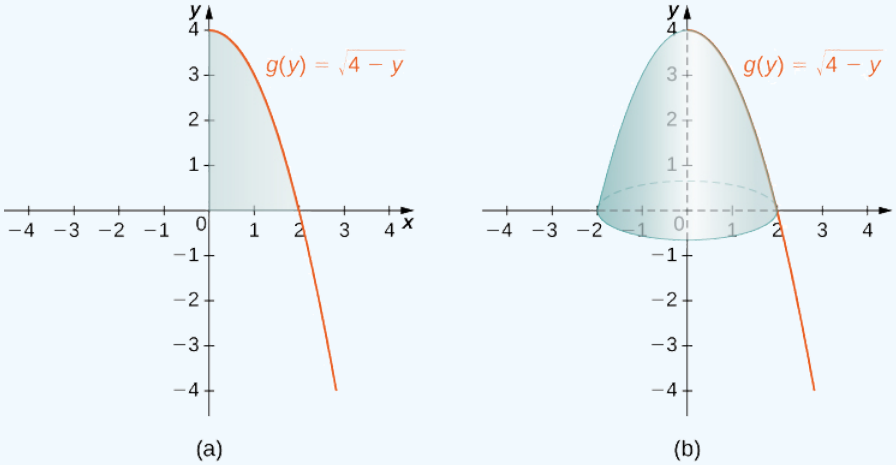


Figure 6.2.11: (a) The region to the left of the function $g(y) = \sqrt{4-y}$ over the y-axis interval $[0, 4]$. (b) The solid of revolution formed by revolving the region about the y-axis.

To find the volume, we integrate with respect to y. We obtain

$$V = \int_c^d \pi [g(y)]^2 \, dy = \int_0^4 \pi \left[\sqrt{4-y}\right]^2 \, dy = \pi \int_0^4 (4-y) \, dy = \pi \left[4y - \frac{y^2}{2}\right]\Big|_0^4 = 8\pi.$$

The volume is 8π units^3.

? Exercise 6.2.4

Use the disk method to find the volume of the solid of revolution generated by rotating the region between the graph of $g(y) = y$ and the y-**axis** over the interval $[1, 4]$ around the y-**axis.**

Hint

Use the procedure from Example 6.2.4.

Answer

21π units^3

The Washer Method

Some solids of revolution have cavities in the middle; they are not solid all the way to the axis of revolution. Sometimes, this is just a result of the way the region of revolution is shaped with respect to the axis of revolution. In other cases, cavities arise when the region of revolution is defined as the region between the graphs of two functions. A third way this can happen is when an axis of revolution other than the x-**axis** or y-**axis** is selected.

When the solid of revolution has a cavity in the middle, the slices used to approximate the volume are not disks, but washers (disks with holes in the center). For example, consider the region bounded above by the graph of the function $f(x) = \sqrt{x}$ and below by the graph of the function $g(x) = 1$ over the interval $[1, 4]$. When this region is revolved around the x-**axis**, the result is a solid with a cavity in the middle, and the slices are washers. The graph of the function and a representative washer are shown in Figure 6.2.12 (a) and (b). The region of revolution and the resulting solid are shown in Figure 6.2.12 (c) and (d).

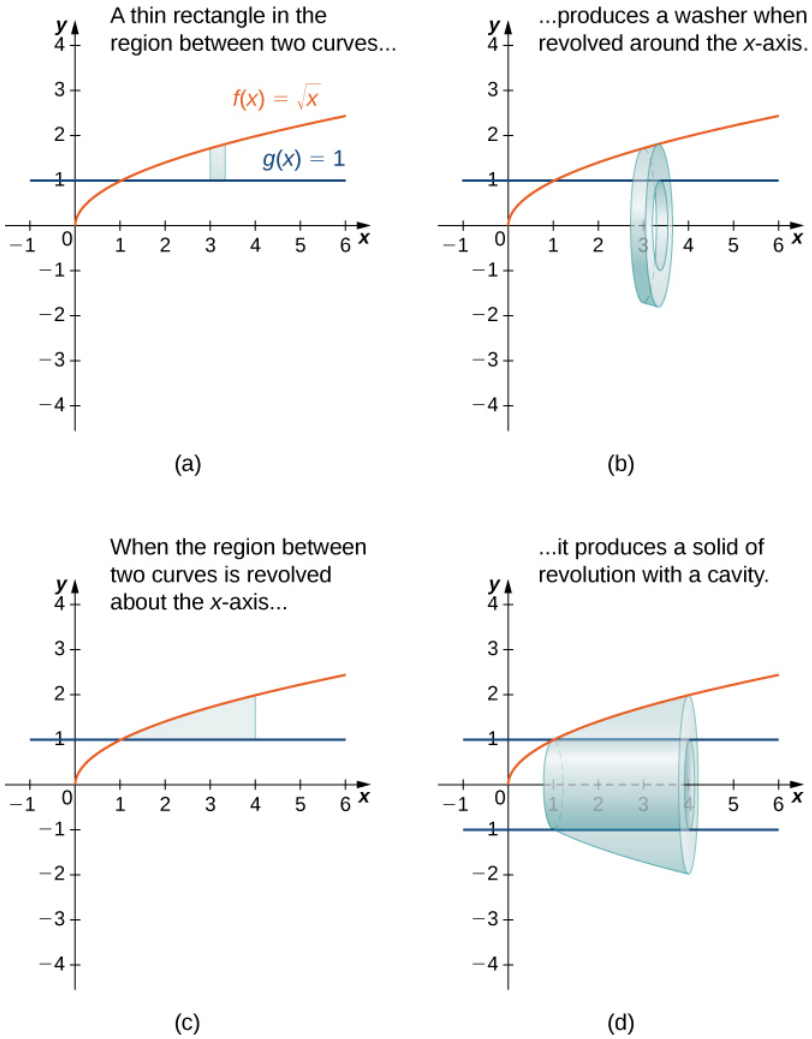


Figure 6.2.12: (a) A thin rectangle in the region between two curves. (b) A representative disk formed by revolving the rectangle about the x**-axis.** (c) The region between the curves over the given interval. (d) The resulting solid of revolution.

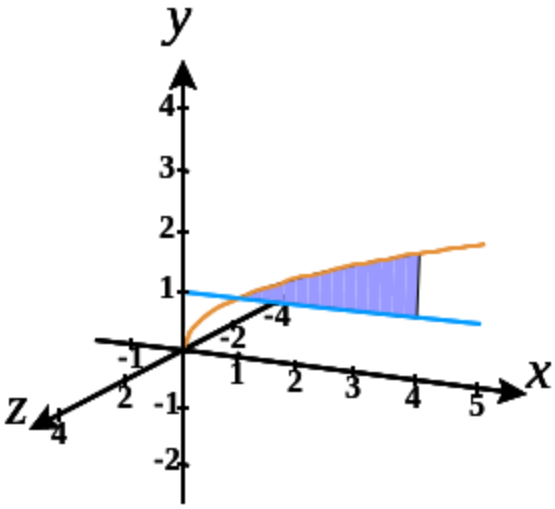


Figure 6.2.12: (e) A dynamic version of this solid of revolution generated using CalcPlot3D.

The cross-sectional area, then, is the area of the outer circle less the area of the inner circle. In this case,

$$A(x) = \pi\left(\sqrt{x}\right)^2 - \pi(1)^2 = \pi(x-1).$$

Then the volume of the solid is

$$V = \int_a^b A(x)\,dx = \int_1^4 \pi(x-1)\,dx = \pi\left[\frac{x^2}{2} - x\right]\Bigg|_1^4 = \frac{9}{2}\pi \text{ units}^3.$$

Generalizing this process gives the **washer method.**

Rule: The Washer Method

Suppose $f(x)$ and $g(x)$ are continuous, nonnegative functions such that $f(x) \geq g(x)$ over $[a, b]$. Let R denote the region bounded above by the graph of $f(x)$, below by the graph of $g(x)$, on the left by the line $x = a$, and on the right by the line $x = b$. Then, the volume of the solid of revolution formed by revolving R around the x-**axis** is given by

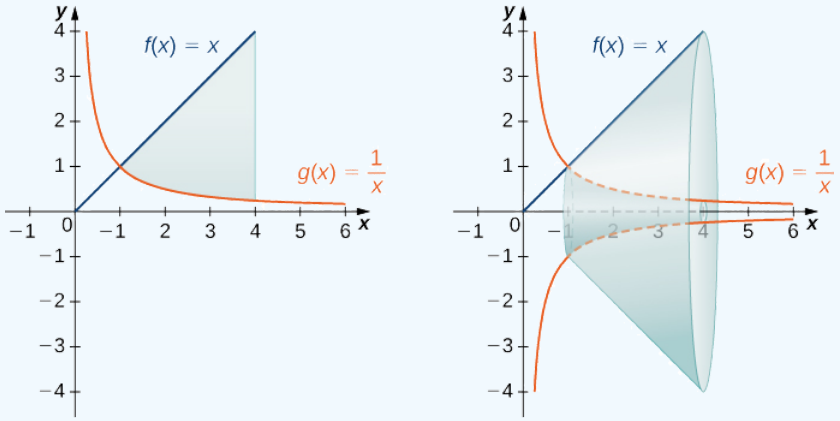
$$V = \int_a^b \pi\left[(f(x))^2 - (g(x))^2\right]\,dx.$$

Example 6.2.5: Using the Washer Method

Find the volume of a solid of revolution formed by revolving the region bounded above by the graph of $f(x) = x$ and below by the graph of $g(x) = 1/x$ over the interval $[1, 4]$ around the x-**axis.**

Solution

The graphs of the functions and the solid of revolution are shown in the following figure.



(a) (b)

Figure 6.2.13: (a) The region between the graphs of the functions $f(x) = x$ and $g(x) = 1/x$ over the interval $[1, 4]$. (b) Revolving the region about the x-**axis** generates a solid of revolution with a cavity in the middle.

We have

$$V = \int_a^b \pi\left[(f(x))^2 - (g(x))^2\right] dx = \pi \int_1^4 \left[x^2 - \left(\frac{1}{x}\right)^2\right] dx$$
$$= \pi \left[\frac{x^3}{3} + \frac{1}{x}\right]\Big|_1^4$$
$$= \frac{81\pi}{4} \text{ units}^3.$$

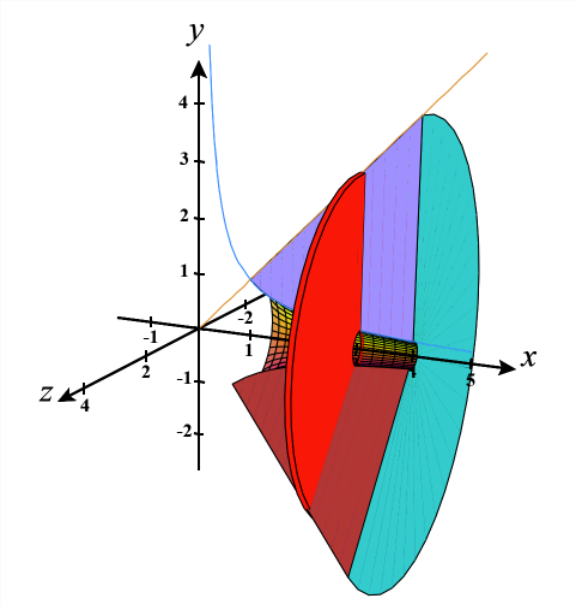


Figure 6.2.13: (c) A dynamic version of this solid of revolution generated using CalcPlot3D.

Exercise 6.2.5

Find the volume of a solid of revolution formed by revolving the region bounded by the graphs of $f(x) = \sqrt{x}$ and $g(x) = 1/x$ over the interval $[1, 3]$ around the x**-axis.**

Hint

Graph the functions to determine which graph forms the upper bound and which graph forms the lower bound, then use the procedure from Example 6.2.5.

Answer

$\frac{10\pi}{3}$ units^3

As with the disk method, we can also apply the washer method to solids of revolution that result from revolving a region around the y-axis. In this case, the following rule applies.

Rule: The Washer Method for Solids of Revolution around the y-axis

Suppose $u(y)$ and $v(y)$ are continuous, nonnegative functions such that $v(y) \leq u(y)$ for $y \in [c, d]$. Let Q denote the region bounded on the right by the graph of $u(y)$, on the left by the graph of $v(y)$, below by the line $y = c$, and above by the line $y = d$. Then, the volume of the solid of revolution formed by revolving Q around the y**-axis** is given by

$$V = \int_c^d \pi \left[(u(y))^2 - (v(y))^2 \right] \, dy.$$

Rather than looking at an example of the washer method with the y**-axis** as the axis of revolution, we now consider an example in which the axis of revolution is a line other than one of the two coordinate axes. The same general method applies, but you may have to visualize just how to describe the cross-sectional area of the volume.

Example 6.2.6:

Find the volume of a solid of revolution formed by revolving the region bounded above by $f(x) = 4 - x$ and below by the x**-axis** over the interval $[0, 4]$ around the line $y = -2$.

Solution

The graph of the region and the solid of revolution are shown in the following figure.

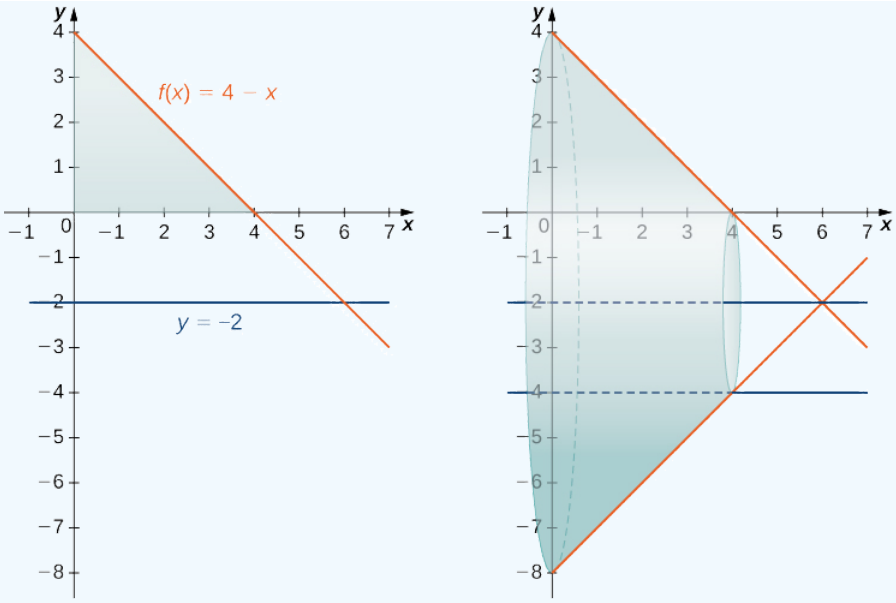


Figure 6.2.14: (a) The region between the graph of the function $f(x) = 4 - x$ and the x-**axis** over the interval $[0, 4]$. (b) Revolving the region about the line $y = -2$ generates a solid of revolution with a cylindrical hole through its middle.

We can't apply the volume formula to this problem directly because the axis of revolution is not one of the coordinate axes. However, we still know that the area of the cross-section is the area of the outer circle less the area of the inner circle. Looking at the graph of the function, we see the radius of the outer circle is given by $f(x) + 2$, which simplifies to

$$f(x) + 2 = (4 - x) + 2 = 6 - x.$$

The radius of the inner circle is $g(x) = 2$. Therefore, we have

$$\begin{aligned} V &= \int_0^4 \pi \left[(6-x)^2 - (2)^2\right] \, dx \\ &= \pi \int_0^4 (x^2 - 12x + 32) \, dx = \pi \left[\frac{x^3}{3} - 6x^2 + 32x\right]\Big|_0^4 \\ &= \frac{160\pi}{3} \text{ units}^3. \end{aligned}$$

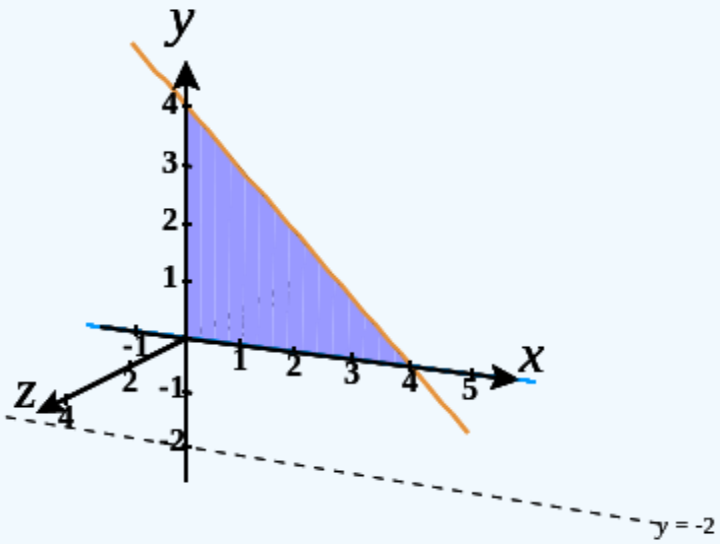


Figure 6.2.14: (c) A dynamic version of this solid of revolution generated using CalcPlot3D.

? Exercise 6.2.6

Find the volume of a solid of revolution formed by revolving the region bounded above by the graph of $f(x) = x + 2$ and below by the x**-axis** over the interval $[0, 3]$ around the line $y = -1$.

Hint

Use the procedure from Example 6.2.6.

Answer

60π units^3

Key Concepts

- Definite integrals can be used to find the volumes of solids. Using the slicing method, we can find a volume by integrating the cross-sectional area.
- For solids of revolution, the volume slices are often disks and the cross-sections are circles. The method of disks involves applying the method of slicing in the particular case in which the cross-sections are circles, and using the formula for the area of a circle.
- If a solid of revolution has a cavity in the center, the volume slices are washers. With the method of washers, the area of the inner circle is subtracted from the area of the outer circle before integrating.

Key Equations

- **Disk Method along the x-axis**

$$V = \int_a^b \pi \big[f(x)\big]^2 \, dx$$

- **Disk Method along the y-axis**

$$V = \int_c^d \pi \big[g(y)\big]^2 \, dy$$

- **Washer Method**

$$V = \int_a^b \pi \left[(f(x))^2 - (g(x))^2\right] \, dx$$

Glossary

cross-section
the intersection of a plane and a solid object

disk method
a special case of the slicing method used with solids of revolution when the slices are disks

slicing method
a method of calculating the volume of a solid that involves cutting the solid into pieces, estimating the volume of each piece, then adding these estimates to arrive at an estimate of the total volume; as the number of slices goes to infinity, this estimate becomes an integral that gives the exact value of the volume

solid of revolution
a solid generated by revolving a region in a plane around a line in that plane

washer method
a special case of the slicing method used with solids of revolution when the slices are washers

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6.2E: Exercises for Section 6.2

1) Derive the formula for the volume of a sphere using the slicing method.

2) Use the slicing method to derive the formula for the volume of a cone.

3) Use the slicing method to derive the formula for the volume of a tetrahedron with side length a.

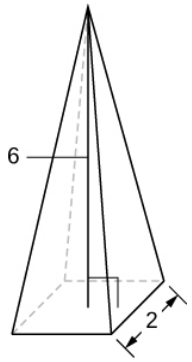
4) Use the disk method to derive the formula for the volume of a trapezoidal cylinder.

5) Explain when you would use the disk method versus the washer method. When are they interchangeable?

Volumes by Slicing

For exercises 6 - 10, draw a typical slice and find the volume using the slicing method for the given volume.

6) A pyramid with height 6 units and square base of side 2 units, as pictured here.



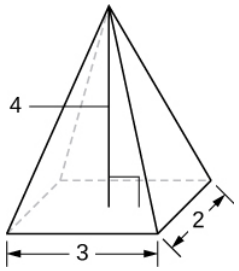
Solution:

Here the cross-sections are squares taken perpendicular to the y-axis.
We use the vertical cross-section of the pyramid through its center to obtain an equation relating x and y.
Here this would be the equation, $y = 6 - 6x$. Since we need the dimensions of the square at each y-level, we solve this equation for x to get, $x = 1 - \frac{y}{6}$.
This is half the distance across the square cross-section at the y-level, so the side length of the square cross-section is, $s = 2\left(1 - \frac{y}{6}\right)$.
Thus, we have the area of a cross-section is,

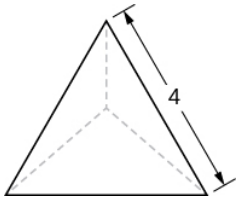
$$A(y) = \left[2\left(1 - \tfrac{y}{6}\right)\right]^2 = 4\left(1 - \tfrac{y}{6}\right)^2.$$

$$\begin{aligned}
\text{Then,}\quad V &= \int_0^6 4\left(1 - \tfrac{y}{6}\right)^2 dy \\
&= -24\int_1^0 u^2\, du, \quad \text{where } u = 1 - \tfrac{y}{6},\ \text{so } du = -\tfrac{1}{6}\, dy, \quad \Longrightarrow \quad -6\, du = dy \\
&= 24\int_0^1 u^2\, du = 24\frac{u^3}{3}\Big|_0^1 \\
&= 8u^3\Big|_0^1 \\
&= 8\left(1^3 - 0^3\right) \quad = \quad 8 \text{ units}^3
\end{aligned}$$

7) A pyramid with height 4 units and a rectangular base with length 2 units and width 3 units, as pictured here.



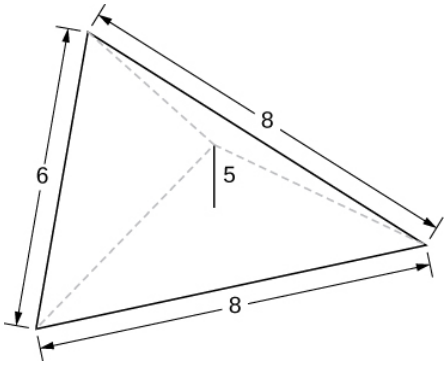
8) A tetrahedron with a base side of 4 units,as seen here.



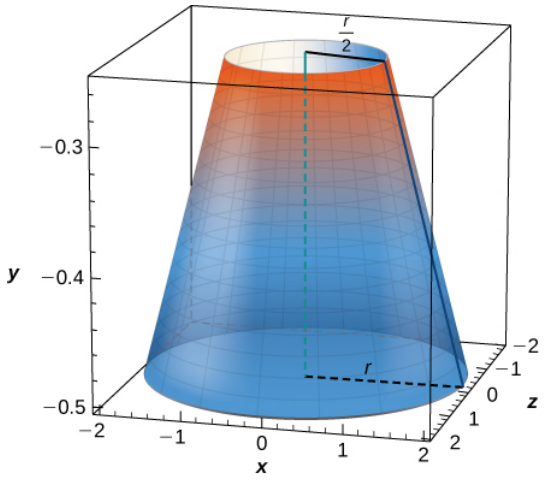
Answer

$V = \frac{32}{3\sqrt{2}} = \frac{16\sqrt{2}}{3}$ units^3

9) A pyramid with height 5 units, and an isosceles triangular base with lengths of 6 units and 8 units, as seen here.



10) A cone of radius r and height h has a smaller cone of radius $r/2$ and height $h/2$ removed from the top, as seen here. The resulting solid is called a frustum.



Answer

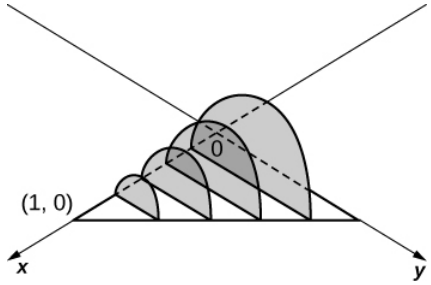
$V = \frac{7\pi}{12} hr^2$ units^3

For exercises 11 - 16, draw an outline of the solid and find the volume using the slicing method.

11) The base is a circle of radius a. The slices perpendicular to the base are squares.

12) The base is a triangle with vertices $(0,0), (1,0)$, and $(0,1)$. Slices perpendicular to the xy-plane are semicircles.

Answer

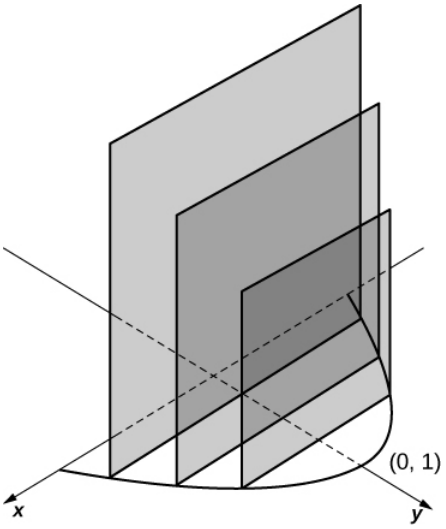


$$V = \int_0^1 \frac{\pi(1-x)^2}{8}\,dx \quad = \quad \frac{\pi}{24} \text{ units}^3$$

13) The base is the region under the parabola $y = 1 - x^2$ in the first quadrant. Slices perpendicular to the xy-plane are squares.

14) The base is the region under the parabola $y = 1 - x^2$ and above the x-**axis**. Slices perpendicular to the y-**axis** are squares.

Answer

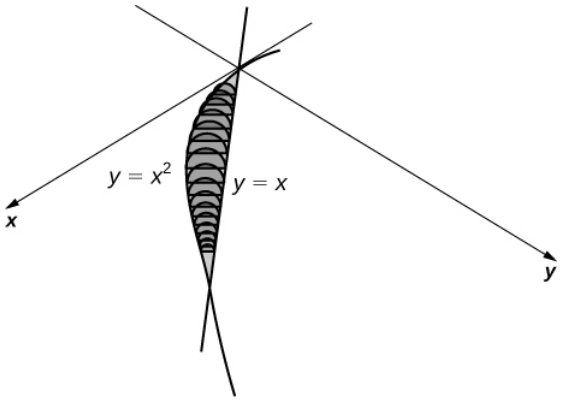


$$V = \int_0^1 4(1-y)\,dy \quad = \quad 2 \text{ units}^3$$

15) The base is the region enclosed by $y = x^2$ and $y = 9$. Slices perpendicular to the x-axis are right isosceles triangles.

16) The base is the area between $y = x$ and $y = x^2$. Slices perpendicular to the x-axis are semicircles.

Answer



$$V = \int_0^1 \frac{\pi}{8}(x - x^2)^2 \, dx \quad = \quad \frac{\pi}{240} \text{ units}^3$$

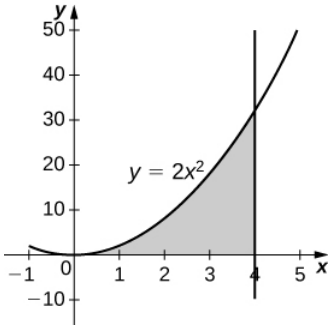
Disk and Washer Method

For exercises 17 - 24, draw the region bounded by the curves. Then, use the disk or washer method to find the volume when the region is rotated around the x-axis.

17) $x + y = 8, \quad x = 0$, and $y = 0$

18) $y = 2x^2, \quad x = 0, \quad x = 4$, and $y = 0$

Answer

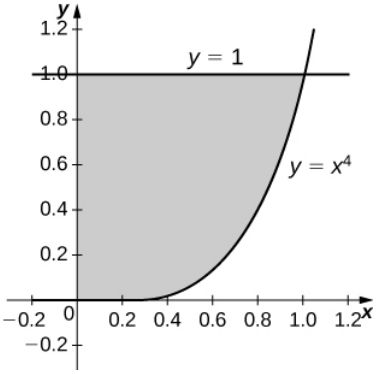


$$V = \int_0^4 4\pi x^4 \, dx \quad = \quad \frac{4096\pi}{5} \text{ units}^3$$

19) $y = e^x + 1, \quad x = 0, \quad x = 1$, and $y = 0$

20) $y = x^4, \quad x = 0$, and $y = 1$

Answer

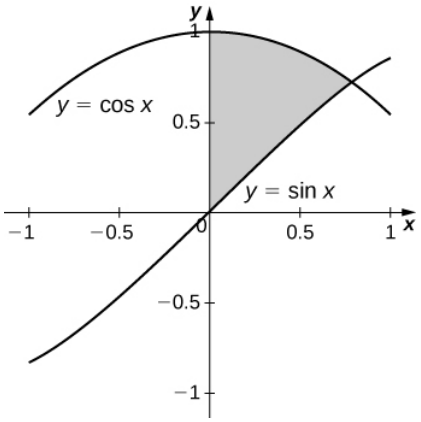


$$V = \int_0^1 \pi\left(1^2 - \left(x^4\right)^2\right)\,dx = \int_0^1 \pi\left(1 - x^8\right)\,dx \quad = \quad \frac{8\pi}{9} \text{ units}^3$$

21) $y = \sqrt{x}, \quad x = 0, \quad x = 4$, and $y = 0$

22) $y = \sin x, \quad y = \cos x$, and $x = 0$

Answer

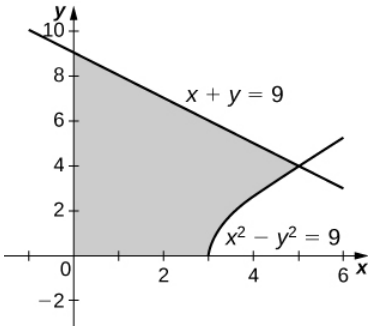


$$V = \int_0^{\pi/4} \pi\left(\cos^2 x - \sin^2 x\right)\,dx = \int_0^{\pi/4} \pi \cos 2x\,dx \quad = \quad \frac{\pi}{2} \text{ units}^3$$

23) $y = \dfrac{1}{x}, \quad x = 2$, and $y = 3$

24) $x^2 - y^2 = 9$ and $x + y = 9, \quad y = 0$ and $x = 0$

Answer



$V = 207\pi$ units^3

For exercises 25 - 32, draw the region bounded by the curves. Then, find the volume when the region is rotated around the y -axis.

25) $y = 4 - \dfrac{1}{2}x, \quad x = 0$, and $y = 0$

26) $y = 2x^3, \quad x = 0, \quad x = 1$, and $y = 0$

Answer